

OPTIMUM: OPTimising IMmunisation Using Mixed schedules

Clinical Study Report - Statistical Analyses

OPTIMUM study team

2026-08-28

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List of Abbreviations

AE	adverse event
AESI	adverse event of special interest
aP	acellular pertussis
CHW	The Children's Hospital at Westmead
DT	diphtheria toxoid
GMC	geometric mean concentration
GMFR	geometric mean fold rise
GMR	geometric mean ratio
GSD	geometric standard deviation
FHA	filamentous haemagglutinin
FIM2/3	fimbriae agglutinogens 2–3
HBsAg	hepatitis B surface antigen
Hib-PRP	capsular polysaccharide of Hib
IgE	immunoglobulin E
IgG	immunoglobulin G
IMP	investigational medicinal product
ITT	intention-to-treat
kU/L	kilounits per litre
MedDRA	Medical Dictionary for Regulatory Activities
MCRI	Murdoch Children's Research Institute
OFC	oral food challenge
PCH	Perth Children's Hospital
PnP	pneumococcal polysaccharide
PRN	pertactin
PT	anti-pertussis toxin
SAE	serious adverse event
SCHN	Sydney Children's Hospital Randwick
SPT	skin prick test
TT	anti-tetanus toxoid

wP whole-cell pertussis

1 Summary

The overall objectives of this study were to determine if, compared to the existing pertussis vaccine schedule of three priming doses of acellular pertussis (aP) vaccine, a single dose of whole-cell pertussis (wP) vaccine at approximately 2 months of age, followed by two doses of aP vaccine at approximately 4 and 6 months of age:

1. Protects against the development of IgE-mediated food allergy.
2. Protects against the development of an atopic immunophenotype, as evidenced by a reduction in the proportion of titres above the assay limit of detection or a reduction in the mean fold-rise in the log titres of total IgE, anti-tetanus toxoid (TT) IgE, and food-associated IgE in response to the third pertussis-containing vaccine at approximately age 6 months (stage 1).
3. Is associated with a protective pertussis antibody response, as evidenced by no reduction in the proportion of children who have at least a 4-fold rise in anti-pertussis toxin (PT) IgG antibody responses and no reduction in the mean fold-rise in PT-IgG, to a fourth pertussis-containing vaccine at approximately age 18 months (stage 1 and stage 2, Perth Children's Hospital (PCH)/The Kids Research Institute Australia¹ study site.)
4. Is associated with an acceptable reactogenicity profile (stage 1 and stage 2, PCH/The Kids Research Institute Australia study site).

1. (Section 4) There was little evidence of a substantial benefit of wP compared to aP at 2 months of age on protection against the development of IgE-mediated food allergy. For the primary outcome, the estimated difference [posterior median (95% CrI)] in IgE-mediated food allergy risk for wP compared to aP was 0.01 (-0.02, 0.04), that is anywhere from a 2%pt. reduction to a 4%pt. increase were plausible differences given the data.

For the secondary outcome of new-onset eczema by 6 and 12 months of age the estimated difference in risk for wP compared to aP was -0.01 (-0.04, 0.02) and -0.01 (-0.04, 0.03) respectively. For the secondary outcome of skin prick test positivity at >1mm and ≥3mm relative to negative control the estimated difference in risk for wP compared to aP was 0.01 (-0.04, 0.06) and 0.01 (-0.04, 0.05) respectively, both showing little evidence of reduction under the wP schedule compared to aP.

2. (Section 7.2) For total IgE the estimated [posterior median (95% CrI)] geometric mean ratio for wP compared to aP at 6-months was 0.99 (0.64, 1.51) and at 7-months was 0.92 (0.61, 1.40) equating to a geometric mean fold-rise ratio of 0.94 (0.72, 1.22). These estimates were imprecise, but show little evidence of a reduction in geometric mean concentrations.

For TT-IgE, geometric mean ratios could not be estimated precisely given a low number of detectable concentrations. Similarly for egg white IgE and whole egg IgE with 6-month geometric mean ratios estimated as 0.49 (0.07, 2.98) and 0.98 (0.21, 4.28) respectively, and 7-month ratios as 0.65 (0.16, 2.46) and 0.73 (0.17, 2.94) respectively.

The proportion of concentrations above detection limits were similar between the groups with the largest estimated reduction being in egg white IgE with an estimated difference in quantitation of -0.09 (-0.23, 0.04) at 6-months and -0.10 (-0.26, 0.05) at 7-months of age.

¹Formerly known as Telethon Kids Institute

3. (Section 7.3) The estimated geometric mean ratio [posterior median (95% CrI)] for PT IgG at 18-months and 19-months of age were 1.19 (0.94, 1.51) and 1.69 (1.33, 2.14) respectively, indicating very low probability of a reduction in IgG, and a probable increase. The corresponding geometric mean fold-rise ratio was 1.42 (1.09, 1.84).

There was low probability of a reduction in the probability of a 4-fold rise in PT IgG from 18-months to 21-35 days after vaccination with estimated difference [posterior median (95% CrI)] of 0.02 (-0.02, 0.07).

4. For descriptive summaries of solicited local reactions and solicited systemic reactions see Section 6 and for safety reporting summaries see Section 5.

2 Introduction

2.1 Purpose

This report summarises the data and results of the analyses for main outcomes of the OPTIMUM trial. The report includes results for:

- participant disposition
- participant baseline characteristics
- skin prick tests (SPTs)
- IgE-mediated food allergy
- eczema
- safety
- reactogenicity
- parental acceptability
- immunogenicity

For full details of the trial processes, objectives, and outcomes refer to the protocol¹ and statistical analysis plan².

2.2 Interventions

Participants were randomised 1:1 to receive either:

- Combined Diphtheria (D)-Tetanus (T)-acellular Pertussis (aP), Hepatitis B (HepB), Inactivated Poliovirus (IPV) and *Haemophilus influenzae* type b (Hib) vaccine (DTaP-HepB-IPV-Hib) administered at approximately 2 months (referred to as 3-aP + aP or aP for short), together with the vaccines recommended by the National Immunisation Program, or
- Combined Diphtheria (D)-Tetanus (T)-whole-cell Pertussis (wP), Hepatitis B (HepB) and *Haemophilus influenzae* type b (Hib) vaccine (DTwP-HepB-Hib) in place of DTaP-HepB-IPV-Hib at approximately 2 months (referred to as 1-wP/2-aP + aP or wP for short), together with the vaccines recommended by the National Immunisation Program.

2.3 Trial design

Originally, a maximum sample size of 3,000 participants was planned. Under the adaptive design, as outlined in the protocol¹ and statistical analysis plan², we estimated the trial had 85% power to detect a reduction in IgE-mediated food allergy from 10% to 7% with type I error of 5%. If δ denotes the difference in probability of IgE-mediated food allergy under each intervention, then trial success required that δ was very likely to be less than zero, specifically that $\Pr(\delta < 0) > 0.95$.

The maximum sample size was subsequently revised down to 2,000 participants due to feasibility constraints. The trial was stopped early due to an inability to procure investigational medicinal product (IMP) with 972 participants randomised.

2.4 Trial outcomes

A summary of trial outcomes is provided below.

2.4.1 Primary outcome (1)

IgE-mediated food allergy with evidence of food sensitisation on SPT by 12 months old and confirmed (where necessary) by medically supervised oral food challenge(s) (OFC). For the purpose of this trial, if either of the following conditions are met, the study participant was considered to have reached the primary endpoint:

1. Unequivocal evidence of IgE-mediated food allergy, defined as:
 - i. A positive OFC with evidence of sensitisation on SPT to the food of interest *OR*
 - ii. History of clinician-confirmed food anaphylaxis, with symptoms affecting at least two of the following: skin, gastrointestinal tract, respiratory tract, cardiovascular system; *AND* evidence of sensitisation on SPT to the food of interest.
2. Highly probable IgE-mediated food allergy defined as a history of food allergic reaction (consistent with PRACTALL criteria³), *AND* evidence of sensitisation on SPT to the food of interest.

We report on the number and proportion who satisfy this composite definition, and the number and proportion satisfying each of the three sub-components as per a hierarchy in order of 1.i, 1.ii, 2.

2.4.2 Secondary outcomes (2)

2.4.2.1 Clinical

- a. A history of parent-reported clinician-diagnosed new onset eczema by **6 months** *AND* a positive SPT to any allergen by approximately 12 months old.
- b. A history of parent-reported clinician-diagnosed new onset eczema by **12 months** *AND* a positive SPT to any allergen by approximately 12 months old.
- c. An SPT more than 1mm greater than (**>1mm**) the negative control to at least one allergen by approximately 12 months of age
- d. An SPT at least 3mm greater than (**≥3mm**) than the negative control to at least one allergen by approximately 12 months of age

2.4.2.2 Reactogenicity

Solicited and unsolicited adverse events following immunisation were collected on diary cards in the 7 days after administration of pertussis-containing vaccines for the first 300 participants at PCH. Solicited local adverse events included injection site pain, erythema, induration and swelling;

the solicited systemic adverse events recorded were fever $\geq 38^{\circ}\text{C}$ (axillary), drowsiness, irritability, anorexia, vomiting, diarrhoea and restlessness.

The definition and intensity grading for fever, diarrhoea and local reactions (except for immunisation site pain) followed the Brighton Collaboration guidelines⁴⁻⁸. For other solicited adverse events, intensity grading scales based on impact on daily activities were used (grade 0 or absent; grade 1 if easily tolerated; grade 2, sufficiently discomforting to interfere with normal everyday activities and grade 3, if prevents normal everyday activities or requires significant medical intervention).

Parental acceptability was also measured using a 5-point Likert scale for whether the parent would be willing to have another child have the combination of vaccinations at the given age.

2.4.2.3 Immunological outcomes

2.4.2.3.1 IgE

Total, TT and food allergen-specific IgE were measured in plasma at the age of 6, 7 and 19 months (stage 1 only), using ImmunoCAP total and specific IgE assays (Thermo Fisher Scientific/Phadia, Uppsala, Sweden).

Interest was in the geometric mean concentrations and geometric mean fold-rise following the third pertussis containing vaccine (i.e. from 6 to 7 months of age) and the proportion of IgE concentrations above the limit of detection (≥ 0.01 kU/L) on each occasion.

2.4.2.3.2 IgG

Vaccine antigen-specific IgG concentrations were measured immediately before and 21–35 days after the 6-month (stage 1 only) and 18-month vaccine doses. These antibody levels were measured using a multiplex fluorescent bead assay⁹⁻¹². For stage 1, sera were tested for IgG against *B. pertussis* antigens (PT, filamentous haemagglutinin (FHA), pertactin (PRN), fimbriae agglutinogens 2–3 (FIM2/3)), TT, capsular polysaccharide of Hib (Hib-PRP), diphtheria toxoid (DT), hepatitis B surface antigen (HBsAg), and each of the 13-valent pneumococcal vaccine serotypes (PnPs). For stage 2, sera were only tested for a subset of antigens and the 18 and/or 19-month ages.

Interest was in the geometric mean concentrations, seroresponse (13vPCV serotype-specific IgG ≥ 0.35 $\mu\text{g/mL}$, TT and DT ≥ 0.1 IU/mL, FHA, FIM2/3, PRN, and PT ≥ 5 IU/mL, HBsAg ≥ 10 mIU/mL, Hib-PRP ≥ 0.15 $\mu\text{g/mL}$ at 6 and 7 months old and ≥ 1 $\mu\text{g/mL}$ at 18 and 19 months old), and geometric mean fold-rise following the 18-month vaccination and the proportion of infants with at least a 4-fold rise, particularly for PT.

2.4.3 Additional outcome details

The protocol defined allergen panel for skin prick tests included cow's milk, hen's egg, peanut, cashew, house dust mite (*Dermatophagoides pteronyssinus*), cat, rye grass (*Lolium perenne*), a positive control (10 mg/mL histamine) and negative control (saline). In addition to these, further allergens

could also be tested at parental request or at discretion of the site principal investigator (PI). When the outcomes say “at least one allergen” [i.e. for eczema for positive SPT] it was meant any allergen, including any additional allergens tested beyond the standard panel.

For SPT’s, the specification of “by approximately 12-months old” was taken to mean “up to 18-months old”. Under the original protocol, SPTs were scheduled to occur at 18-months of age, but following an early revision, scheduled SPTs were revised to occur at 12-months of age. Unscheduled SPTs could still occur up to 18 months of age. Exploratory analyses investigated the magnitude of SPT results in addition to sensitisation.

For both IgE-mediated food allergy and eczema, sensitisation on (positive) SPT was defined to be a result more than 1mm greater than negative control by 18 months of age. For IgE-mediated food allergy, the positive SPT needed to match the specific food allergen.

For eczema, the age requirement (by 6 or 12 months of age) was determined by the “age at which allergy first occurred” (`outageval`). To be considered “new-onset” eczema, this age at which allergy first occurred needed to be after the infants enrolment date. Infants with clinician-diagnosed eczema first occurring prior to enrolment were to be excluded as cases which were not “new-onset eczema”.

2.5 Analyses

Unless otherwise stated, all summaries and analyses group participants according to their assigned treatment group and use all available data irrespective of protocol deviations or other intercurrent events (ITT).

Statistical analyses were conducted using Bayesian inference on assumed probabilistic models. The probabilistic model defines what observations are possible and how observations are related. Bayesian inference allocates and reallocates (in light of observed data) credibility across the possibilities. The result is a probability distribution over quantities of interest summarising uncertainty in their values where more plausible values have higher probability. These distributions are generally summarised by their median, 95% equal-tailed credible intervals (bounds for the central portion of the distribution which values contribute 95% of the probability), and probability of direction of effect.

2.5.1 Primary outcome

The primary analysis was to assume independent Binomial models for occurrence of IgE-mediated food allergy under each treatment group with uniform priors on the outcome probabilities.

i Primary model

Let θ_{WP} and θ_{AP} denote the probability of the outcome amongst infants who are assigned to the respective strategy and $\delta = \theta_{WP} - \theta_{AP}$ the difference in probability. If n_{WP} and n_{AP} infants

were assigned to each respective strategy, with y_{wP} and y_{aP} events observed, then from our assumed model we estimate that $\theta_j \sim \text{Beta}(1 + y_j, 1 + n_j - y_j)$ quantifies our uncertainty and δ the difference between these.

A secondary analysis was to assume a logistic regression model for occurrence of IgE-mediated food allergy adjusting for baseline factors: site, sex, breastfeeding status, birth order, parental income, mode of delivery, and family history of atopic disease.

2.5.2 Secondary outcomes (2) - clinical

Analyses of secondary clinical outcomes are analogous to those for the primary outcome, both the primary and secondary models.

2.5.3 Geometric mean concentrations

Both the IgE and IgG concentrations were analysed assuming multivariate normal models on the $\log_{10}$ transformed concentrations with an unstructured covariance matrix shared across all participants. Analyses were undertaken on the scale of the lab-reported units.

The geometric mean ratio (GMR) of wP relative to aP on each occasion was calculated for each allergen and antigen type. Additionally, geometric mean fold rise ratios (GMFRR) of wP relative to aP are calculated for relevant sequential occasions.

In the analysis of allergen-specific IgE concentrations, some values were reported as 0.00 kU/L (6-month and 7-month) or 0.000 kU/L (19-month). These values were treated as left-censored at 0.005 kU/L and 0.0005 kU/L respectively to satisfy the model condition that concentrations were above 0 for all participants.

All models adjusted for the same covariates as specified for the primary outcome. Effects were allowed to vary by study visit.

i Concentration model

For infants $i = 1, \dots, n$ measured on occasions $j = 1, \dots, J$ (at most $J = 4$: 6-month, 7-month, 18-month, and 19-month), with model covariates $p = 1, \dots, P$, the pre-specified model for concentration Y_i was

$$\begin{aligned}
\log_{10}(Y_i) &\sim \text{Normal}_J(\mu_i, \Sigma) \\
\mu_{ij} &= \alpha_j + x_{ij}^T \beta_j \\
\Sigma &= \text{diag}(\sigma) \Omega \text{diag}(\sigma) \\
\alpha_j &\sim \text{Normal}(0, 10^2), \quad j = 1, \dots, J \\
\beta_{jp} &\sim \text{Normal}(0, 10^2), \quad j = 1, \dots, J, \quad p = 1, \dots, P \\
\Omega &\sim \text{LKJ}(1) \\
\sigma_j &\sim \text{Cauchy}^+(0, 10), \quad j = 1, \dots, J.
\end{aligned}$$

2.5.4 Binary outcomes (quantitation, seropositivity, four-fold rise)

For binary outcomes, logistic regression models were assumed. If the outcome consists of multiple responses per participant then a participant-specific random intercept was included in the model. All models adjusted for the same covariates as specified for the primary outcome. Effects were allowed to vary by study visit.

i Binary outcome model

Denote by Y_{ij} the binary outcome for participant i on occasion j , then the model was

$$\begin{aligned}
Y_{ij} &\sim \text{Bernoulli}(\pi_{ij}) \\
\pi_{ij} &= \text{logit}^{-1}(\eta_{ij}) \\
\eta_{ij} &= \alpha + \omega_j + z_i \theta_j + \sum_{p=1}^P x_{ip} \beta_{jp} + \gamma_i \\
\gamma | \tau^2 &\sim \text{Normal}(0, \tau^2) \\
\omega_1 &= 0 \\
\alpha, \omega_{2:J}, \theta, \beta &\sim \text{Student-t}(4, 0, 1.75^2) \\
\tau &\sim \text{Exponential}(1)
\end{aligned}$$

2.5.5 Subgroup Analyses

No subgroup analyses were pre-specified in the protocol or SAP. Exploratory post-hoc subgroup analyses were performed and summarised for the primary outcome (IgE-mediated food allergy) according to:

- participating site of study enrolment
- season of birth (Autumn, Winter, Spring, Summer)
- family history of atopy
- family history of food allergy
- one or more parent's born in Asia

For these analyses logistic regression models were estimated with terms for assigned treatment group, the grouping factor of interest (as above), and their interaction. Weakly informative priors were specified on the overall treatment effect and on the group level deviations.

2.5.6 Quantity Definitions

Geometric mean concentration: For concentrations $x_1, \dots, x_n$, the geometric mean concentration (GMC) is just the geometric mean of the concentrations, i.e.

$$\text{GMC} = \exp \left(n^{-1} \sum_{i=1}^n \ln x_i \right).$$

Under a log-normal model, where $\ln X \sim \text{Normal}(\mu, \sigma^2)$, the geometric mean of X is just $\exp(\mu)$.

Geometric standard deviation: The geometric standard deviation (GSD) of a collection of values $x_1, \dots, x_n$ is

$$\text{GSD} = \exp \left(\sqrt{n^{-1} \sum_{i=1}^n \left(\ln \frac{x_i}{\text{GM}} \right)^2} \right)$$

where GM is the geometric mean of $x_1, \dots, x_n$.

Under a log-normal model, where $\ln X \sim \text{Normal}(\mu, \sigma^2)$, the geometric standard deviation of X is just $\exp(\sigma)$.

Geometric mean ratio: the geometric mean ratio (GMR) between two groups is the ratio of the geometric mean concentration in each group. For example, if GMC_{wP} is the GMC under the wP schedule, and GMC_{aP} under the aP schedule, then the GMR of wP compared to aP is

$$\text{GMR} = \frac{\text{GMC}_{\text{wP}}}{\text{GMC}_{\text{aP}}}.$$

Geometric mean fold-rise: the fold-rise is the ratio of a subject's pre and post-vaccination immunogenicity value. That is, if x_{i1} was their pre-vaccination concentration and x_{i2} their post-vaccination concentration, then their fold-rise was x_{i2}/x_{i1} . The geometric mean fold-rise (GMFR) is the geometric mean of a collection of fold-rises which also equals the ratio of the pre and post geometric mean concentrations, that is

$$\text{GMFR} = \exp \left(n^{-1} \sum_{i=1}^n \ln \frac{x_{i2}}{x_{i1}} \right) = \frac{\text{GMC}_2}{\text{GMC}_1}.$$

Geometric mean fold-rise ratio: the geometric mean fold-rise ratio (GMFRR) between two groups is the ratio of the geometric mean fold-rise in each group which also equals the ratio of the pre and post geometric mean ratios between the two groups. For example, if GMFR_{wP} is the GMFR under the wP schedule, and GMFR_{aP} under the aP schedule, then the GMFRR of wP compared to aP is

$$\text{GMFRR} = \frac{\text{GMFR}_{\text{wp}}}{\text{GMFR}_{\text{ap}}} = \frac{\text{GMR}_2}{\text{GMR}_1}.$$

2.5.7 Software

Summaries and analyses were performed in R (4.5.3)¹³. Where necessary, Bayesian inference was performed using Stan (2.38.0)¹⁴ via cmdstanr (0.9.0)¹⁵.

3 Study Participants - Disposition and Characteristics

3.1 Enrolment

Participants were enrolled at 4 sites in Australia. There were 972 participants randomised, 486 per treatment group. The first participant was randomised on 2018-03-07 and the final participant was randomised on 2023-12-22 Figure 1.

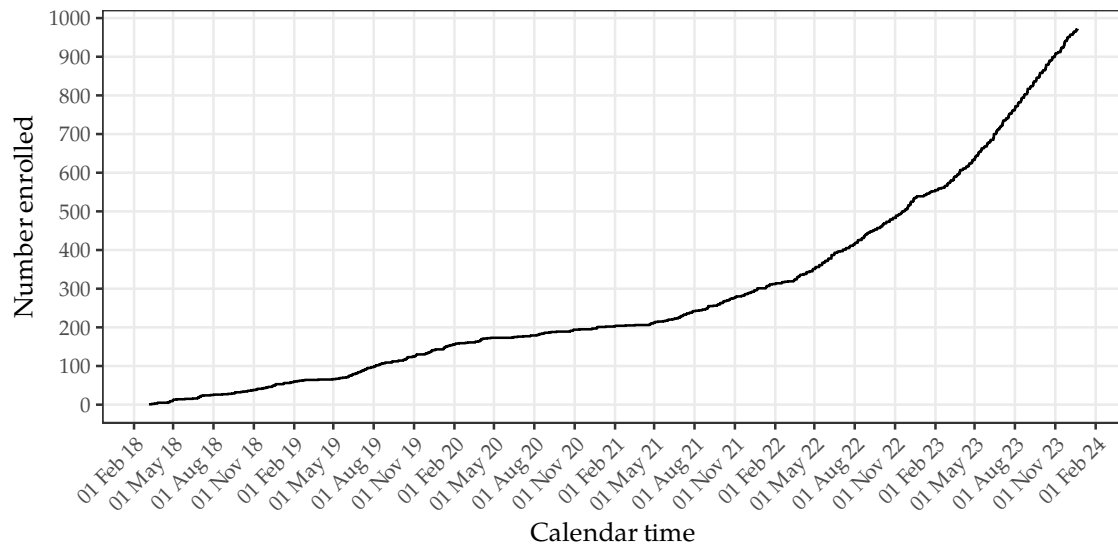


Figure 1: Cumulative trial enrolment by calendar time.

3.2 Study completion

The study completion status as reported in the database is summarised in Table 1 and by site in Table 2. Most of the *early termination - consent withdrawal* and *early termination - protocol violation/deviation* were due to the first 300 PCH participants who were scheduled for the additional blood collections, a number of whom withdrew consent for blood collection at the 18-month and 19-month visits (Table 2). Data was generally still collected for these participants with the exception of those blood collections, and so these participants may not have been “early termination” in the sense of having withdrawn from participating in the trial. There were 17 participants where contact was lost at or before 15 months of age (based on discdat) (Table 3).

Table 1: Number of early study withdrawals and associated reasons.

Participant status	Reason	aP N = 486	wP N = 486	Overall N = 972
Completed protocol	—	446 (91.8%)	453 (93.2%)	899 (92.5%)
Early termination	Consent withdrawal	12 (2.5%)	12 (2.5%)	24 (2.5%)
Early termination	Lost to follow-up	6 (1.2%)	5 (1.0%)	11 (1.1%)
Early termination	Moved from study area	7 (1.4%)	6 (1.2%)	13 (1.3%)
Early termination	Protocol violation/deviation ¹	15 (3.1%)	10 (2.1%)	25 (2.6%)

¹In most cases, this was due to no blood collection at visit 3 resulting in no visit 4. Only applicable at PCH.

Table 2: Number of early study withdrawals and associated reasons by study site.

Participant status	Reason	aP N = 486	wP N = 486	Overall N = 972
CHW				
Completed protocol	—	30 (100.0%)	29 (100.0%)	59 (100.0%)
MCRI				
Completed protocol	—	216 (98.2%)	216 (98.6%)	432 (98.4%)
Early termination	Consent withdrawal	2 (0.9%)	— (—)	2 (0.5%)
Early termination	Lost to follow-up	1 (0.5%)	1 (0.5%)	2 (0.5%)
Early termination	Moved from study area	1 (0.5%)	2 (0.9%)	3 (0.7%)
PCH				
Completed protocol	—	172 (83.5%)	181 (87.4%)	353 (85.5%)
Early termination	Consent withdrawal	10 (4.9%)	12 (5.8%)	22 (5.3%)
Early termination	Lost to follow-up	4 (1.9%)	2 (1.0%)	6 (1.5%)
Early termination	Moved from study area	5 (2.4%)	2 (1.0%)	7 (1.7%)
Early termination	Protocol violation/deviation ¹	15 (7.3%)	10 (4.8%)	25 (6.1%)
SCH				
Completed protocol	—	28 (93.3%)	27 (87.1%)	55 (90.2%)
Early termination	Lost to follow-up	1 (3.3%)	2 (6.5%)	3 (4.9%)
Early termination	Moved from study area	1 (3.3%)	2 (6.5%)	3 (4.9%)

¹In most cases, this was due to no blood collection at visit 3 resulting in no visit 4

Table 3: Listing of infants with last contact occurring at or prior to 15 months of age.

Record ID	Treatment	Withdrawal reason	Date of birth	Age last contact	Attend 12-month visit	Attend 18-month visit
5785-110	wP	Lost to follow-up	2022-05-08	1	FALSE	FALSE
5785-132	aP	Consent withdrawal	2022-06-25	3	FALSE	FALSE
30	wP	Consent withdrawal	2018-08-06	6	FALSE	FALSE
5785-404	aP	Consent withdrawal	2023-09-06	8	FALSE	FALSE
59	aP	Lost to follow-up	2018-12-18	9	FALSE	FALSE
4629-177	wP	Consent withdrawal	2023-01-28	11	FALSE	FALSE
141	aP	Consent withdrawal	2019-10-21	12	FALSE	FALSE
4629-150	aP	Moved from study area	2022-10-25	12	TRUE	FALSE
5785-433	wP	Moved from study area	2023-10-23	12	TRUE	FALSE
8644-18	aP	Moved from study area	2023-04-07	12	FALSE	FALSE
4629-184	aP	Moved from study area	2023-02-15	13	FALSE	FALSE
8644-22	wP	Lost to follow-up	2023-05-01	13	FALSE	FALSE
8644-28	wP	Lost to follow-up	2023-05-22	13	FALSE	FALSE
4629-65	wP	Consent withdrawal	2021-04-07	14	FALSE	FALSE
8644-12	wP	Moved from study area	2023-03-28	14	TRUE	FALSE
75	aP	Consent withdrawal	2019-04-18	15	FALSE	FALSE
4629-232	wP	Consent withdrawal	2023-07-31	15	FALSE	FALSE

3.3 Protocol deviations

Full deviations are documented in the site deviation log. The following reports only on those of particular relevance to the analyses.

3.3.1 Treatment administration

All participants received the first scheduled vaccination at the first visit.

For one participant the pharmacy reportedly dispensed the alternative study vaccine to what the participant had been randomly allocated. The IMP was monitored for all participants, and this was the only documented occasion that this was reported to have occurred. This error was documented in the protocol deviations log [subject 02-333, record id 4629-184]. This participant was analysed as assigned rather than as treated. Given there was only one such case, the analyses were not repeated with their treatment group switched to as treated.

3.3.2 Completed visits

The majority of participants attended each study visit Table 4, Table 5.

Table 4: Number and proportion of infants who attended each visit in stage 1 (based on presence of a reported “visit date”). 6-month + 72 hour was an optional visit.

Visit	aP N = 75	wP N = 75	Overall N = 150
4-month	75 (100%)	75 (100%)	150 (100%)
6-month	75 (100%)	74 (99%)	149 (99%)
6-month + 72 hrs	32 (43%)	25 (33%)	57 (38%)
7-month	73 (97%)	74 (99%)	147 (98%)
12-month	70 (93%)	73 (97%)	143 (95%)
18-month	72 (96%)	71 (95%)	143 (95%)
19-month	70 (93%)	69 (92%)	139 (93%)

Table 5: Number and proportion of infants who attended each visit in stage 2 (based on presence of a reported “visit date”).

Visit	aP N = 411	wP N = 411	Overall N = 822
12-month	404 (98%)	400 (97%)	804 (98%)
18-month	398 (97%)	399 (97%)	797 (97%)
19-month ¹	46 (11%)	58 (14%)	104 (13%)

¹19-month visit was for first 150 at PCH only: 46/75 (61%) aP and 58/75 (77%) wP of PCH attended.

Table 6: Amongst stage 1 participants who attended the visit, what percentage were reportedly in the protocolised visit window (windyn). Cells are y/n (p%) where y is number in window, n number who attended, and p percentage of attendees in window.

Visit age	aP	wP
4-month	73/75 (97%)	75/75 (100%)
6-month	75/75 (100%)	73/74 (99%)
6-month + 72 hrs	32/32 (100%)	25/25 (100%)
7-month	71/73 (97%)	71/74 (96%)
12-month	61/70 (87%)	63/73 (86%)
18-month	67/72 (93%)	70/71 (99%)
19-month	63/70 (90%)	62/69 (90%)

Table 7: Amongst stage 2 participants who attended the visit, what percentage were reportedly (windyn) in the protocolised visit window. Cells are y/n (p%) where y is number in window, n number who attended, and p percentage of attendees in window.

Visit age	aP	wP
12-month	399/404 (99%)	394/400 (98%)
18-month	396/398 (99%)	395/399 (99%)
19-month	46/46 (100%)	55/58 (95%)

3.3.3 Food and household questionnaires

The majority of food and household questionnaires were completed.

Table 8: Number and proportion of infants who completed food and household questionnaire for each occasion in stage 1.

Visit	aP N = 75	wP N = 75	Overall N = 150
6-week	75 (100%)	75 (100%)	150 (100%)
4-month	75 (100%)	75 (100%)	150 (100%)
6-month	75 (100%)	74 (99%)	149 (99%)
7-month	73 (97%)	74 (99%)	147 (98%)
9-month	74 (99%)	74 (99%)	148 (99%)
12-month	70 (93%)	73 (97%)	143 (95%)
15-month	70 (93%)	73 (97%)	143 (95%)
18-month	71 (95%)	71 (95%)	142 (95%)
19-month	70 (93%)	69 (92%)	139 (93%)

Table 9: Number and proportion of infants who completed food and household questionnaire for each occasion in stage 2.

Visit	aP N = 411	wP N = 411	Overall N = 822
6-week	411 (100%)	411 (100%)	822 (100%)
6-month	401 (98%)	405 (99%)	806 (98%)
9-month	400 (97%)	406 (99%)	806 (98%)
12-month	404 (98%)	404 (98%)	808 (98%)
15-month	384 (93%)	394 (96%)	778 (95%)
18-month	399 (97%)	401 (98%)	800 (97%)

3.4 Baseline characteristics

The baseline characteristics of all participants by their randomly assigned treatment group are reported in Table 10. Characteristics were similar between the groups.

Table 10: Baseline characteristics of trial participants.

Characteristic	N	aP N = 486 ¹	wP N = 486 ¹	Overall N = 972 ¹
Protocol stage	972			
1		75 (15%)	75 (15%)	150 (15%)
2		411 (85%)	411 (85%)	822 (85%)
Participating site	972			
CHW		30 (6%)	29 (6%)	59 (6%)
MCRI		220 (45%)	219 (45%)	439 (45%)
PCH		206 (42%)	207 (43%)	413 (42%)
SCH		30 (6%)	31 (6%)	61 (6%)
Gender	972			
Female		221 (45%)	242 (50%)	463 (48%)
Male		265 (55%)	244 (50%)	509 (52%)
Age at enrolment (days)	972	47 (44, 50)	47 (44, 50)	47 (44, 50)
Combined parental income status	971			
< \$18,000		1 (0%)	0 (0%)	1 (0%)
\$18,000 to \$37,000		1 (0%)	4 (1%)	5 (1%)
\$37,001 to \$87,000		22 (5%)	20 (4%)	42 (4%)
\$87,001 to \$180,000		197 (41%)	222 (46%)	419 (43%)
> \$180,000		264 (54%)	240 (49%)	504 (52%)
(Missing)		1	0	1
Birth country of mother	959			
Australia		372 (78%)	378 (79%)	750 (78%)
United Kingdom		17 (4%)	17 (4%)	34 (4%)
New Zealand		14 (3%)	10 (2%)	24 (3%)
Malaysia		4 (1%)	13 (3%)	17 (2%)
Ireland		5 (1%)	6 (1%)	11 (1%)
China (excludes SARs and Taiwan)		4 (1%)	6 (1%)	10 (1%)
Other		64 (13%)	49 (10%)	113 (12%)
(Missing)		6	7	13
Birth country of father	953			
Australia		362 (76%)	350 (74%)	712 (75%)
United Kingdom		28 (6%)	21 (4%)	49 (5%)
New Zealand		13 (3%)	20 (4%)	33 (3%)
Ireland		8 (2%)	10 (2%)	18 (2%)
India		7 (1%)	4 (1%)	11 (1%)
Other		60 (13%)	70 (15%)	130 (14%)
(Missing)		8	11	19
Mother's ethnicity²	972			
European Caucasian		425 (87%)	405 (83%)	830 (85%)
Asian		46 (9%)	68 (14%)	114 (12%)
Indian subcontinent		10 (2%)	17 (3%)	27 (3%)
South American		7 (1%)	4 (1%)	11 (1%)
Indigenous Australia		6 (1%)	1 (0%)	7 (1%)
Pacific Islander		2 (0%)	4 (1%)	6 (1%)
Black African		0 (0%)	2 (0%)	2 (0%)
Father's ethnicity²	972			
European Caucasian		433 (89%)	414 (85%)	847 (87%)
Asian		34 (7%)	44 (9%)	78 (8%)
Indian subcontinent		13 (3%)	15 (3%)	28 (3%)
Pacific Islander		6 (1%)	11 (2%)	17 (2%)
South American		7 (1%)	4 (1%)	11 (1%)
Indigenous Australia		3 (1%)	3 (1%)	6 (1%)
Black African		1 (0%)	3 (1%)	4 (0%)
Infant's ethnicity²	972			

European Caucasian		449 (92%)	442 (91%)	891 (92%)
Asian		51 (10%)	76 (16%)	127 (13%)
Indian subcontinent		17 (3%)	21 (4%)	38 (4%)
South American		11 (2%)	6 (1%)	17 (2%)
Pacific Islander		4 (1%)	10 (2%)	14 (1%)
Indigenous Australia		8 (2%)	5 (1%)	13 (1%)
Black African		1 (0%)	4 (1%)	5 (1%)
First born	972	278 (57%)	273 (56%)	551 (57%)
Feeding status	972			
Both breastfed and formula-fed		111 (23%)	140 (29%)	251 (26%)
Both breastfed and started on solids		1 (0%)	0 (0%)	1 (0%)
Exclusively breastfed		343 (71%)	321 (66%)	664 (68%)
Exclusively formula-fed		31 (6%)	25 (5%)	56 (6%)
Cat ownership	972			
No		373 (77%)	374 (77%)	747 (77%)
Inside		57 (12%)	62 (13%)	119 (12%)
Outside		2 (0%)	3 (1%)	5 (1%)
Both		54 (11%)	47 (10%)	101 (10%)
Dog ownership	971			
No		239 (49%)	278 (57%)	517 (53%)
Inside		57 (12%)	48 (10%)	105 (11%)
Outside		12 (2%)	10 (2%)	22 (2%)
Both		177 (36%)	150 (31%)	327 (34%)
(Missing)		1	0	1
Maternal gravidity	971			
1		213 (44%)	205 (42%)	418 (43%)
2		144 (30%)	154 (32%)	298 (31%)
3		69 (14%)	70 (14%)	139 (14%)
4		24 (5%)	32 (7%)	56 (6%)
5		16 (3%)	5 (1%)	21 (2%)
6+		19 (4%)	20 (4%)	39 (4%)
(Missing)		1	0	1
Maternal parity	971			
1		284 (59%)	279 (57%)	563 (58%)
2		150 (31%)	166 (34%)	316 (33%)
3		41 (8%)	30 (6%)	71 (7%)
4+		10 (2%)	11 (2%)	21 (2%)
(Missing)		1	0	1
Delivery type	971			
Elective caesarean section		116 (24%)	121 (25%)	237 (24%)
Emergency caesarean section		97 (20%)	108 (22%)	205 (21%)
Forceps/vacuum assisted delivery		69 (14%)	67 (14%)	136 (14%)
Vaginal delivery		204 (42%)	189 (39%)	393 (40%)
(Missing)		0	1	1
Intrapartum Antibiotics?	962	166 (35%)	146 (30%)	312 (32%)
(Missing)		6	4	10
Neonatal systemic antibiotics (first 28 days of life)?	971	41 (8%)	35 (7%)	76 (8%)
(Missing)		1	0	1
Gestational age at delivery (weeks)	972	39 (38, 40)	39 (38, 39)	39 (38, 39)
Weight (grams)	970	3,420 (3,140, 3,750)	3,400 (3,100, 3,760)	3,405 (3,120, 3,752)
(Missing)		2	0	2
Length (cm)	965	51 (49, 52)	50 (49, 52)	50 (49, 52)
(Missing)		4	3	7
Head Circumference (cm)	963	35 (34, 36)	35 (34, 36)	35 (34, 36)
(Missing)		5	4	9
Apgar score (1 minute)	963			
1-7		56 (12%)	65 (13%)	121 (13%)
8		51 (11%)	67 (14%)	118 (12%)
9		372 (78%)	351 (73%)	723 (75%)
10		1 (0%)	0 (0%)	1 (0%)
(Missing)		6	3	9
Apgar score (5 minutes)	963			
1-7		6 (1%)	10 (2%)	16 (2%)

8		14 (3%)	20 (4%)	34 (4%)
9		434 (90%)	429 (89%)	863 (90%)
10		26 (5%)	24 (5%)	50 (5%)
(Missing)		6	3	9
Maternal pertussis vaccination during preceding pregnancy?	971	465 (96%)	459 (95%)	924 (95%)
(Missing)		0	1	1
Maternal pertussis Vaccine administered	909			
Adacel		169 (37%)	161 (35%)	330 (36%)
Boostrix		238 (52%)	240 (53%)	478 (53%)
Unknown		48 (11%)	53 (12%)	101 (11%)
(Missing)		31	32	63
Other maternal pertussis vaccination (last 5 years)	968	240 (50%)	238 (49%)	478 (49%)
(Missing)		2	2	4
Maternal seasonal influenza vaccination during pregnancy?	966	381 (79%)	369 (77%)	750 (78%)
(Missing)		1	5	6
Family history of atopy	972	418 (86%)	415 (85%)	833 (86%)

¹n (%); Median (Q1, Q3)

²Multiple may apply to each participant

Reactogenicity and immunogenicity data was only collected for the first 300 participants at PCH. The baseline characteristics of this cohort of participants by their randomly assigned treatment group are reported in Table 11.

Table 11: Baseline characteristics of trial participants.

Characteristic	N	aP N = 150 ¹	wP N = 150 ¹	Overall N = 300 ¹
Protocol stage	300			
1		75 (50%)	75 (50%)	150 (50%)
2		75 (50%)	75 (50%)	150 (50%)
Gender	300			
Female		70 (47%)	71 (47%)	141 (47%)
Male		80 (53%)	79 (53%)	159 (53%)
Age at enrolment (days)	300	47 (44, 49)	46 (44, 49)	47 (44, 49)
Combined parental income status	300			
< \$18,000		1 (1%)	0 (0%)	1 (0%)
\$18,000 to \$37,000		0 (0%)	1 (1%)	1 (0%)
\$37,001 to \$87,000		8 (5%)	2 (1%)	10 (3%)
\$87,001 to \$180,000		60 (40%)	80 (53%)	140 (47%)
> \$180,000		81 (54%)	67 (45%)	148 (49%)
Birth country of mother	288			
Australia		110 (76%)	112 (78%)	222 (77%)
United Kingdom		6 (4%)	6 (4%)	12 (4%)
New Zealand		6 (4%)	1 (1%)	7 (2%)
Other		22 (15%)	25 (17%)	47 (16%)
(Missing)		6	6	12
Birth country of father	287			
Australia		113 (78%)	108 (76%)	221 (77%)
United Kingdom		10 (7%)	10 (7%)	20 (7%)
Ireland		3 (2%)	7 (5%)	10 (3%)
New Zealand		4 (3%)	6 (4%)	10 (3%)
Other		14 (10%)	12 (8%)	26 (9%)
(Missing)		6	7	13
Mother's ethnicity²	300			
European Caucasian		137 (91%)	128 (85%)	265 (88%)
Asian		11 (7%)	19 (13%)	30 (10%)
Indian subcontinent		3 (2%)	6 (4%)	9 (3%)
South American		1 (1%)	3 (2%)	4 (1%)
Pacific Islander		0 (0%)	1 (1%)	1 (0%)
Black African		0 (0%)	1 (1%)	1 (0%)
Indigenous Australia		0 (0%)	0 (0%)	0 (0%)
Father's ethnicity²	300			
European Caucasian		141 (94%)	141 (94%)	282 (94%)
Asian		7 (5%)	8 (5%)	15 (5%)
Indian subcontinent		4 (3%)	0 (0%)	4 (1%)
Indigenous Australia		1 (1%)	2 (1%)	3 (1%)
Black African		1 (1%)	1 (1%)	2 (1%)
Pacific Islander		0 (0%)	1 (1%)	1 (0%)
South American		0 (0%)	1 (1%)	1 (0%)
Infant's ethnicity²	300			
European Caucasian		145 (97%)	146 (97%)	291 (97%)
Asian		13 (9%)	24 (16%)	37 (12%)
Indian subcontinent		6 (4%)	4 (3%)	10 (3%)
South American		1 (1%)	3 (2%)	4 (1%)
Indigenous Australia		1 (1%)	2 (1%)	3 (1%)
Black African		1 (1%)	2 (1%)	3 (1%)
Pacific Islander		0 (0%)	2 (1%)	2 (1%)
First born	300	85 (57%)	81 (54%)	166 (55%)
Feeding status	300			
Both breastfed and formula-fed		39 (26%)	44 (29%)	83 (28%)
Both breastfed and started on solids		1 (1%)	0 (0%)	1 (0%)

Exclusively breastfed		96 (64%)	98 (65%)	194 (65%)
Exclusively formula-fed		14 (9%)	8 (5%)	22 (7%)
Cat ownership	300			
No		116 (77%)	123 (82%)	239 (80%)
Inside		12 (8%)	8 (5%)	20 (7%)
Outside		2 (1%)	0 (0%)	2 (1%)
Both		20 (13%)	19 (13%)	39 (13%)
Dog ownership	300			
No		66 (44%)	78 (52%)	144 (48%)
Inside		28 (19%)	16 (11%)	44 (15%)
Outside		5 (3%)	3 (2%)	8 (3%)
Both		51 (34%)	53 (35%)	104 (35%)
Maternal gravidity	300			
1		69 (46%)	56 (37%)	125 (42%)
2		44 (29%)	44 (29%)	88 (29%)
3		20 (13%)	28 (19%)	48 (16%)
4		7 (5%)	14 (9%)	21 (7%)
5		7 (5%)	3 (2%)	10 (3%)
6+		3 (2%)	5 (3%)	8 (3%)
Maternal parity	300			
1		88 (59%)	84 (56%)	172 (57%)
2		47 (31%)	52 (35%)	99 (33%)
3		11 (7%)	10 (7%)	21 (7%)
4+		4 (3%)	4 (3%)	8 (3%)
Delivery type	300			
Elective caesarean section		47 (31%)	37 (25%)	84 (28%)
Emergency caesarean section		27 (18%)	35 (23%)	62 (21%)
Forceps/vacuum assisted delivery		25 (17%)	22 (15%)	47 (16%)
Vaginal delivery		51 (34%)	56 (37%)	107 (36%)
Intrapartum Antibiotics?	299	55 (37%)	51 (34%)	106 (35%)
(Missing)		0	1	1
Neonatal systemic antibiotics (first 28 days of life)?	300	17 (11%)	11 (7%)	28 (9%)
Gestational age at delivery (weeks)	300			
34		1 (1%)	1 (1%)	2 (1%)
35		0 (0%)	1 (1%)	1 (0%)
36		3 (2%)	3 (2%)	6 (2%)
37		18 (12%)	18 (12%)	36 (12%)
38		52 (35%)	50 (33%)	102 (34%)
39		54 (36%)	48 (32%)	102 (34%)
40		19 (13%)	23 (15%)	42 (14%)
41		3 (2%)	6 (4%)	9 (3%)
Weight (grams)	299	3,350 (3,047, 3,610)	3,415 (3,123, 3,770)	3,370 (3,090, 3,664)
(Missing)		1	0	1
Length (cm)	299	50 (49, 51)	50 (49, 52)	50 (49, 52)
(Missing)		1	0	1
Head Circumference (cm)	299	35 (34, 36)	35 (34, 36)	35 (34, 36)
(Missing)		1	0	1
Apgar score (1 minute)	298			
1-7		15 (10%)	22 (15%)	37 (12%)
8		12 (8%)	11 (7%)	23 (8%)
9		120 (81%)	117 (78%)	237 (80%)
10		1 (1%)	0 (0%)	1 (0%)
(Missing)		2	0	2
Apgar score (5 minutes)	298			
1-7		0 (0%)	5 (3%)	5 (2%)
8		6 (4%)	7 (5%)	13 (4%)
9		134 (91%)	133 (89%)	267 (90%)
10		8 (5%)	5 (3%)	13 (4%)
(Missing)		2	0	2
Maternal pertussis vaccination during preceding pregnancy?	300	149 (99%)	144 (96%)	293 (98%)
Maternal pertussis Vaccine administered	291			

Adacel		52 (35%)	50 (35%)	102 (35%)
Boostrix		75 (51%)	74 (51%)	149 (51%)
Unknown		20 (14%)	20 (14%)	40 (14%)
(Missing)		3	6	9
Other maternal pertussis vaccination (last 5 years)	300	86 (57%)	87 (58%)	173 (58%)
Maternal seasonal influenza vaccination during pregnancy?	298	126 (85%)	121 (81%)	247 (83%)
(Missing)		1	1	2
Family history of atopy	300	126 (84%)	122 (81%)	248 (83%)

¹n (%); Median (Q1, Q3)

²Multiple may apply to each participant

3.5 Introduction of solid foods

For some foods, there are multiple sources of information from the food and household questionnaire. The survey first asked: “does your child currently eat egg?”, “does your child currently eat/drink cow’s milk?”, and “does your child currently eat peanuts/other nuts?”.

The survey subsequently asked for each of: rice cereal, cow’s milk, fruit, vegetables, meat, fish, egg, and peanut, whether that food had been given for the first time since the last visit, and if so, the date it was first given and the quantity that was given.

3.5.1 Egg, cow’s milk, and nuts

On the food and household questionnaire, parents were asked whether their child *currently* eats egg, cow’s milk, or peanuts/other nuts. Table 12 and Table 13 report on the responses to these questions by treatment group and the visit occasion (scheduled age) for stage 1 and stage 2 participants respectively. Ages are for the scheduled age of assessment, not necessarily the age at which that survey response was obtained. Proportions and denominators only include those participants who responded to the survey and so vary with the scheduled age of assessment.

Table 12: Number and percentage of infants who consume egg, cow’s milk, and peanut by treatment group and visit age, stage 1 participants.

Treatment group and item	6-month N = 75 ¹	7-month N = 75 ¹	9-month N = 75 ¹	12-month N = 75 ¹	15-month N = 75 ¹	18-month N = 75 ¹	19-month N = 75 ¹
aP							
Does your child currently eat egg?	29/75 (39)	52/74 (70)	73/74 (99)	72/72 (100)	70/70 (100)	71/71 (100)	69/70 (99)
Does your child currently eat/drink cow’s milk?	33/75 (44)	51/74 (69)	66/74 (89)	72/72 (100)	70/70 (100)	71/71 (100)	69/70 (99)
Does your child currently eat peanuts/other nuts?	19/75 (25)	43/74 (58)	66/74 (89)	68/72 (94)	70/70 (100)	71/71 (100)	68/70 (97)
wP							
Does your child currently eat egg?	34/74 (46)	51/74 (69)	67/74 (91)	71/74 (96)	71/73 (97)	69/71 (97)	69/69 (100)
Does your child currently eat/drink cow’s milk?	41/74 (55)	63/74 (85)	68/74 (92)	71/74 (96)	71/73 (97)	69/71 (97)	68/69 (99)
Does your child currently eat peanuts/other nuts?	26/74 (35)	42/74 (57)	59/74 (80)	69/74 (93)	69/73 (95)	65/71 (92)	65/69 (94)

¹n/N (%)

Table 13: Number and percentage of infants who consume egg, cow’s milk, and peanut by treatment group and visit age, stage 2 participants.

Treatment group and item	6-month N = 411 ¹	9-month N = 411 ¹	12-month N = 411 ¹	15-month N = 411 ¹	18-month N = 411 ¹
aP					
Does your child currently eat egg?	137/401 (34)	333/400 (83)	364/405 (90)	357/384 (93)	376/400 (94)
Does your child currently eat/drink cow’s milk?	102/401 (25)	253/400 (63)	328/405 (81)	342/384 (89)	363/400 (91)
Does your child currently eat peanuts/other nuts?	122/401 (30)	331/400 (83)	367/405 (91)	369/384 (96)	381/400 (95)
wP					
Does your child currently eat egg?	125/406 (31)	342/406 (84)	359/404 (89)	375/394 (95)	372/401 (93)
Does your child currently eat/drink cow’s milk?	98/406 (24)	260/406 (64)	321/404 (79)	352/394 (89)	361/401 (90)
Does your child currently eat peanuts/other nuts?	106/406 (26)	336/406 (83)	361/404 (89)	378/394 (96)	382/401 (95)

¹n/N (%)

3.5.2 First time foods

Parents were subsequently asked whether any of the specified foods had been introduced since the previous visit, and if so, when they were introduced and what quantity was given. In stage 1, rather than a date of introduction, the survey collected the *age* of introduction in days. In stage 2, only the *date* of introduction was elicited. For stage 1 responses, the reported age in days was used unless no record of introduction existed, in which case the response was treated as right-censored at the date of the last completed food and household questionnaire.

In some cases, the stage 2 date at which the food was introduced (`fefooddat`) contained clear errors (e.g. date was before date of birth or after the survey completion date). To calculate age (in days) at which the food was first introduced we first corrected any obvious errors in dates and then took the difference between the infant's date of birth and the minimum of `fefooddat` (date food was first given) and `fedat` (date survey was completed). Again, not all infants had a record of food introduction. For infants with no responses of yes for the food in question the age of introduction was treated as right censored at the date of the last completed food and household questionnaire.

The `fefooddat` data are reported as documented without cross-checking against other available fields given that it is only possible to cross-check for cow's milk, egg, and nuts. Some inconsistencies were noted between the responses to these items and those from the previous section. For example, where `femilyn` is "Yes" to all surveys from a certain date but `fefoodng2` (has cow's milk been introduced since last visit) is "No" or "N/A" on all questionnaires and `fefooddat2` (date cow's milk was introduced) is missing. Missing dates were treated as right-censored which may not be accurate (e.g. food could have been introduced by xx-months of age but date of introduction was not documented on any questionnaire).

Kaplan-Meier curves for age at which a food was introduced for stage 1 participants stratified by the assigned treatment group are shown in Figure 2. Curves for stage 2 participants are shown in Figure 3.

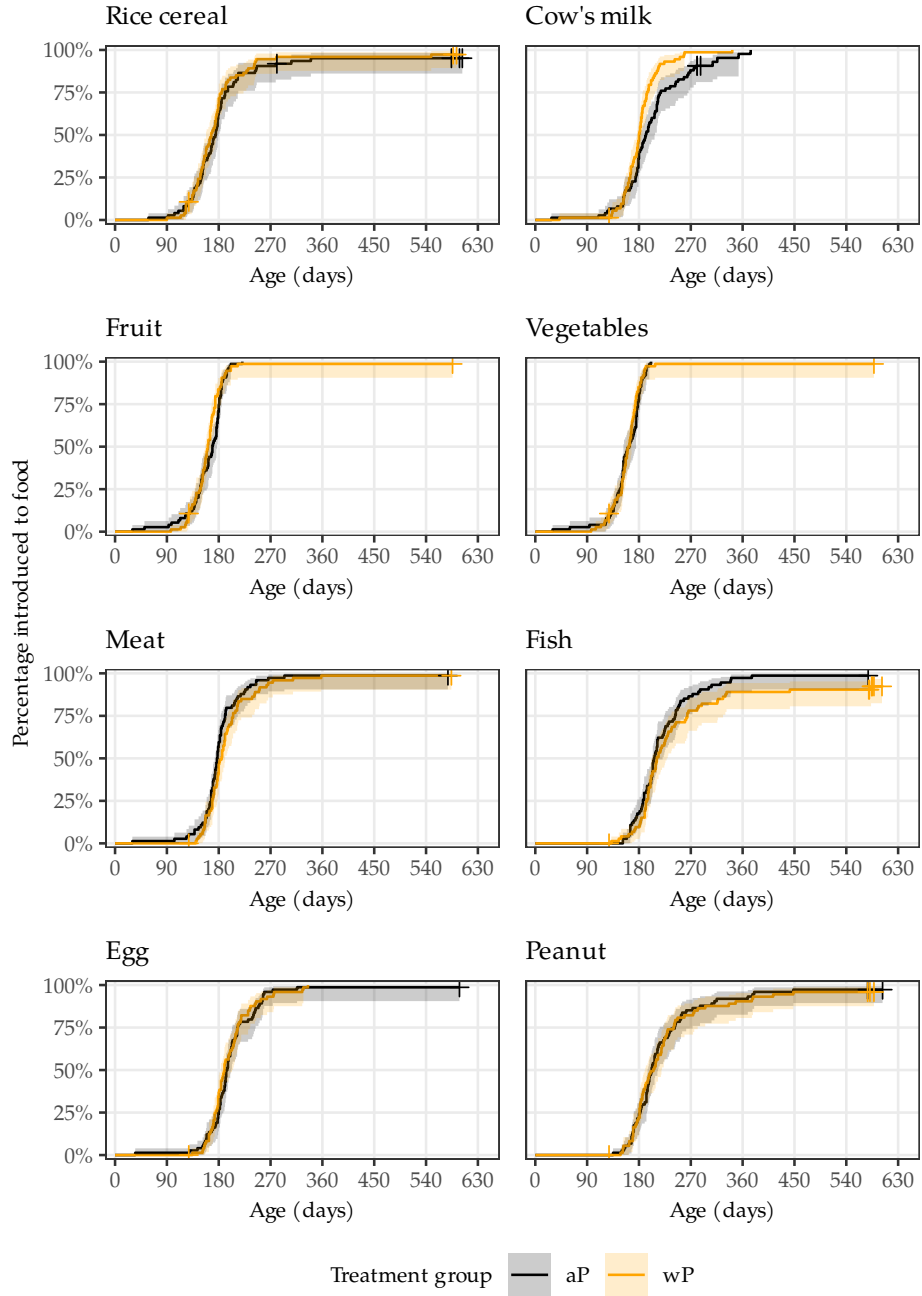


Figure 2: Age of food introduction for stage 1 participants. Censored observations are marked by a cross.

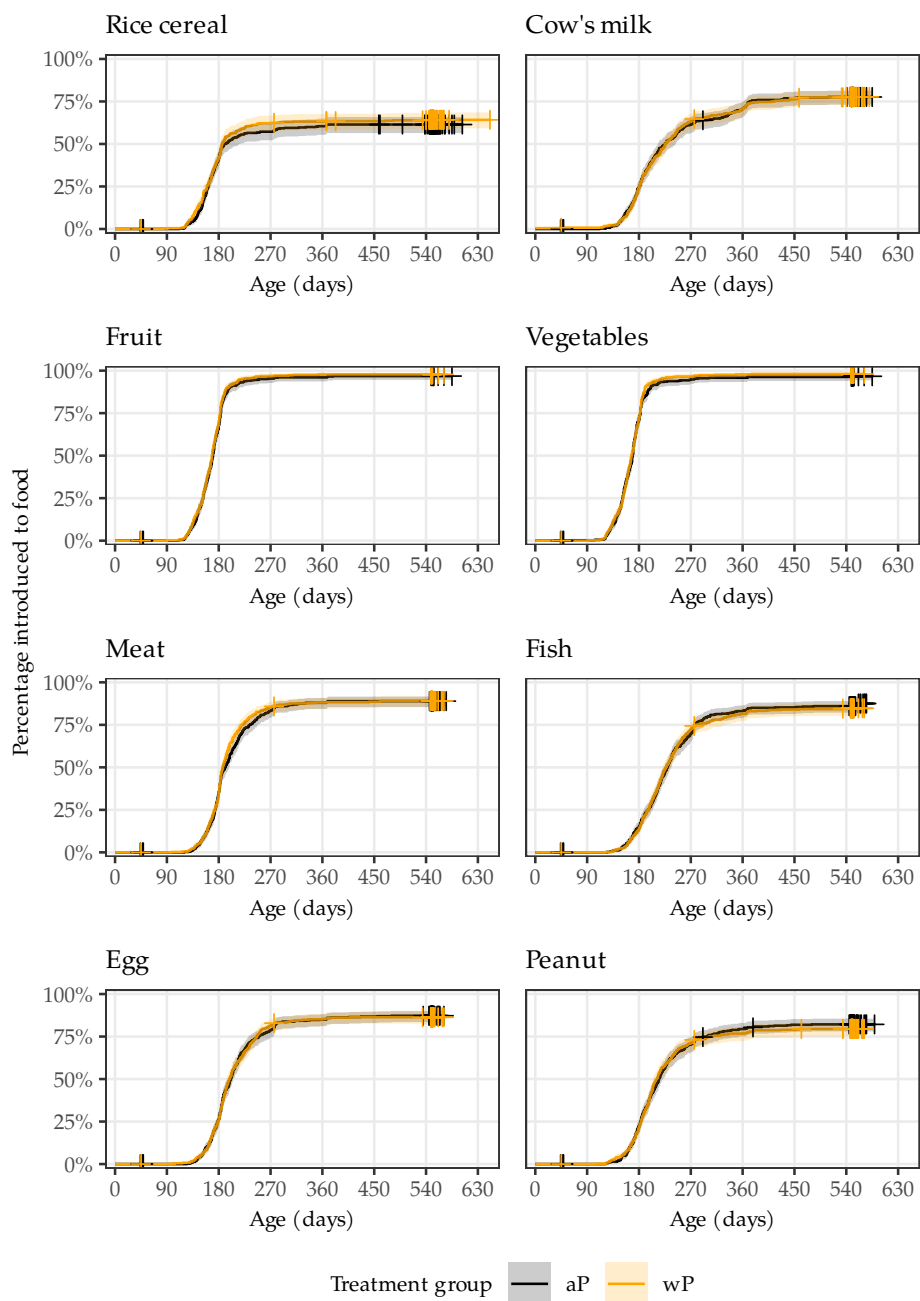


Figure 3: Age of food introduction for stage 2 participants. Censored observations are marked by a cross.

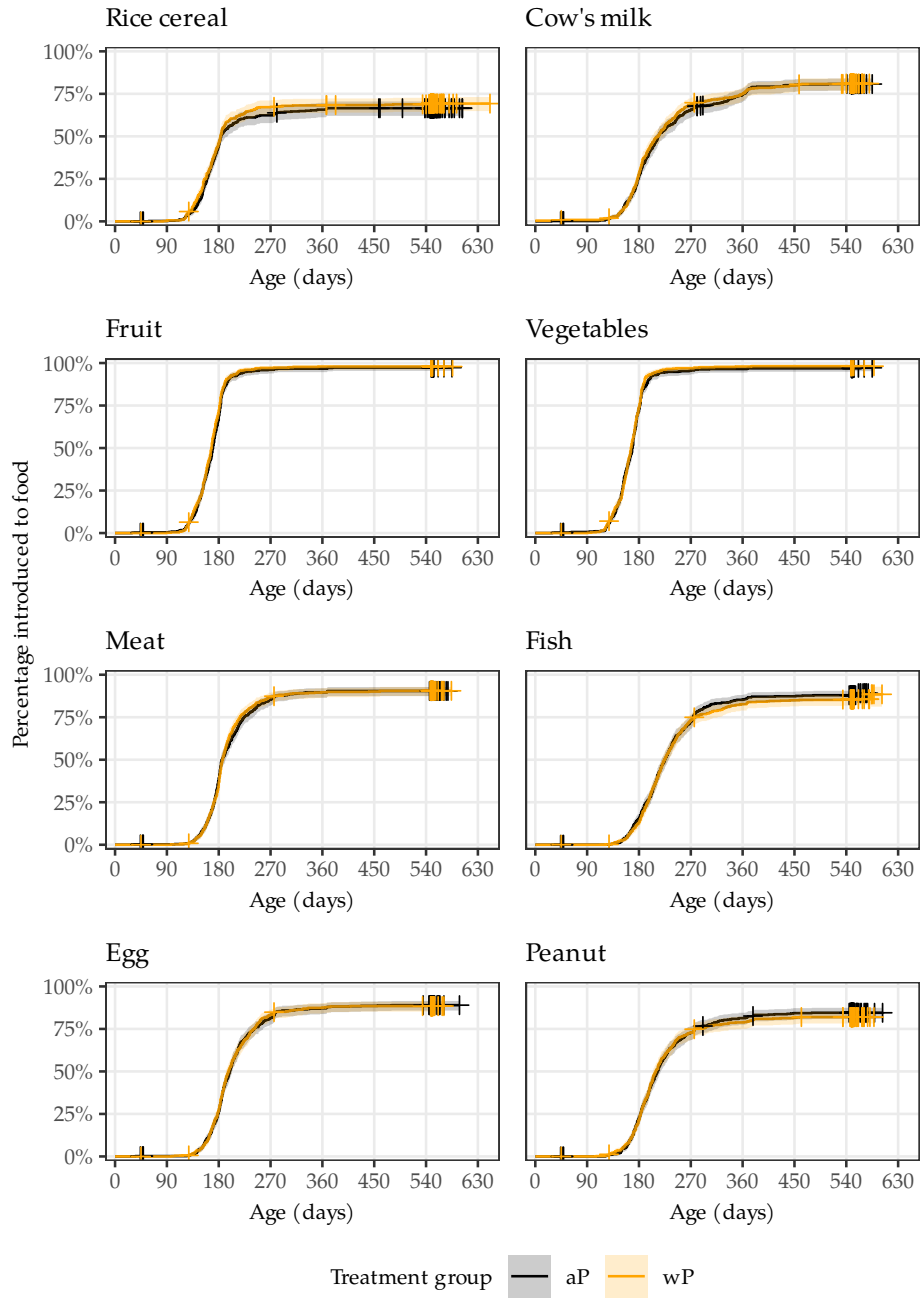


Figure 4: Age of food introduction for all participants combined. Censored observations are marked by a cross.

4 Clinical and mechanistic outcomes

Analyses of outcomes grouped participants according to their *assigned* treatment (Section 3.3.1).

All of the trial outcomes depend upon a positive SPT to either a specific allergen or to any allergen. Therefore, this section first reports on the SPT data, and outcomes 2c and 2d, followed by the primary (IgE-mediated food allergy) and other secondary clinical (eczema) outcomes.

4.1 Skin prick tests

4.1.1 Overview

There were 23 participants with no documented SPTs (Table 14). In many cases this was due to early withdrawal from the study, with the remaining due to withdrawn consent from the parent. All other participants had a record for at least the scheduled SPT, and in a few cases additional unscheduled SPT (Table 15).

Table 14: Listing of infants with no SPT records.

Record ID	Study site	streas	Last contact age (months)
30	PCH	Early termination	6
59	PCH	Early termination	9
141	PCH	Early termination	12
4629-65	PCH	Early termination	14
4629-120	PCH	Early termination	19
4629-177	PCH	Early termination	11
4629-184	PCH	Early termination	13
4629-232	PCH	Early termination	15
4630-15	CHW	Completed protocol	18
5785-110	MCRI	Early termination	1
5785-132	MCRI	Early termination	3
5785-217	MCRI	Completed protocol	18
5785-256	MCRI	Completed protocol	18
5785-291	MCRI	Completed protocol	18
5785-325	MCRI	Completed protocol	18
5785-401	MCRI	Completed protocol	18
5785-404	MCRI	Early termination	8
8644-18	SCHN	Early termination	12
8644-22	SCHN	Early termination	13
8644-28	SCHN	Early termination	13
8644-49	SCHN	Completed protocol	18
8644-53	SCHN	Completed protocol	18
8644-59	SCHN	Completed protocol	18

Table 15: Number and percentage of infants with scheduled and unscheduled SPTs.

Characteristic	aP N = 486	wP N = 486	Overall N = 972
Scheduled SPT	475 (98%)	473 (97%)	948 (98%)
Unscheduled SPTs			
0	461 (95%)	453 (93%)	914 (94%)
1	19 (3.9%)	31 (6.4%)	50 (5.1%)
2	6 (1.2%)	2 (0.4%)	8 (0.8%)

For the outcomes, a requirement was positive SPT by 18 months of age. Age of SPT in months was derived from `birthdat` (date of birth) and `pridat` (date of skin prick test) or in some cases `univdat` (date of unscheduled visit) when `pridat` was missing.

Most SPTs were performed by 13 months of age, although some were delayed, with a few occurring after 18-months of age (Figure 5). There were 5 infants who did have an SPT, but none by 18-months of age (record ID: 3, 25, 34, 36, 5785-10) (Table 16).

Participants who were enrolled under early stage 1 protocol versions had their SPT scheduled for 18 months of age rather than 12 months of age. Delays meant some of these SPT's did not occur until 19 or 20 months of age. For the outcomes, "approximately 12 months of age" was taken to mean "by 18 months of age" which then currently excludes some of these later SPTs.

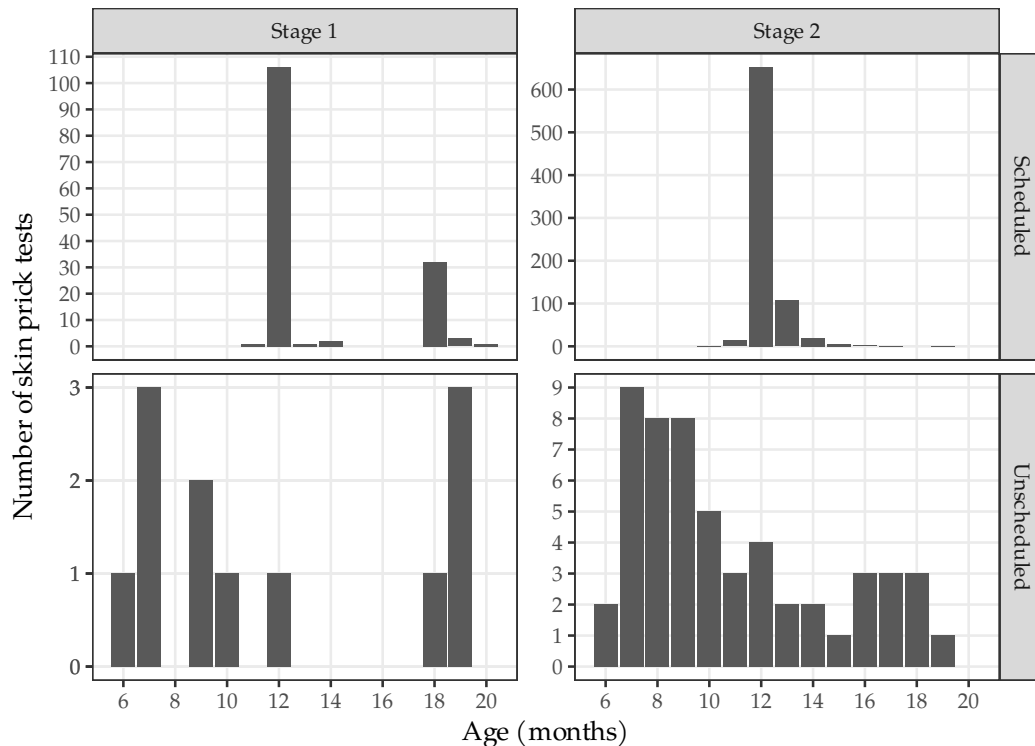


Figure 5: Age at skin prick test by protocol stage and occasion type (scheduled or unscheduled). Y-axis scale varies by panel.

Table 16: Listing of SPTs after 18-months of age.

Record ID	SPT type	SPT occasion	Age (months)
3	Scheduled	1	19
17	Unscheduled	2	19
25	Scheduled	1	19
34	Scheduled	1	19
36	Scheduled	1	20
68	Unscheduled	2	19
4629-17	Unscheduled	2	19
5785-10	Scheduled	1	19

4.1.2 Skin prick test sensitisation

Table 17 reports on the number and percentage of infants who had a SPT result of >1mm and ≥ 3 mm relative to negative control for at least one tested allergen (outcomes 2c and 2d) and also the number with any SPT result >0mm. Any infants with no SPT records by 18 months of age are reported as missing.

Table 17: Number and percentage of infants with a positive SPT by 18 months of age. Infants with no SPT records counted as missing.

Outcome	aP N = 486	wP N = 486	Overall N = 972
Any SPT >0mm	93 (20%)	97 (21%)	190 (20%)
Missing	13	15	28
Any SPT >1mm vs. neg control	81 (17%)	86 (18%)	167 (18%)
Missing	13	15	28
Any SPT ≥ 3 mm vs neg. control	58 (12%)	61 (13%)	119 (13%)
Missing	13	15	28

Table 18 provides a breakdown of the specific allergens from the standard panel which satisfied the sensitisation criteria. Table 19 reports on any other tested allergens beyond the standard panel.

Whole egg had a higher proportion of missingness than other allergens from the standard panel. This was usually because either egg white or egg yolk specifically was assessed and reported in the additional allergens instead of whole egg.

Table 18: Number and percentage of infants with a positive SPT (scheduled only) by 18 months of age by tested allergen (for those included in the standard panel). Each cell gives positive/tested (%) [missing result%].

Allergen	aP N = 486	wP N = 486	Overall N = 972
Wheal size >0mm			
<i>D.pteronyssinus</i>	19/471 (4.0%) [3.1%]	17/470 (3.6%) [3.3%]	36/941 (3.8%) [3.2%]
Cat dander	11/471 (2.3%) [3.1%]	16/471 (3.4%) [3.1%]	27/942 (2.9%) [3.1%]
Perennial ryegrass	9/471 (1.9%) [3.1%]	9/471 (1.9%) [3.1%]	18/942 (1.9%) [3.1%]
Whole egg	42/455 (9.2%) [6.4%]	47/451 (10%) [7.2%]	89/906 (9.8%) [6.8%]
Cashew	16/471 (3.4%) [3.1%]	16/470 (3.4%) [3.3%]	32/941 (3.4%) [3.2%]
Cow's milk	18/469 (3.8%) [3.5%]	16/471 (3.4%) [3.1%]	34/940 (3.6%) [3.3%]
Peanut	23/471 (4.9%) [3.1%]	24/469 (5.1%) [3.5%]	47/940 (5.0%) [3.3%]
Wheal size >1mm vs. neg control			
<i>D.pteronyssinus</i>	14/470 (3.0%) [3.3%]	15/470 (3.2%) [3.3%]	29/940 (3.1%) [3.3%]
Cat dander	9/470 (1.9%) [3.3%]	10/471 (2.1%) [3.1%]	19/941 (2.0%) [3.2%]
Perennial ryegrass	6/470 (1.3%) [3.3%]	6/471 (1.3%) [3.1%]	12/941 (1.3%) [3.2%]
Whole egg	40/454 (8.8%) [6.6%]	44/451 (9.8%) [7.2%]	84/905 (9.3%) [6.9%]
Cashew	14/470 (3.0%) [3.3%]	12/470 (2.6%) [3.3%]	26/940 (2.8%) [3.3%]
Cow's milk	17/468 (3.6%) [3.7%]	15/471 (3.2%) [3.1%]	32/939 (3.4%) [3.4%]
Peanut	20/470 (4.3%) [3.3%]	20/469 (4.3%) [3.5%]	40/939 (4.3%) [3.4%]
Wheal size ≥3mm vs neg. control			
<i>D.pteronyssinus</i>	10/470 (2.1%) [3.3%]	14/470 (3.0%) [3.3%]	24/940 (2.6%) [3.3%]
Cat dander	2/470 (0.4%) [3.3%]	5/471 (1.1%) [3.1%]	7/941 (0.7%) [3.2%]
Perennial ryegrass	3/470 (0.6%) [3.3%]	2/471 (0.4%) [3.1%]	5/941 (0.5%) [3.2%]
Whole egg	29/454 (6.4%) [6.6%]	31/451 (6.9%) [7.2%]	60/905 (6.6%) [6.9%]
Cashew	11/470 (2.3%) [3.3%]	8/470 (1.7%) [3.3%]	19/940 (2.0%) [3.3%]
Cow's milk	14/468 (3.0%) [3.7%]	10/471 (2.1%) [3.1%]	24/939 (2.6%) [3.4%]
Peanut	12/470 (2.6%) [3.3%]	13/469 (2.8%) [3.5%]	25/939 (2.7%) [3.4%]

Table 19: Number and percentage of infants with a positive SPT (scheduled only) by 18 months of age by tested allergen for additional allergens which were tested. Each cell gives positive/tested (%). Totals restricted to only those infants with at least one additional allergen tested

Allergen ¹	aP N = 84	wP N = 82	Overall N = 166
Wheal size >0mm			
Sesame	1/62 (1.6%)	0/57 (0%)	1/119 (0.8%)
Egg white	3/17 (18%)	6/21 (29%)	9/38 (24%)
Egg yolk	4/17 (24%)	4/19 (21%)	8/36 (22%)
Pistachio	0/1 (0%)	1/2 (50%)	1/3 (33%)
Soy	1/2 (50%)	0/1 (0%)	1/3 (33%)
Avocado	0/1 (0%)	0/1 (0%)	0/2 (0%)
Banana	0/0 (-%)	0/2 (0%)	0/2 (0%)
Dog	1/2 (50%)	0/0 (-%)	1/2 (50%)
Eggplant	0/1 (0%)	0/1 (0%)	0/2 (0%)
Kiwi fruit	0/2 (0%)	0/0 (-%)	0/2 (0%)
<i>D. farinae</i>	0/0 (-%)	0/1 (0%)	0/1 (0%)
Garlic	0/1 (0%)	0/0 (-%)	0/1 (0%)
King prawn	0/0 (-%)	0/1 (0%)	0/1 (0%)

Mango	0/1 (0%)	0/0 (-%)	0/1 (0%)
Onion	0/1 (0%)	0/0 (-%)	0/1 (0%)
Ovomucoid	0/0 (-%)	0/1 (0%)	0/1 (0%)
Pinenut	0/1 (0%)	0/0 (-%)	0/1 (0%)
Pork meat	0/0 (-%)	0/1 (0%)	0/1 (0%)
Raspberry	0/1 (0%)	0/0 (-%)	0/1 (0%)
Sesame seed	0/0 (-%)	0/1 (0%)	0/1 (0%)
Shrimp	1/1 (100%)	0/0 (-%)	1/1 (100%)
Tahini	0/0 (-%)	0/1 (0%)	0/1 (0%)
Tomato	0/0 (-%)	0/1 (0%)	0/1 (0%)
Wheat	0/1 (0%)	0/0 (-%)	0/1 (0%)
Whole wheat	0/1 (0%)	0/0 (-%)	0/1 (0%)

Wheal size >1mm vs. neg control

Sesame	1/62 (1.6%)	0/57 (0%)	1/119 (0.8%)
Egg white	3/17 (18%)	4/21 (19%)	7/38 (18%)
Egg yolk	4/17 (24%)	3/19 (16%)	7/36 (19%)
Pistachio	0/1 (0%)	1/2 (50%)	1/3 (33%)
Soy	1/2 (50%)	0/1 (0%)	1/3 (33%)
Avocado	0/1 (0%)	0/1 (0%)	0/2 (0%)
Banana	0/0 (-%)	0/2 (0%)	0/2 (0%)
Dog	1/2 (50%)	0/0 (-%)	1/2 (50%)
Eggplant	0/1 (0%)	0/1 (0%)	0/2 (0%)
Kiwi fruit	0/2 (0%)	0/0 (-%)	0/2 (0%)
<i>D. farinae</i>	0/0 (-%)	0/1 (0%)	0/1 (0%)
Garlic	0/1 (0%)	0/0 (-%)	0/1 (0%)
King prawn	0/0 (-%)	0/1 (0%)	0/1 (0%)
Mango	0/1 (0%)	0/0 (-%)	0/1 (0%)
Onion	0/1 (0%)	0/0 (-%)	0/1 (0%)
Ovomucoid	0/0 (-%)	0/1 (0%)	0/1 (0%)
Pinenut	0/1 (0%)	0/0 (-%)	0/1 (0%)
Pork meat	0/0 (-%)	0/1 (0%)	0/1 (0%)
Raspberry	0/1 (0%)	0/0 (-%)	0/1 (0%)
Sesame seed	0/0 (-%)	0/1 (0%)	0/1 (0%)
Shrimp	1/1 (100%)	0/0 (-%)	1/1 (100%)
Tahini	0/0 (-%)	0/1 (0%)	0/1 (0%)
Tomato	0/0 (-%)	0/1 (0%)	0/1 (0%)
Wheat	0/1 (0%)	0/0 (-%)	0/1 (0%)
Whole wheat	0/1 (0%)	0/0 (-%)	0/1 (0%)

Wheal size ≥3mm vs neg. control

Sesame	1/62 (1.6%)	0/57 (0%)	1/119 (0.8%)
Egg white	3/17 (18%)	3/21 (14%)	6/38 (16%)
Egg yolk	2/17 (12%)	0/19 (0%)	2/36 (5.6%)
Pistachio	0/1 (0%)	1/2 (50%)	1/3 (33%)
Soy	1/2 (50%)	0/1 (0%)	1/3 (33%)
Avocado	0/1 (0%)	0/1 (0%)	0/2 (0%)
Banana	0/0 (-%)	0/2 (0%)	0/2 (0%)
Dog	1/2 (50%)	0/0 (-%)	1/2 (50%)
Eggplant	0/1 (0%)	0/1 (0%)	0/2 (0%)
Kiwi fruit	0/2 (0%)	0/0 (-%)	0/2 (0%)
<i>D. farinae</i>	0/0 (-%)	0/1 (0%)	0/1 (0%)
Garlic	0/1 (0%)	0/0 (-%)	0/1 (0%)
King prawn	0/0 (-%)	0/1 (0%)	0/1 (0%)
Mango	0/1 (0%)	0/0 (-%)	0/1 (0%)
Onion	0/1 (0%)	0/0 (-%)	0/1 (0%)
Ovomucoid	0/0 (-%)	0/1 (0%)	0/1 (0%)

Pinenut	0/1 (0%)	0/0 (-%)	0/1 (0%)
Pork meat	0/0 (-%)	0/1 (0%)	0/1 (0%)
Raspberry	0/1 (0%)	0/0 (-%)	0/1 (0%)
Sesame seed	0/0 (-%)	0/1 (0%)	0/1 (0%)
Shrimp	1/1 (100%)	0/0 (-%)	1/1 (100%)
Tahini	0/0 (-%)	0/1 (0%)	0/1 (0%)
Tomato	0/0 (-%)	0/1 (0%)	0/1 (0%)
Wheat	0/1 (0%)	0/0 (-%)	0/1 (0%)
Whole wheat	0/1 (0%)	0/0 (-%)	0/1 (0%)

¹As documented in free-text field

In some cases, egg was split into egg white or whole egg. The following table combines the above two tables, grouping together any “egg” components which were tested.

Table 20: Number and percentage of infants with a positive SPT (scheduled or unscheduled) by 18 months of age by tested allergen for all allergens which were tested. Each cell gives positive/tested (%).

Allergen	aP N = 486	wP N = 486	Overall N = 972
Wheal size >0mm			
<i>D.pteronyssinus</i>	19/471 (4.0%) [3.1%]	18/470 (3.8%) [3.3%]	37/941 (3.9%) [3.2%]
Cat dander	11/471 (2.3%) [3.1%]	17/471 (3.6%) [3.1%]	28/942 (3.0%) [3.1%]
Perennial ryegrass	9/471 (1.9%) [3.1%]	10/471 (2.1%) [3.1%]	19/942 (2.0%) [3.1%]
Egg	46/472 (9.7%) [2.9%]	55/471 (12%) [3.1%]	101/943 (11%) [3.0%]
Cashew	17/471 (3.6%) [3.1%]	17/470 (3.6%) [3.3%]	34/941 (3.6%) [3.2%]
Cow's milk	20/470 (4.3%) [3.3%]	20/471 (4.2%) [3.1%]	40/941 (4.3%) [3.2%]
Peanut	23/471 (4.9%) [3.1%]	25/469 (5.3%) [3.5%]	48/940 (5.1%) [3.3%]
Wheal size >1mm vs. neg control			
<i>D.pteronyssinus</i>	14/470 (3.0%) [3.3%]	15/470 (3.2%) [3.3%]	29/940 (3.1%) [3.3%]
Cat dander	9/470 (1.9%) [3.3%]	10/471 (2.1%) [3.1%]	19/941 (2.0%) [3.2%]
Perennial ryegrass	6/470 (1.3%) [3.3%]	6/471 (1.3%) [3.1%]	12/941 (1.3%) [3.2%]
Egg	44/471 (9.3%) [3.1%]	50/471 (11%) [3.1%]	94/942 (10.0%) [3.1%]
Cashew	15/470 (3.2%) [3.3%]	12/470 (2.6%) [3.3%]	27/940 (2.9%) [3.3%]
Cow's milk	18/469 (3.8%) [3.5%]	17/471 (3.6%) [3.1%]	35/940 (3.7%) [3.3%]
Peanut	20/470 (4.3%) [3.3%]	21/469 (4.5%) [3.5%]	41/939 (4.4%) [3.4%]
Wheal size ≥3mm vs neg. control			
<i>D.pteronyssinus</i>	10/470 (2.1%) [3.3%]	14/470 (3.0%) [3.3%]	24/940 (2.6%) [3.3%]
Cat dander	2/470 (0.4%) [3.3%]	5/471 (1.1%) [3.1%]	7/941 (0.7%) [3.2%]
Perennial ryegrass	3/470 (0.6%) [3.3%]	2/471 (0.4%) [3.1%]	5/941 (0.5%) [3.2%]
Egg	33/471 (7.0%) [3.1%]	35/471 (7.4%) [3.1%]	68/942 (7.2%) [3.1%]
Cashew	12/470 (2.6%) [3.3%]	9/470 (1.9%) [3.3%]	21/940 (2.2%) [3.3%]
Cow's milk	15/469 (3.2%) [3.5%]	14/471 (3.0%) [3.1%]	29/940 (3.1%) [3.3%]
Peanut	12/470 (2.6%) [3.3%]	14/469 (3.0%) [3.5%]	26/939 (2.8%) [3.4%]

4.1.3 Skin prick test sensitisation model

Table 21 summarises the observed counts and estimated model parameters for skin prick test sensitisation under each treatment group and by the size requirement (see Section 2.4.2 numbers 2c and 2d), and their estimated differences (the posterior densities are shown in Figure 6). The results between the analyses (unadjusted and adjusted, Section 2.5.1) were near identical. The most plausible values for the difference in skin prick test sensitisation at >1mm [2.c] between wP and aP were 1.13 (-3.73, 6.00) [posterior median, (95% credible interval)]. The probability of a reduction in sensitisation under wP compared to aP was 0.32. At ≥ 3 mm [2.d] they were 0.68 (-3.56, 4.93) and 0.38.

Table 21: Posterior summaries for risk and difference in risk of positive skin prick test by size.

	aP		wP		Difference
	y/n (%)	Median (95% CrI)	y/n (%)	Median (95% CrI)	Median (95% CrI), Pr(<0)
Unadjusted					
>0mm	93/473 (20%)	19.7 (16.3, 23.5)	97/471 (21%)	20.7 (17.2, 24.5)	0.9 (-4.2, 6.0), 0.36
>1mm vs ctrl. [2.c]	81/473 (17%)	17.2 (14.0, 20.8)	86/471 (18%)	18.4 (15.0, 22.0)	1.1 (-3.7, 6.0), 0.32
≥ 3 mm vs ctrl. [2.d]	58/473 (12%)	12.4 (9.6, 15.5)	61/471 (13%)	13.1 (10.2, 16.3)	0.7 (-3.6, 4.9), 0.38
Adjusted					
>0mm	93/473 (20%)	19.8 (16.4, 23.4)	97/471 (21%)	20.6 (17.2, 24.4)	0.9 (-4.2, 5.9), 0.37
>1mm vs ctrl. [2.c]	81/473 (17%)	17.3 (14.0, 20.8)	86/471 (18%)	18.2 (15.0, 21.7)	1.0 (-3.8, 5.6), 0.35
≥ 3 mm vs ctrl. [2.d]	58/473 (12%)	12.4 (9.7, 15.5)	61/471 (13%)	12.9 (10.1, 16.1)	0.5 (-3.7, 4.7), 0.40

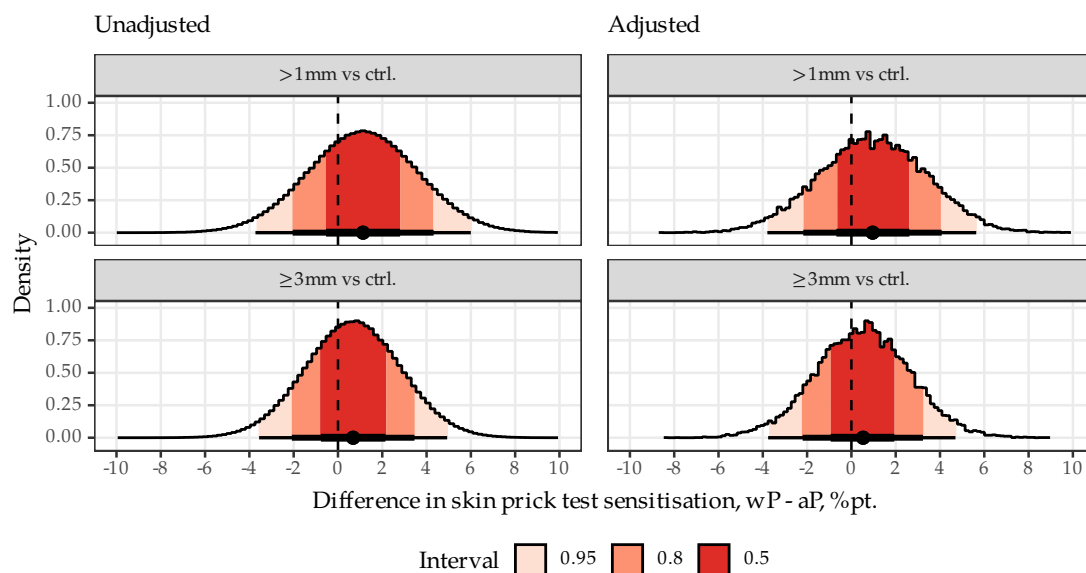


Figure 6: Posterior density for difference in risk of skin prick test sensitisation by size and model.

4.1.4 Skin prick test magnitude

Table 22 and Figure 7 report on the magnitude of the SPT results (in millimetres) from scheduled SPTs, in particular, amongst infants with SPT sensitisation. There was little evidence of a difference in sensitisation and, given the small number of infants with any sensitisation, in difference in SPT magnitudes, whether overall or conditional on sensitisation.

Table 22: SPT sensitisation and magnitude by assigned treatment for scheduled SPTs.

	aP			wP		
	y/n (p%) ¹	E[y] ²	E[y y>0] ³	y/n (p%) ¹	E[y] ²	E[y y>0] ³
D.pteronyssinus	19/471 (4.0%)	0.12	3.1	17/470 (3.6%)	0.12	3.4
Cat dander	11/471 (2.3%)	0.05	2.3	16/471 (3.4%)	0.08	2.3
Perennial ryegrass	9/471 (1.9%)	0.04	2.3	9/471 (1.9%)	0.04	2.1
Whole egg	42/455 (9.2%)	0.31	3.4	47/451 (10.4%)	0.34	3.3
Cashew	16/471 (3.4%)	0.12	3.7	16/470 (3.4%)	0.09	2.7
Cow's milk	18/469 (3.8%)	0.19	5.1	16/471 (3.4%)	0.12	3.6
Peanut	23/471 (4.9%)	0.17	3.4	24/469 (5.1%)	0.18	3.4

¹y: number with any sensitisation (>0mm), n: number of scheduled SPT results, p: percentage

²E[y]: mean of SPT results (mm) amongst all infants

³E[y|y>0]: mean of SPT results (mm) amongst infants with any sensitisation

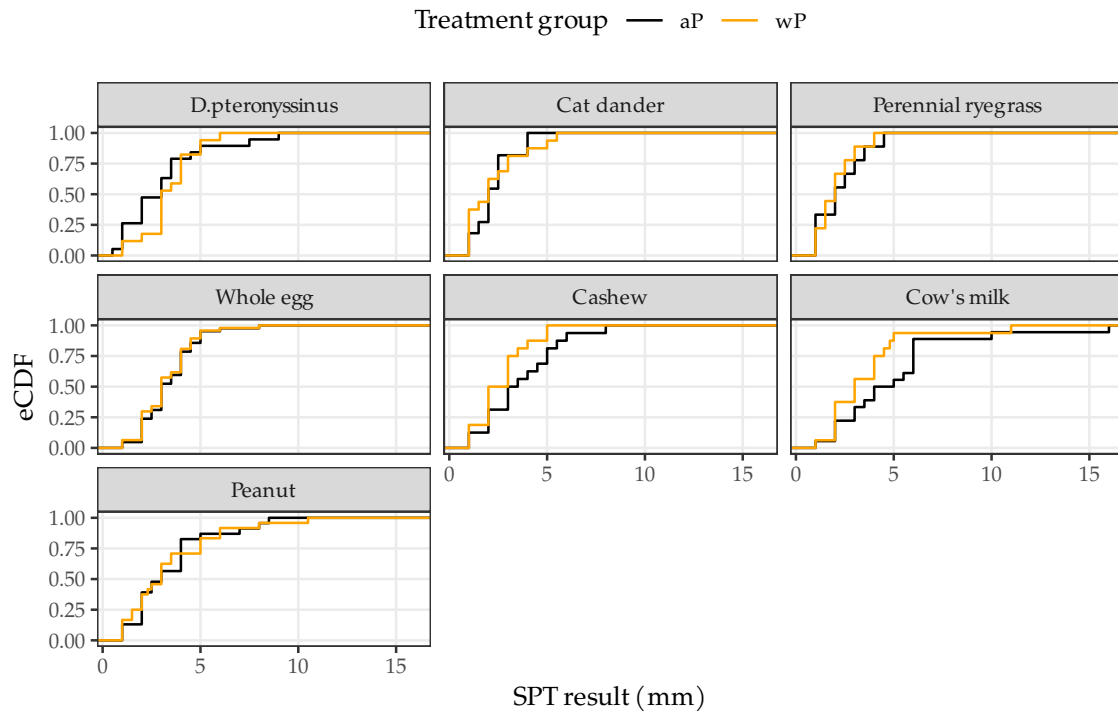


Figure 7: SPT magnitude amongst infants with any sensitisation (>0mm).

As a post-hoc exploratory analysis of SPT results, hurdle log-normal models were estimated for each allergen on the standard panel for scheduled SPTs. These models conditioned on assigned treatment

only. The estimate differences in mean SPT result, difference in probability of sensitisation, and difference in mean SPT result conditional on sensitisation are reported in Figure 8.

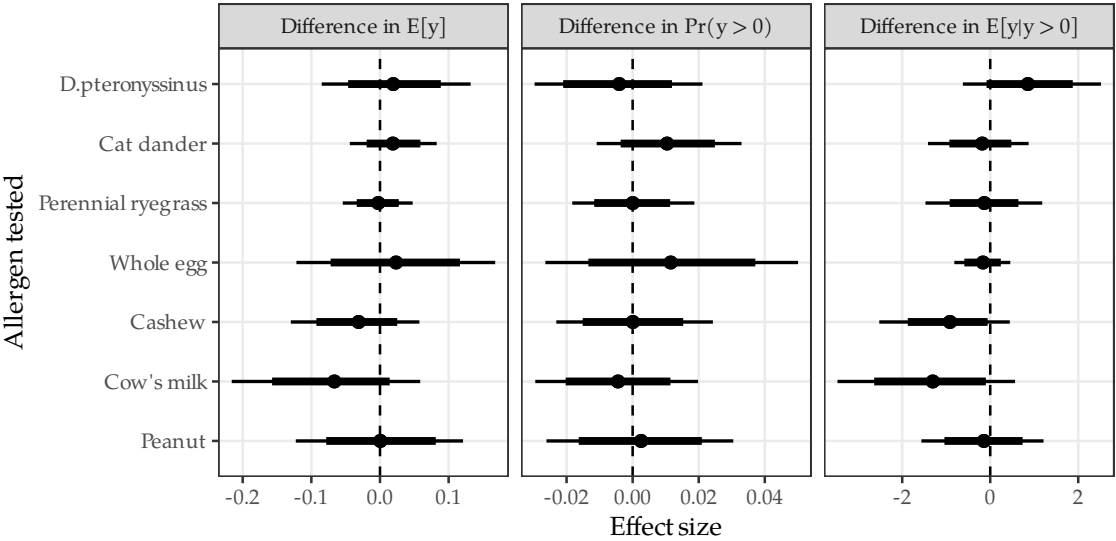


Figure 8: Posterior summaries for model estimated difference in mean SPT result, sensitisation probability, and mean SPT result conditional on sensitisation, wP vs aP.

4.2 IgE-mediated food allergy

4.2.1 Overview

The primary source of information for IgE-mediated food allergy was the `outcome_report` form. Reported outcomes were cross checked against the food and household questionnaire form (`food_and_household_questionnaire`) and the oral food challenge form (`food_challenge`).

Age at first reaction was derived from `outageval`. If this was missing then `birthdat` (date of birth) and `outdiagdat` (clinician-diagnosis date) was used to derive age in months.

In some cases participants may have withdrawn from the study without sufficient follow-up for IgE-mediated food allergy to have been identified. The report makes no allowance for “missing” IgE-mediated food allergy status, with absence of clinician diagnosed allergy (with or without adequate follow-up) counted as not meeting the primary outcome. Only 17 participants had an age of last contact 15 months or younger and may have been those most likely to have insufficient follow-up.

The following sections provide a summary of the number of infants who met each criteria. The infants with an outcome reported are line listed in the appendix (Section 8.1) along with the reported allergens.

4.2.2 Oral food challenge

There were 31 outcome records for food allergy diagnoses resulting from oral food challenge amongst 23 infants. One reaction (record ID: 8644-34) reportedly first occurred after 18 months of age and all others were by 18 months of age. All records had a matching positive skin prick test to the reported allergen and so 22 infants met the primary outcome according to the definition (1.i).

4.2.3 Anaphylaxis

There were 6 outcome records for food allergy diagnoses resulting from anaphylaxis amongst 6 infants. Two of these infants (record ID: 68, 5785-283) were diagnosed after 18 months of age with first reported reaction occurring after 18 months of age. One infant (record ID: 8644-34) had no skin prick test for the reported allergens. So of the 6 infants with anaphylaxis reported, 3 did not meet the criteria for the primary outcome according to the definition (1.ii) leaving 3 who met the condition.

4.2.4 Highly probable IgE-mediated food allergy

There were 56 outcome records for food allergy diagnoses resulting from highly probable IgE-mediated allergy amongst 44 infants. For one infant (record ID: 68), the age at first reaction and age at diagnosis were after 18 months of age. Four infants (record ID: 75, 5785-239, 5785-243, 5785-331) had no positive SPT to the reported allergen. One infant (record ID: 5785-7) had no

date of allergy confirmation of clinician and did not meet the outcome due to a negative food challenge. Of the 44 infants with IgE-mediated food allergy reported, 6 did not meet the primary outcome according to the definition (2.) leaving 38 infants who met the condition.

4.2.5 Primary outcome (IgE-mediated food allergy)

Overall 57/972 (6%) infants met the primary outcome, 26/486 (5%) who were assigned to aP and 31/486 (6%) who were assigned to wP.

Table 23 reports on the number of participants who met each of the primary outcome criteria (1.i, 1.ii, and 2, see Section 2.4.1) as outlined in the sections above by their assigned treatment group.

Table 24 reports the total number of infants who met the primary outcome, and the primary reason grouped hierarchically according to the ordering in the definition (i.e. OFC → Anaphylaxis → IgE-mediated) (irrespective of the timing of those criteria).

Table 25 reports the number of infants who met the primary outcome by the food allergy. Each infant may meet the primary outcome for more than one food.

Table 23: Number and percentage of infants who met each primary outcome condition and treatment group.

Outcome	aP N = 486	wP N = 486	Overall N = 972
Met condition 1.i.	13 (2.7%)	9 (1.9%)	22 (2.3%)
Met condition 1.ii.	2 (0.4%)	1 (0.2%)	3 (0.3%)
Met condition 2.	14 (2.9%)	24 (4.9%)	38 (3.9%)

Table 24: Number and percentage of infants who met the primary outcome by primary reason and treatment group.

Outcome	aP N = 486	wP N = 486	Overall N = 972
Met primary outcome	26 (5.3%)	31 (6.4%)	57 (5.9%)
Reason			
Criteria 1.i	13 (2.7%)	9 (1.9%)	22 (2.3%)
Criteria 1.ii	1 (0.2%)	1 (0.2%)	2 (0.2%)
Criteria 2.	12 (2.5%)	21 (4.3%)	33 (3.4%)

Table 25: Number and percentage of infants who met the primary outcome by the reported food. The same infant may meet the primary outcome for more than one allergen.

IgE-mediated food allergy	aP N = 486	wP N = 486	Overall N = 972
None	460 (95%)	455 (94%)	915 (94%)
Whole egg	16 (3.3%)	17 (3.5%)	33 (3.4%)
Peanut	6 (1.2%)	12 (2.5%)	18 (1.9%)
Cow's milk	8 (1.6%)	8 (1.6%)	16 (1.6%)
Cashew	2 (0.4%)	3 (0.6%)	5 (0.5%)
Egg white	2 (0.4%)	1 (0.2%)	3 (0.3%)
Egg yolk	2 (0.4%)	1 (0.2%)	3 (0.3%)
Almond	1 (0.2%)	0 (0%)	1 (0.1%)
Beef	1 (0.2%)	0 (0%)	1 (0.1%)
Coconut	1 (0.2%)	0 (0%)	1 (0.1%)
Hazelnut	0 (0%)	1 (0.2%)	1 (0.1%)
Pecan	1 (0.2%)	0 (0%)	1 (0.1%)
Sesame	1 (0.2%)	0 (0%)	1 (0.1%)
Soy	1 (0.2%)	0 (0%)	1 (0.1%)

4.2.6 Primary outcome model

Table 26 summarises the estimated model parameters for the probability of IgE-mediated food allergy under each treatment group, and their difference. The posterior densities for the difference in risk of IgE-mediated food allergy (wP vs aP) are shown in Figure 9.

The results between the primary analysis (unadjusted) and secondary (adjusted) analyses were near identical. Given the data, under the assumed model the most plausible values for the difference in IgE-mediated food allergy between wP and aP were 0.01 (-0.02, 0.04) [posterior median, (95% credible interval)]. The probability of a reduction in IgE-mediated food allergy under wP compared to aP was 0.25.

A pre-specified stopping rule was to assess the predicted probability of success (PPoS) up to the planned maximum sample size of 2,000 infants (1,000 per arm). Success required that the probability of a reduction in IgE-mediated food allergy exceed 0.95. Calculation of PPoS used the posterior predictive distribution under each arm to impute outcomes for future participants. If $PPoS < 0.05$ then the trial was to be stopped for futility. At the time of stopping, 486 participants had been assigned to each arm with 514 further infants required to reach 1,000 per arm. At this stage, $PPoS \approx 0.006 < 0.05$ and would have met the futility criteria.

Table 26: Posterior summaries for risk and difference in risk of IgE-mediated food allergy.

	aP		wP		Difference
	y/n (%)	Median (95% CrI)	y/n (%)	Median (95% CrI)	Median (95% CrI), Pr(<0)
Unadjusted	26/486 (5.3%)	5.5 (3.7, 7.7)	31/486 (6.4%)	6.5 (4.5, 8.9)	1.0 (-2.0, 4.0), 0.25
Adjusted	26/486 (5.3%)	5.4 (3.7, 7.6)	31/486 (6.4%)	6.4 (4.5, 8.7)	1.0 (-1.9, 3.9), 0.26

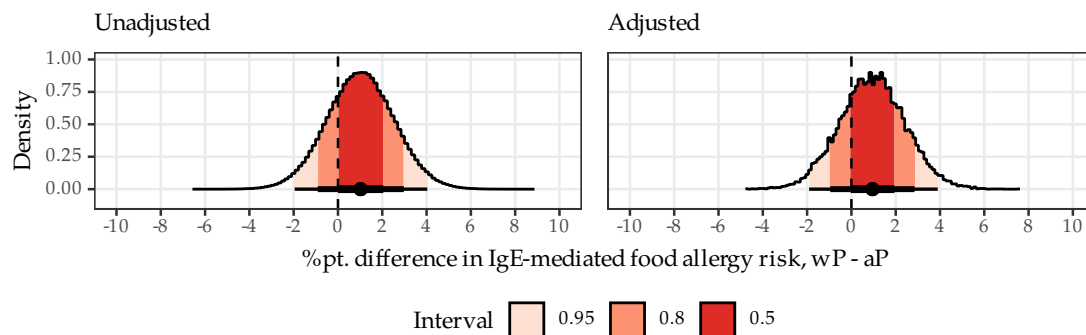


Figure 9: Posterior density for difference in risk of IgE-mediated food allergy by model. Values in regions with high density are those most plausible given the data and model assumptions.

4.2.7 Oral food challenge model

Table 27: Posterior summaries for risk and difference in risk of positive oral food challenge.

	aP		wP		Difference
	y/n (%)	Median (95% CrI)	y/n (%)	Median (95% CrI)	Median (95% CrI), Pr(<0)
Unadjusted	13/486 (2.7%)	2.8 (1.6, 4.5)	9/486 (1.9%)	2.0 (1.0, 3.5)	-0.8 (-2.8, 1.1), 0.80
Adjusted	13/486 (2.7%)	2.7 (1.5, 4.4)	9/486 (1.9%)	1.9 (1.0, 3.3)	-0.8 (-2.7, 1.0), 0.82

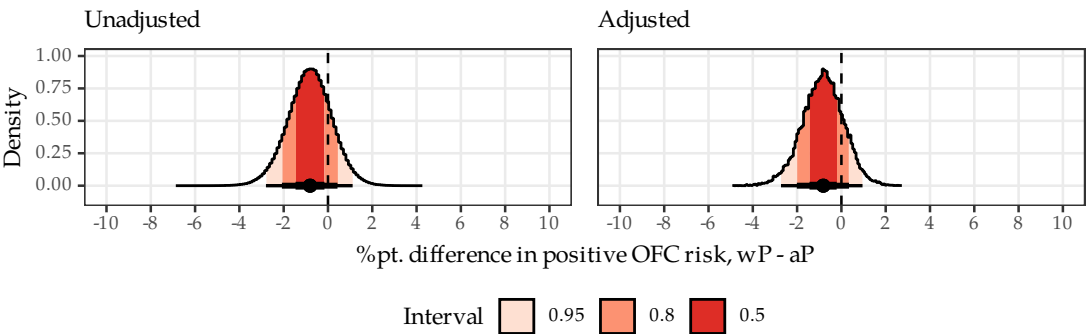


Figure 10: Posterior density for difference in risk of positive OFC by model. Values in regions with high density are those most plausible given the data and model assumptions.

4.2.8 Subgroup analyses

4.2.8.1 Site

Table 28: Number and percentage of infants who met each primary outcome condition by treatment group and participating site.

Outcome	MCRI			PCH			SCH			CHW		
	aP N = 220	wP N = 219	Overall N = 439	aP N = 206	wP N = 207	Overall N = 413	aP N = 30	wP N = 31	Overall N = 61	aP N = 30	wP N = 29	Overall N = 59
Met condition 1.i.	11 (5.0%)	5 (2.3%)	16 (3.6%)	1 (0.5%)	1 (0.5%)	2 (0.5%)	0 (0%)	2 (6.5%)	2 (3.3%)	1 (3.3%)	1 (3.4%)	2 (3.4%)
Met condition 1.ii.	1 (0.5%)	0 (0%)	1 (0.2%)	1 (0.5%)	1 (0.5%)	2 (0.5%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Met condition 2.	6 (2.7%)	11 (5.0%)	17 (3.9%)	5 (2.4%)	11 (5.3%)	16 (3.9%)	1 (3.3%)	1 (3.2%)	2 (3.3%)	2 (6.7%)	1 (3.4%)	3 (5.1%)

Table 29: Number and percentage of infants who met the primary outcome by primary reason, treatment group, and participating site.

Outcome	MCRI			PCH			SCH			CHW		
	aP N = 220	wP N = 219	Overall N = 439	aP N = 206	wP N = 207	Overall N = 413	aP N = 30	wP N = 31	Overall N = 61	aP N = 30	wP N = 29	Overall N = 59
Met primary outcome	16 (7.3%)	14 (6.4%)	30 (6.8%)	6 (2.9%)	13 (6.3%)	19 (4.6%)	1 (3.3%)	2 (6.5%)	3 (4.9%)	3 (10%)	2 (6.9%)	5 (8.5%)
Reason												
Criteria 1.i	11 (5.0%)	5 (2.3%)	16 (3.6%)	1 (0.5%)	1 (0.5%)	2 (0.5%)	0 (0%)	2 (6.5%)	2 (3.3%)	1 (3.3%)	1 (3.4%)	2 (3.4%)
Criteria 1.ii	0 (0%)	0 (0%)	0 (0%)	1 (0.5%)	1 (0.5%)	2 (0.5%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Criteria 2.	5 (2.3%)	9 (4.1%)	14 (3.2%)	4 (1.9%)	11 (5.3%)	15 (3.6%)	1 (3.3%)	0 (0%)	1 (1.6%)	2 (6.7%)	1 (3.4%)	3 (5.1%)

4.2.8.2 Season of birth

Table 30: Number and percentage of infants who met each primary outcome condition by treatment group and season of birth.

Outcome	Autumn			Winter			Spring			Summer		
	aP N = 133	wP N = 138	Overall N = 271	aP N = 144	wP N = 144	Overall N = 288	aP N = 135	wP N = 133	Overall N = 268	aP N = 74	wP N = 71	Overall N = 145
Met condition 1.i.	6 (4.5%)	3 (2.2%)	9 (3.3%)	5 (3.5%)	4 (2.8%)	9 (3.1%)	1 (0.7%)	1 (0.8%)	2 (0.7%)	1 (1.4%)	1 (1.4%)	2 (1.4%)
Met condition 1.ii.	0 (0%)	0 (0%)	0 (0%)	2 (1.4%)	0 (0%)	2 (0.7%)	0 (0%)	1 (0.8%)	1 (0.4%)	0 (0%)	0 (0%)	0 (0%)
Met condition 2.	3 (2.3%)	10 (7.2%)	13 (4.8%)	8 (5.6%)	8 (5.6%)	16 (5.6%)	2 (1.5%)	4 (3.0%)	6 (2.2%)	1 (1.4%)	2 (2.8%)	3 (2.1%)

Table 31: Number and percentage of infants who met the primary outcome by primary reason, treatment group, and season of birth.

Outcome	Autumn			Winter			Spring			Summer		
	aP N = 133	wP N = 138	Overall N = 271	aP N = 144	wP N = 144	Overall N = 288	aP N = 135	wP N = 133	Overall N = 268	aP N = 74	wP N = 71	Overall N = 145
Met primary outcome	8 (6.0%)	12 (8.7%)	20 (7.4%)	13 (9.0%)	11 (7.6%)	24 (8.3%)	3 (2.2%)	6 (4.5%)	9 (3.4%)	2 (2.7%)	2 (2.8%)	4 (2.8%)
Reason												
Criteria 1.i	6 (4.5%)	3 (2.2%)	9 (3.3%)	5 (3.5%)	4 (2.8%)	9 (3.1%)	1 (0.7%)	1 (0.8%)	2 (0.7%)	1 (1.4%)	1 (1.4%)	2 (1.4%)
Criteria 1.ii	0 (0%)	0 (0%)	0 (0%)	1 (0.7%)	0 (0%)	1 (0.3%)	0 (0%)	1 (0.8%)	1 (0.4%)	0 (0%)	0 (0%)	0 (0%)
Criteria 2.	2 (1.5%)	9 (6.5%)	11 (4.1%)	7 (4.9%)	7 (4.9%)	14 (4.9%)	2 (1.5%)	4 (3.0%)	6 (2.2%)	1 (1.4%)	1 (1.4%)	2 (1.4%)

4.2.8.3 Family history of atopy

Family history of atopy was derived from responses on the corresponding baseline form. If an infant had any parent or sibling with a reported history of asthma, eczema, allergic rhinitis, or food allergy, then they were considered to have a family history of atopy.

Table 32: Number and percentage of infants who met each primary outcome condition by treatment group and family history of atopy (yes or no).

Outcome	Yes			No		
	aP N = 418	wP N = 415	Overall N = 833	aP N = 68	wP N = 71	Overall N = 139
Met condition 1.i.	13 (3.1%)	6 (1.4%)	19 (2.3%)	0 (0%)	3 (4.2%)	3 (2.2%)
Met condition 1.ii.	2 (0.5%)	1 (0.2%)	3 (0.4%)	0 (0%)	0 (0%)	0 (0%)
Met condition 2.	12 (2.9%)	20 (4.8%)	32 (3.8%)	2 (2.9%)	4 (5.6%)	6 (4.3%)

Table 33: Number and percentage of infants who met the primary outcome by primary reason, treatment group, and family history of atopy (yes or no).

Outcome	Yes			No		
	aP N = 418	wP N = 415	Overall N = 833	aP N = 68	wP N = 71	Overall N = 139
Met primary outcome	24 (5.7%)	25 (6.0%)	49 (5.9%)	2 (2.9%)	6 (8.5%)	8 (5.8%)
Reason						
Criteria 1.i	13 (3.1%)	6 (1.4%)	19 (2.3%)	0 (0%)	3 (4.2%)	3 (2.2%)
Criteria 1.ii	1 (0.2%)	1 (0.2%)	2 (0.2%)	0 (0%)	0 (0%)	0 (0%)
Criteria 2.	10 (2.4%)	18 (4.3%)	28 (3.4%)	2 (2.9%)	3 (4.2%)	5 (3.6%)

4.2.8.4 Family history of food allergy

Family history of food allergy was derived from responses on the corresponding baseline form. If an infant had any parent or sibling with a reported history of food allergy, then they were considered to have a family history of food allergy.

Table 34: Number and percentage of infants who met each primary outcome condition by treatment group and family history of atopy (yes or no).

Outcome	No			Yes		
	aP N = 365	wP N = 364	Overall N = 729	aP N = 121	wP N = 122	Overall N = 243
Met condition 1.i.	7 (1.9%)	7 (1.9%)	14 (1.9%)	6 (5.0%)	2 (1.6%)	8 (3.3%)
Met condition 1.ii.	0 (0%)	1 (0.3%)	1 (0.1%)	2 (1.7%)	0 (0%)	2 (0.8%)
Met condition 2.	6 (1.6%)	13 (3.6%)	19 (2.6%)	8 (6.6%)	11 (9.0%)	19 (7.8%)

Table 35: Number and percentage of infants who met the primary outcome by primary reason, treatment group, and family history of atopy (yes or no).

Outcome	No			Yes		
	aP N = 365	wP N = 364	Overall N = 729	aP N = 121	wP N = 122	Overall N = 243
Met primary outcome	13 (3.6%)	19 (5.2%)	32 (4.4%)	13 (11%)	12 (9.8%)	25 (10%)
Reason						
Criteria 1.i	7 (1.9%)	7 (1.9%)	14 (1.9%)	6 (5.0%)	2 (1.6%)	8 (3.3%)
Criteria 1.ii	0 (0%)	1 (0.3%)	1 (0.1%)	1 (0.8%)	0 (0%)	1 (0.4%)
Criteria 2.	6 (1.6%)	11 (3.0%)	17 (2.3%)	6 (5.0%)	10 (8.2%)	16 (6.6%)

4.2.8.5 Parental country of birth

Whether an infant had one or more parent born in East Asia was derived from the mother's country of birth (bcmoth) and the father's country of birth (bcfath) classified according to the Standard Australian Classification of Countries (SACC). If either of these was unknown (being missing or reported as "other") than the infant's status was considered "unknown".

Table 36: Number and percentage of infants who met each primary outcome condition by treatment group and whether at least one parent was born in East Asia. Unknown cases occurred where either the mother or father's country of birth was not known (either missing or reported as "other").

Outcome	No			Yes			Unknown		
	aP	wP	Overall	aP	wP	Overall	aP	wP	Overall
	N = 406	N = 389	N = 795	N = 36	N = 48	N = 84	N = 44	N = 49	N = 93
Met condition 1.i.	10 (2.5%)	8 (2.1%)	18 (2.3%)	2 (5.6%)	0 (0%)	2 (2.4%)	1 (2.3%)	1 (2.0%)	2 (2.2%)
Met condition 1.ii.	2 (0.5%)	1 (0.3%)	3 (0.4%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Met condition 2.	12 (3.0%)	13 (3.3%)	25 (3.1%)	2 (5.6%)	10 (21%)	12 (14%)	0 (0%)	1 (2.0%)	1 (1.1%)

Table 37: Number and percentage of infants who met the primary outcome by primary reason, treatment group, and whether at least one parent was born in East Asia. Unknown cases occurred where either the mother or father's country of birth was not known (either missing or reported as "other").

Outcome	No			Yes			Unknown		
	aP	wP	Overall	aP	wP	Overall	aP	wP	Overall
	N = 406	N = 389	N = 795	N = 36	N = 48	N = 84	N = 44	N = 49	N = 93
Met primary outcome Reason	21 (5.2%)	20 (5.1%)	41 (5.2%)	4 (11%)	10 (21%)	14 (17%)	1 (2.3%)	1 (2.0%)	2 (2.2%)
Criteria 1.i	10 (2.5%)	8 (2.1%)	18 (2.3%)	2 (5.6%)	0 (0%)	2 (2.4%)	1 (2.3%)	1 (2.0%)	2 (2.2%)
Criteria 1.ii	1 (0.2%)	1 (0.3%)	2 (0.3%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Criteria 2.	10 (2.5%)	11 (2.8%)	21 (2.6%)	2 (5.6%)	10 (21%)	12 (14%)	0 (0%)	0 (0%)	0 (0%)

4.2.8.6 Subgroup forest plot

For each subgroup we estimated the difference in risk of IgE-mediated food allergy for wP compared to aP within each grouping level. The estimated risk differences for each factor are reported in Figure 11.

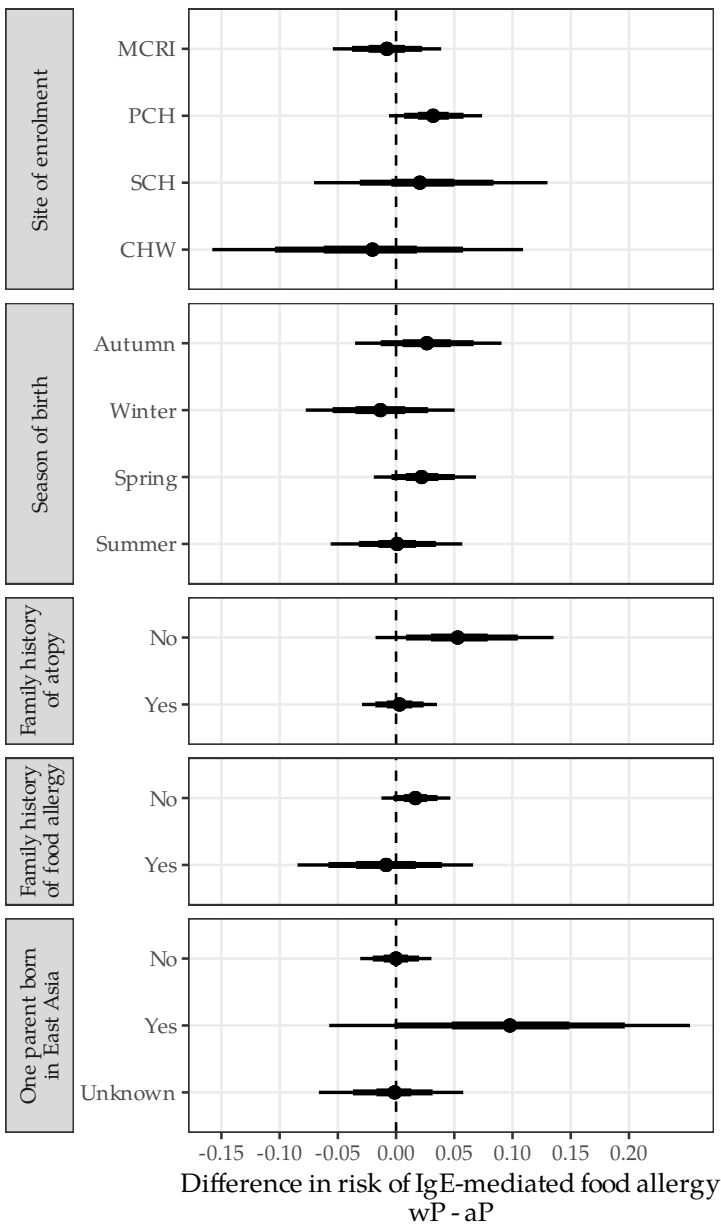


Figure 11: Subgroup specific estimates of difference in risk of IgE-mediated food allergy, wP versus aP.

4.3 Eczema

4.3.1 Overview

The primary source of information for eczema was the outcome_report form. The timing was based on the documented age at which eczema first occurred. Reported outcomes were cross checked against the food and household questionnaire form and medical history forms. Infants whose first reaction date for clinician-diagnosed eczema was prior to enrolment were considered to have “pre-existing” eczema and were excluded from the analysis set. There were 8 such infants.

The number and percentage of infants with any clinician-diagnosed eczema is reported in Table 38. For the protocol defined outcomes [2a. and 2b.], the eczema diagnosis needed to be new-onset and accompanied by a positive SPT to at least one allergen by approximately 12 months of age. These are reported in Table 39.

Table 38: Number and percentage of infants with any clinician-diagnosed eczema.

Outcome	aP N = 486	wP N = 486	Overall N = 972
Eczema	164 (34%)	158 (33%)	322 (33%)
Eczema by 6-months	89 (18%)	84 (17%)	173 (18%)
Eczema by 12-months	147 (30%)	135 (28%)	282 (29%)
Eczema by 18-months	163 (34%)	158 (33%)	321 (33%)

Table 39: Number and percentage of infants with new-onset clinician-diagnosed eczema. To be new-onset the first reaction date needed to be after the infants randomisation date. Those with missing were due to eczema diagnosis being present but no SPT records being available.

Outcome	aP N = 484	wP N = 480	Overall N = 964
New-onset cases			
Eczema	162 (33%)	152 (32%)	314 (33%)
Eczema by 6-months	87 (18%)	78 (16%)	165 (17%)
Eczema by 12-months	145 (30%)	129 (27%)	274 (28%)
Eczema by 18-months	161 (33%)	152 (32%)	313 (32%)
New-onset cases with positive SPT			
Eczema	50 (10%)	48 (10%)	98 (10%)
Missing	2	2	4
Eczema by 6-months	34 (7.0%)	28 (5.9%)	62 (6.5%)
Missing	1	2	3
Eczema by 12-months	47 (9.8%)	43 (9.0%)	90 (9.4%)
Missing	2	2	4
Eczema by 18-months	50 (10%)	48 (10%)	98 (10%)
Missing	2	2	4

4.3.2 Eczema model

Table 40 summarises the estimated model parameters for the probability of eczema by age under each treatment group, and their difference (the posterior densities are shown in Figure 12). The results between the analyses (unadjusted and adjusted, Section 2.5.1) were near identical. The most plausible values for the difference in eczema at 6 months [2.a] between wP and aP were -0.01 (-0.04, 0.02) [posterior median, (95% credible interval)]. The probability of a reduction in new-onset eczema under wP compared to aP was 0.77. At 12m [2.b] they were -0.01 (-0.04, 0.03) and 0.65.

Table 40: Posterior summaries for risk and difference in risk of eczema by age of onset.

	aP		wP		Difference
	y/n (%)	Median (95% CrI)	y/n (%)	Median (95% CrI)	Median (95% CrI), Pr(<0)
Unadjusted					
6m	34/483 (7.0%)	7.2 (5.1, 9.7)	28/478 (5.9%)	6.0 (4.1, 8.3)	-1.2 (-4.3, 2.0), 0.77
12m	47/482 (9.8%)	9.9 (7.4, 12.7)	43/478 (9.0%)	9.1 (6.8, 11.9)	-0.7 (-4.5, 3.0), 0.65
18m	50/482 (10.4%)	10.5 (8.0, 13.4)	48/478 (10.0%)	10.2 (7.7, 13.1)	-0.3 (-4.2, 3.5), 0.57
Adjusted					
6m	34/483 (7.0%)	7.2 (5.1, 9.6)	28/478 (5.9%)	5.8 (4.0, 8.1)	-1.3 (-4.3, 1.7), 0.80
12m	47/482 (9.8%)	9.8 (7.4, 12.7)	43/478 (9.0%)	9.0 (6.7, 11.7)	-0.8 (-4.5, 2.7), 0.68
18m	50/482 (10.4%)	10.5 (8.1, 13.4)	48/478 (10.0%)	10.0 (7.6, 12.8)	-0.6 (-4.3, 3.2), 0.61

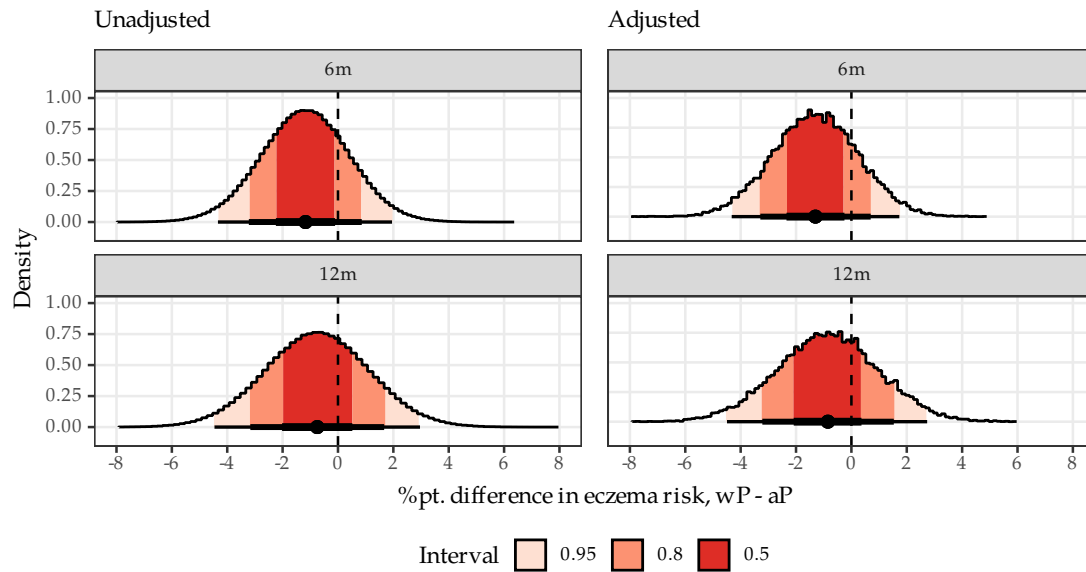


Figure 12: Posterior density for difference in risk of eczema by age and model. Values in regions with high density are those most plausible given the data and model assumptions.

5 Safety

5.1 Adverse events

There were 155 adverse events (135 unsolicited, 2 adverse events of special interest, and 18 serious adverse events) documented during the trial. Most adverse events were for infants randomised at PCH (149). Most of these events were due to the closer follow-up as part of stage 1 (Table 41) and for the first 150 participants recruited as part of stage 2 (Table 42). There were 6 adverse events for infants randomised at a site other than PCH (Table 43). A listing of the related AEs is given in Table 44.

There was a higher number of adverse events in the wP group compared to aP. There was also a higher number of infants who experienced at least one AE (Table 45). Some of the difference in total AEs was due to infants with multiple events. For both groups only a few AEs were documented as related to the blinded study vaccine.

A summary of the number and proportion of infants randomised at PCH with any reported adverse event by the MedDRA preferred term is given in Table 48. A summary of the total number of events is given in Table 49.

Table 41: Total number of adverse events (event rate) by study stage and site.

	aP (events/participants)	wP (events/participants)
Stage 1		
PCH	36/75	67/75
Stage 2		
PCH	17/131	29/132
CHW	2/30	1/29
MCRI	0/220	1/219
SCH	1/30	1/31

Table 42: Number of adverse events amongst stage 2 PCH participants for the first 150 and for the remaining participants.

	aP (events/participants)	wP (events/participants)
First 150 at PCH	16/75	25/75
Remaining PCH participants	1/56	4/57

Table 43: Adverse event listing of infants randomised at a site other than PCH or at PCH, but not as part of stage 1 or the first 150 of stage 2.

Record ID	Site	Treatment	Preferred term	Days after visit 1	Type of event	Severity	Relationship to study vaccine	Type of medical advice sought
4629-158	PCH	aP	Gastroenteritis NOS	19	SAE	Moderate	Not related	Hospitalisation
4629-175	PCH	wP	Irritability postvaccinal	0	SAE	Moderate	Related	Hospitalisation
4629-187	PCH	wP	COVID-19	224	Unsolicited	Mild	Not related	GP visit
4629-235	PCH	wP	Bronchiolitis	191	SAE	Moderate	Not related	Hospitalisation
4629-235	PCH	wP	Bronchiolitis	209	SAE	Moderate	Not related	Hospitalisation
4630-19	CHW	aP	Pertussis	479	AESI	Moderate	Not related	GP visit
4630-48	CHW	wP	Benign rolandic epilepsy	154	SAE	Moderate	Not related	Hospitalisation
4630-55	CHW	aP	Respiratory syncytial virus infection	505	Unsolicited	Severe	Not related	Hospitalisation
5785-140	MCRI	wP	Gastroesophageal reflux	3	SAE	Severe	Not related	Hospitalisation
8644-9	SCH	aP	Herniotomy	29	SAE	Moderate	Not related	Hospitalisation
8644-10	SCH	wP	Acute bronchiolitis	42	SAE	Severe	Not related	Hospitalisation

Table 44: Listing of adverse events designated as “Related”.

Record ID	Treatment	Preferred term	Days after visit 1	Type of event	Severity	Relationship to study vaccine	Type of medical advice sought
20	wP	Injection site haematoma	0	Unsolicited	Mild	Related	—
35	wP	Injection site haematoma	0	Unsolicited	Mild	Related	—
70 ¹	aP	Injection site haematoma	104	Unsolicited	Mild	Related	—
70 ¹	aP	Injection site haematoma	104	Unsolicited	Mild	Related	—
4629-75	wP	Rash trunk	1	Unsolicited	Mild	Related	—
4629-83	aP	Rash trunk	1	Unsolicited	Mild	Related	—
4629-101	wP	Injection site hematoma	—	Unsolicited	Mild	Related	—
4629-175	wP	Irritability postvaccinal	0	SAE	Moderate	Related	Hospitalisation

¹One was for left leg and the other was for right leg injection site

Table 45: Number and proportion of infants with any adverse events and the number of adverse events per infant for the first 300 participants at PCH.

Characteristic	aP N = 150	wP N = 150	Overall N = 300
Any AE	41 (27%)	67 (45%)	108 (36%)
Any SAE	4 (3%)	4 (3%)	8 (3%)
Any related AE	2 (1%)	4 (3%)	6 (2%)
Any related SAE	0 (0%)	0 (0%)	0 (0%)
Any severe AE	4 (3%)	2 (1%)	6 (2%)
Number of AEs			
0	109 (73%)	83 (55%)	192 (64%)
1	32 (21%)	50 (33%)	82 (27%)
2	7 (5%)	14 (9%)	21 (7%)
3	2 (1%)	1 (1%)	3 (1%)
4	0 (0%)	1 (1%)	1 (0%)
5+	0 (0%)	1 (1%)	1 (0%)
Number of SAEs			
0	146 (97%)	146 (97%)	292 (97%)
1	4 (3%)	3 (2%)	7 (2%)
2	0 (0%)	0 (0%)	0 (0%)
3	0 (0%)	1 (1%)	1 (0%)
Number of related AEs			
0	148 (99%)	146 (97%)	294 (98%)
1	1 (1%)	4 (3%)	5 (2%)
2	1 (1%)	0 (0%)	1 (0%)

Table 46: Number and proportion of infants with any adverse events and the number of adverse events per infant for remaining PCH participants and participants at a site other than PCH.

Characteristic	aP N = 336	wP N = 336	Overall N = 672
Any AE	4 (1%)	6 (2%)	10 (1%)
Any SAE	2 (1%)	5 (1%)	7 (1%)
Any related AE	0 (0%)	1 (0%)	1 (0%)
Any related SAE	0 (0%)	1 (0%)	1 (0%)
Any severe AE	1 (0%)	2 (1%)	3 (0%)
Number of AEs			
0	332 (99%)	330 (98%)	662 (99%)
1	4 (1%)	5 (1%)	9 (1%)
2	0 (0%)	1 (0%)	1 (0%)
3	0 (0%)	0 (0%)	0 (0%)
4	0 (0%)	0 (0%)	0 (0%)
5+	0 (0%)	0 (0%)	0 (0%)
Number of SAEs			
0	334 (99%)	331 (99%)	665 (99%)
1	2 (1%)	4 (1%)	6 (1%)
2	0 (0%)	1 (0%)	1 (0%)
3	0 (0%)	0 (0%)	0 (0%)
Number of related AEs			
0	336 (100%)	335 (100%)	671 (100%)
1	0 (0%)	1 (0%)	1 (0%)
2	0 (0%)	0 (0%)	0 (0%)

Table 47: Number and proportion of infants with any adverse events and the number of adverse events per infant for all participants stratified by receipt of intensive monitoring.

	First 300 PCH	Remaining
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Characteristic	aP N = 150	wP N = 150	Overall N = 300	aP N = 336	wP N = 336	Overall N = 672
Any AE	41 (27%)	67 (45%)	108 (36%)	4 (1%)	6 (2%)	10 (1%)
Any SAE	4 (3%)	4 (3%)	8 (3%)	2 (1%)	5 (1%)	7 (1%)
Any related AE	2 (1%)	4 (3%)	6 (2%)	0 (0%)	1 (0%)	1 (0%)
Any related SAE	0 (0%)	0 (0%)	0 (0%)	0 (0%)	1 (0%)	1 (0%)
Any severe AE	4 (3%)	2 (1%)	6 (2%)	1 (0%)	2 (1%)	3 (0%)
Number of AEs						
0	109 (73%)	83 (55%)	192 (64%)	332 (99%)	330 (98%)	662 (99%)
1	32 (21%)	50 (33%)	82 (27%)	4 (1%)	5 (1%)	9 (1%)
2	7 (5%)	14 (9%)	21 (7%)	0 (0%)	1 (0%)	1 (0%)
3	2 (1%)	1 (1%)	3 (1%)	0 (0%)	0 (0%)	0 (0%)
4	0 (0%)	1 (1%)	1 (0%)	0 (0%)	0 (0%)	0 (0%)
5+	0 (0%)	1 (1%)	1 (0%)	0 (0%)	0 (0%)	0 (0%)
Number of SAEs						
0	146 (97%)	146 (97%)	292 (97%)	334 (99%)	331 (99%)	665 (99%)
1	4 (3%)	3 (2%)	7 (2%)	2 (1%)	4 (1%)	6 (1%)
2	0 (0%)	0 (0%)	0 (0%)	0 (0%)	1 (0%)	1 (0%)
3	0 (0%)	1 (1%)	1 (0%)	0 (0%)	0 (0%)	0 (0%)
Number of related AEs						
0	148 (99%)	146 (97%)	294 (98%)	336 (100%)	335 (100%)	671 (100%)
1	1 (1%)	4 (3%)	5 (2%)	0 (0%)	1 (0%)	1 (0%)
2	1 (1%)	0 (0%)	1 (0%)	0 (0%)	0 (0%)	0 (0%)

Table 48: Proportion of infants with any event by MeDRA preferred terms, first 300 at PCH only.

Preferred term	aP N = 150	wP N = 150	Overall N = 300
Number of patients with event	41 (27%)	67 (45%)	108 (36%)
Teething	9 (6.0%)	12 (8.0%)	21 (7.0%)
Viral upper respiratory tract infection	5 (3.3%)	7 (4.7%)	12 (4.0%)
Upper respiratory tract infection	4 (2.7%)	7 (4.7%)	11 (3.7%)
Injection site haematoma	3 (2.0%)	5 (3.3%)	8 (2.7%)
Rash	0 (0%)	5 (3.3%)	5 (1.7%)
Conjunctivitis	2 (1.3%)	2 (1.3%)	4 (1.3%)
Teething pain	1 (0.7%)	3 (2.0%)	4 (1.3%)
Constipation	0 (0%)	3 (2.0%)	3 (1.0%)
Gastroenteritis	1 (0.7%)	2 (1.3%)	3 (1.0%)
Gastroesophageal reflux disease	1 (0.7%)	2 (1.3%)	3 (1.0%)
Arthropod sting	0 (0%)	2 (1.3%)	2 (0.7%)
Bronchiolitis	0 (0%)	2 (1.3%)	2 (0.7%)
Dry skin	0 (0%)	2 (1.3%)	2 (0.7%)
Gastroenteritis viral	1 (0.7%)	1 (0.7%)	2 (0.7%)
Hospitalisation	1 (0.7%)	1 (0.7%)	2 (0.7%)
Rash trunk	1 (0.7%)	1 (0.7%)	2 (0.7%)
Tonsillitis	1 (0.7%)	1 (0.7%)	2 (0.7%)
Urticaria	1 (0.7%)	1 (0.7%)	2 (0.7%)
Vaccination site rash	1 (0.7%)	1 (0.7%)	2 (0.7%)
Vessel puncture site haematoma	1 (0.7%)	1 (0.7%)	2 (0.7%)
Viral rash	0 (0%)	2 (1.3%)	2 (0.7%)
Wheezing	1 (0.7%)	1 (0.7%)	2 (0.7%)
Abdominal distension	0 (0%)	1 (0.7%)	1 (0.3%)
Abdominal pain	0 (0%)	1 (0.7%)	1 (0.3%)
Arthropod bite	0 (0%)	1 (0.7%)	1 (0.3%)
Balanoposthitis	0 (0%)	1 (0.7%)	1 (0.3%)
Constipation neonatal	1 (0.7%)	0 (0%)	1 (0.3%)
Croup	0 (0%)	1 (0.7%)	1 (0.3%)
Dacryostenosis congenital	0 (0%)	1 (0.7%)	1 (0.3%)
Dermatitis	0 (0%)	1 (0.7%)	1 (0.3%)
Dermatitis contact	1 (0.7%)	0 (0%)	1 (0.3%)
Dermatitis diaper	0 (0%)	1 (0.7%)	1 (0.3%)
Ear infection	1 (0.7%)	0 (0%)	1 (0.3%)
Escherichia urinary tract infection	0 (0%)	1 (0.7%)	1 (0.3%)
Flatulence	0 (0%)	1 (0.7%)	1 (0.3%)
Gastroenteritis NOS	1 (0.7%)	0 (0%)	1 (0.3%)
Gingival pain	1 (0.7%)	0 (0%)	1 (0.3%)
Ileocolic intussusception	1 (0.7%)	0 (0%)	1 (0.3%)
Inflammation at site of injection	0 (0%)	1 (0.7%)	1 (0.3%)
Injection site hematoma	0 (0%)	1 (0.7%)	1 (0.3%)
Investigation	0 (0%)	1 (0.7%)	1 (0.3%)
Irritability	1 (0.7%)	0 (0%)	1 (0.3%)
Laryngomalacia	1 (0.7%)	0 (0%)	1 (0.3%)
Lip injury	0 (0%)	1 (0.7%)	1 (0.3%)
Oral candidiasis	0 (0%)	1 (0.7%)	1 (0.3%)
Periorbital swelling	0 (0%)	1 (0.7%)	1 (0.3%)
Procedural pain	1 (0.7%)	0 (0%)	1 (0.3%)
Puffy eyes	1 (0.7%)	0 (0%)	1 (0.3%)
Rash on leg	0 (0%)	1 (0.7%)	1 (0.3%)
Rectal haemorrhage	0 (0%)	1 (0.7%)	1 (0.3%)
Runny nose	0 (0%)	1 (0.7%)	1 (0.3%)
Scratch	1 (0.7%)	0 (0%)	1 (0.3%)
Seborrhoeic dermatitis	1 (0.7%)	0 (0%)	1 (0.3%)
Shaking of hands	0 (0%)	1 (0.7%)	1 (0.3%)
Sleep disorder	1 (0.7%)	0 (0%)	1 (0.3%)
Sore throat	0 (0%)	1 (0.7%)	1 (0.3%)
Upper respiratory infection	0 (0%)	1 (0.7%)	1 (0.3%)
Upper respiratory tract infection viral NOS	0 (0%)	1 (0.7%)	1 (0.3%)

Urinary tract infection	1 (0.7%)	0 (0%)	1 (0.3%)
Vessel puncture site pain	1 (0.7%)	0 (0%)	1 (0.3%)
Viral infection	0 (0%)	1 (0.7%)	1 (0.3%)

Table 49: Number of events by MedDRA preferred terms, first 300 at PCH only.

Preferred term	aP N = 150	wP N = 150	Overall N = 300
Total number of events	52	92	144
Teething	11	14	25
Viral upper respiratory tract infection	5	7	12
Upper respiratory tract infection	4	7	11
Injection site haematoma	4	5	9
Rash	0	5	5
Conjunctivitis	2	2	4
Teething pain	1	3	4
Constipation	0	3	3
Gastroenteritis	1	2	3
Gastroesophageal reflux disease	1	2	3
Abdominal distension	0	2	2
Arthropod bite	0	2	2
Arthropod sting	0	2	2
Bronchiolitis	0	2	2
Dry skin	0	2	2
Gastroenteritis viral	1	1	2
Hospitalisation	1	1	2
Rash trunk	1	1	2
Tonsillitis	1	1	2
Urticaria	1	1	2
Vaccination site rash	1	1	2
Vessel puncture site haematoma	1	1	2
Viral rash	0	2	2
Wheezing	1	1	2
Abdominal pain	0	1	1
Balanoposthitis	0	1	1
Constipation neonatal	1	0	1
Croup	0	1	1
Dacryostenosis congenital	0	1	1
Dermatitis	0	1	1
Dermatitis contact	1	0	1
Dermatitis diaper	0	1	1
Ear infection	1	0	1
Escherichia urinary tract infection	0	1	1
Flatulence	0	1	1
Gastroenteritis NOS	1	0	1
Gingival pain	1	0	1
Ileocolic intussusception	1	0	1
Inflammation at site of injection	0	1	1
Injection site hematoma	0	1	1
Investigation	0	1	1
Irritability	1	0	1
Laryngomalacia	1	0	1
Lip injury	0	1	1
Oral candidiasis	0	1	1
Periorbital swelling	0	1	1
Procedural pain	1	0	1
Puffy eyes	1	0	1
Rash on leg	0	1	1
Rectal haemorrhage	0	1	1
Runny nose	0	1	1
Scratch	1	0	1
Seborrhoeic dermatitis	1	0	1

Shaking of hands	0	1	1
Sleep disorder	1	0	1
Sore throat	0	1	1
Upper respiratory infection	0	1	1
Upper respiratory tract infection viral NOS	0	1	1
Urinary tract infection	1	0	1
Vessel puncture site pain	1	0	1
Viral infection	0	1	1

5.2 Serious adverse events

A listing of serious adverse events are reported in Table 50. These are any events reported either in the SAE reporting log (sae_reporting_log, stage 2 only) or on the adverse events form (adverse_events) with aetype = "SAE".

Table 50: Listing of serious adverse events. Missing values were due to no adverse event record for corresponding SAE log record.

Record ID	Treatment	MedDRA code	Severity	Type of medical advice sought	Relationship to study vaccine?	Days after visit 1
22	wP	Bronchiolitis	Mild	Hospitalisation	Not Related	144
24	wP	Escherichia urinary tract infection	Moderate	Hospitalisation	Not Related	43
24	wP	Hospitalisation	Mild	Hospitalisation	Not Related	95
24	wP	Investigation	Mild	Hospitalisation	Not Related	190
60	wP	Bronchiolitis	Moderate	Hospitalisation	Not Related	49
65	aP	Hospitalisation	Mild	Hospitalisation	Not Related	7
94	wP	Rectal haemorrhage	Moderate	Hospitalisation	Not Related	7
4629-2	aP	Ileocolic intussusception	Severe	Hospitalisation	Not Related	189
4629-5	aP	Laryngomalacia	Mild	Hospitalisation	Not Related	48
4629-91	aP	Urinary tract infection	Moderate	Hospitalisation	Not Related	46
4629-158	aP	Gastroenteritis NOS	Moderate	Hospitalisation	Not Related	19
4629-175	wP	Irritability postvaccinal	Moderate	Hospitalisation	Related	0
4629-235	wP	Bronchiolitis	Moderate	Hospitalisation	Not Related	191
4629-235	wP	Bronchiolitis	Moderate	Hospitalisation	Not Related	209
4630-48	wP	Benign rolandic epilepsy	Moderate	Hospitalisation	Not Related	154
5785-140	wP	Gastrooesophageal reflux	Severe	Hospitalisation	Not Related	3
8644-9	aP	Herniotomy	Moderate	Hospitalisation	Not Related	29
8644-10	wP	Acute bronchiolitis	Severe	Hospitalisation	Not Related	42

6 Reactogenicity

6.1 Diary cards

Combined data from stage 1 and stage 2 diary card responses are reported in the following sections. During stage 1, diary cards were provided following 6-week, 4-month, 6-month and 18-month vaccinations. During stage 2, at PCH only, for the first 153 participants, diary cards were provided following 6-week and 18-month vaccinations. Given the different collection schedules for the protocol versions, the denominators for the 6-week and 18-month responses differ to those for the 4-month and 6-month responses.

The number and percentage of returned diary cards for stage 1 are shown in Table 51 and stage 2 in Table 52.

Table 51: Number and proportion of parents from stage 1 who returned diary cards following each vaccination occasion.

Vaccination	aP N = 75	wP N = 75	Overall N = 150
6-week	72 (96%)	74 (99%)	146 (97%)
4-month	73 (97%)	71 (95%)	144 (96%)
6-month	68 (91%)	69 (92%)	137 (91%)
18-month	62 (83%)	62 (83%)	124 (83%)

Table 52: Number and proportion of parents from stage 2 who returned diary cards following each vaccination occasion.

Vaccination	aP N = 77	wP N = 76	Overall N = 153
6-week	73 (95%)	63 (83%)	136 (89%)
18-month	47 (61%)	62 (82%)	109 (71%)

6.2 Parental acceptability

Following vaccination, parent's were asked whether they would be willing to have another child have the combination of vaccinations at the given age. There were 3 occasions where a parent disagreed or strongly disagreed.

Table 53: Number and percentage of parents giving each response to the given statement following 6-week, 4-month, 6-month, and 18-month vaccination, by assigned treatment group. Unknown is due to either a failure to return the diary card, or non-response to the item.

Characteristic	6-week		4-month		6-month		18-month	
	aP	wP	aP	wP	aP	wP	aP	wP
	N = 152	N = 151	N = 75	N = 75	N = 75	N = 75	N = 152	N = 151
I would be willing to have another child have this combination of immunisations at this age								
Strongly disagree	0 (0%)	1 (0.8%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	1 (1.0%)
Disagree	0 (0%)	0 (0%)	0 (0%)	0 (0%)	1 (1.5%)	0 (0%)	0 (0%)	0 (0%)
Neither agree or disagree	6 (4.3%)	3 (2.3%)	3 (4.2%)	1 (1.4%)	4 (5.9%)	0 (0%)	3 (3.3%)	3 (2.9%)
Agree	30 (21%)	44 (33%)	11 (15%)	11 (15%)	13 (19%)	16 (24%)	22 (24%)	28 (27%)
Strongly agree	105 (74%)	84 (64%)	58 (81%)	59 (83%)	50 (74%)	50 (76%)	67 (73%)	73 (70%)
Unknown	11	19	3	4	7	9	60	46

6.3 Solicited systemic reactions

The number and percentage of participants who experienced any severe reaction over the 7 days following vaccination, and the highest reaction grade by type over those 7 days are reported in Table 54, Table 55, and Figure 13.

The subsequent sections report on the daily reaction grades following each vaccination occasion by the type of reaction.

Table 54: Experience of any grade 3 intensity systemic reaction following vaccination (day 0 to 6) by assigned treatment. Excludes participants with missing diary cards or non-response for all days.

	aP N = 152	wP N = 151
6-week	16/145 (11%)	24/137 (18%)
4-month	12/73 (16%)	9/71 (13%)
6-month	5/68 (7%)	5/69 (7%)
18-month	14/107 (13%)	12/121 (10%)

Table 55: Highest intensity grade for systemic reactions after each pertussis-containing vaccine dose.

	aP					wP				
	None	1	2	3	Missing	None	1	2	3	Missing
6-week										
Decreased feeding	91 (66)	39 (28)	5 (4)	2 (1)	15	58 (43)	54 (40)	21 (16)	2 (1)	16
Diarrhoea	67 (49)	54 (39)	15 (11)	1 (1)	15	80 (59)	46 (34)	9 (7)	0 (0)	16
Drowsiness	49 (36)	76 (56)	10 (7)	1 (1)	16	36 (27)	62 (46)	33 (24)	4 (3)	16
Irritability	21 (15)	64 (47)	39 (28)	13 (9)	15	11 (8)	51 (38)	55 (41)	18 (13)	16
Restlessness	31 (23)	67 (49)	29 (21)	10 (7)	15	36 (27)	42 (31)	45 (33)	12 (9)	16
Vomiting	91 (66)	32 (23)	12 (9)	2 (1)	15	88 (65)	35 (26)	12 (9)	0 (0)	16
4-month										
Decreased feeding	52 (72)	17 (24)	3 (4)	0 (0)	3	51 (72)	18 (25)	1 (1)	1 (1)	4
Diarrhoea	51 (71)	17 (24)	4 (6)	0 (0)	3	51 (72)	19 (27)	1 (1)	0 (0)	4
Drowsiness	42 (58)	19 (26)	8 (11)	3 (4)	3	33 (46)	34 (48)	4 (6)	0 (0)	4
Irritability	16 (22)	24 (33)	24 (33)	8 (11)	3	18 (25)	29 (41)	16 (23)	8 (11)	4
Restlessness	30 (42)	27 (38)	11 (15)	4 (6)	3	30 (42)	31 (44)	7 (10)	3 (4)	4
Vomiting	57 (79)	12 (17)	3 (4)	0 (0)	3	53 (75)	16 (23)	2 (3)	0 (0)	4
6-month										
Decreased feeding	50 (75)	13 (19)	4 (6)	0 (0)	8	54 (78)	13 (19)	2 (3)	0 (0)	6
Diarrhoea	59 (88)	5 (7)	3 (4)	0 (0)	8	66 (96)	2 (3)	1 (1)	0 (0)	6
Drowsiness	47 (70)	18 (27)	2 (3)	0 (0)	8	52 (75)	17 (25)	0 (0)	0 (0)	6
Irritability	24 (36)	25 (38)	13 (20)	4 (6)	9	28 (41)	29 (42)	7 (10)	5 (7)	6
Restlessness	36 (54)	22 (33)	7 (10)	2 (3)	8	47 (68)	16 (23)	6 (9)	0 (0)	6
Vomiting	55 (82)	11 (16)	1 (1)	0 (0)	8	58 (84)	10 (14)	1 (1)	0 (0)	6
18-month										
Decreased feeding	70 (67)	26 (25)	7 (7)	2 (2)	47	71 (60)	35 (29)	10 (8)	3 (3)	32
Diarrhoea	77 (74)	23 (22)	2 (2)	2 (2)	48	101 (85)	14 (12)	4 (3)	0 (0)	32
Drowsiness	72 (69)	23 (22)	8 (8)	2 (2)	47	79 (66)	31 (26)	6 (5)	3 (3)	32
Irritability	24 (23)	44 (42)	30 (29)	7 (7)	47	29 (25)	48 (41)	30 (25)	11 (9)	33
Restlessness	57 (54)	33 (31)	11 (10)	4 (4)	47	61 (51)	36 (30)	17 (14)	5 (4)	32
Vomiting	96 (91)	3 (3)	4 (4)	2 (2)	47	110 (92)	5 (4)	2 (2)	2 (2)	32

For participants with non-responses for some days, the maximum intensity was taken as the greatest intensity reported on days with responses available. Percentages exclude participants with a missing highest grade (diary card not returned, or non-response for all days).

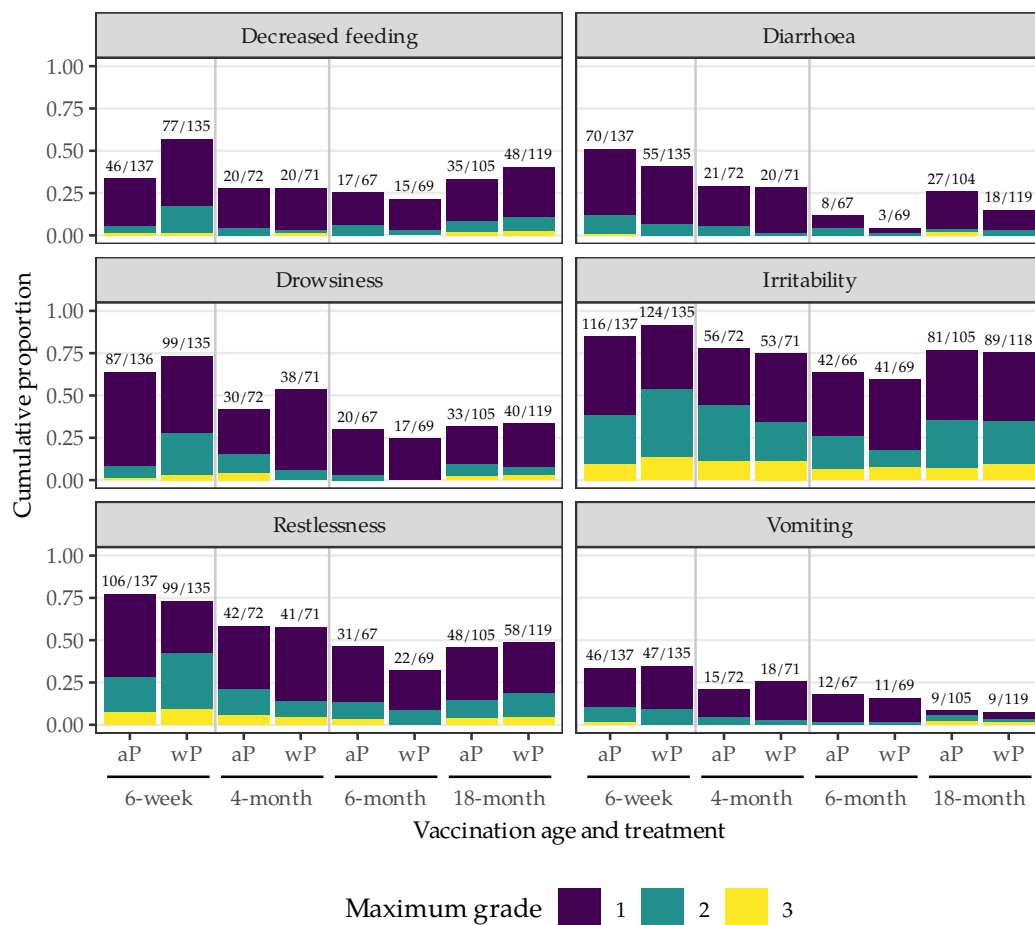


Figure 13: Highest intensity grade for systemic reactions each pertussis-containing vaccine dose. *aP*, acellular pertussis vaccine; *wP*, whole-cell pertussis vaccine; grade 1: easily tolerated; grade 2, interferes with normal everyday activities; grade 3, prevents normal everyday activities or requires significant medical intervention. Above each bar (n/N), we indicate the number of infants reporting systemic reactions grade 1 and above (numerator; n) versus the total number of infants reporting on systemic reactions (none, grade 1, grade 2, or grade 3; denominator; N).

6.3.1 Irritability

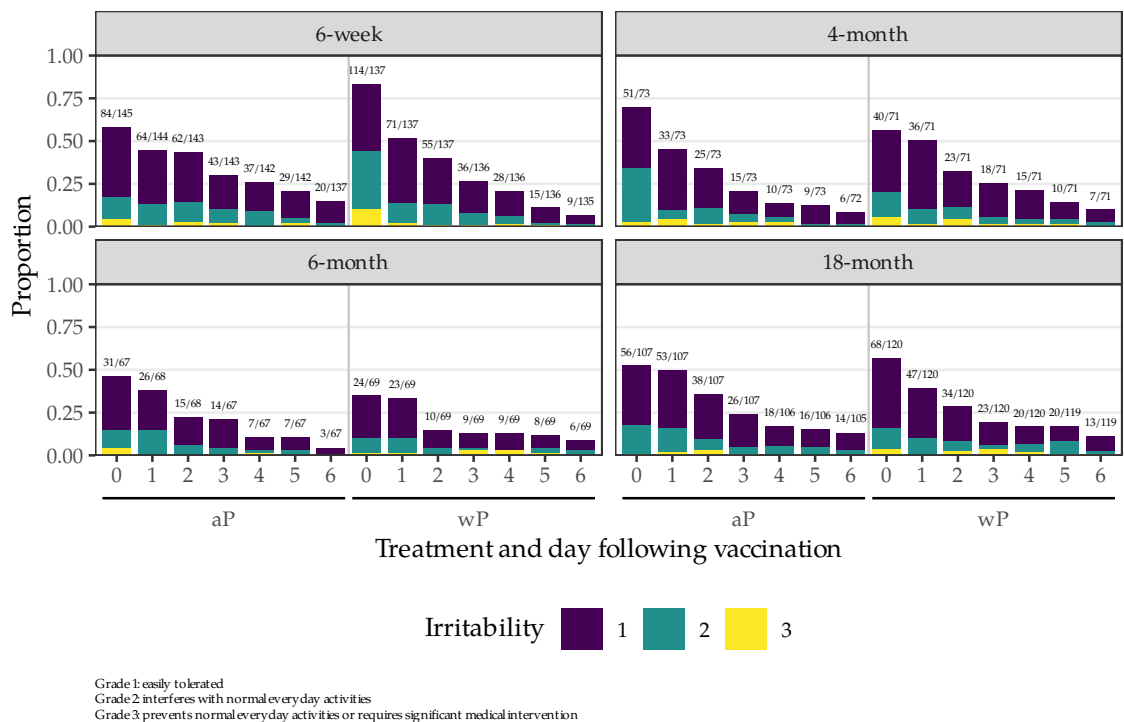


Figure 14: Irritability in the 7 days following each pertussis-containing vaccine dose. *aP*, acellular pertussis vaccine; *wP*, whole-cell pertussis vaccine; grade 1: easily tolerated; grade 2, interferes with normal everyday activities; grade 3, prevents normal everyday activities or requires significant medical intervention. Above each bar (n/N), we indicate the number of infants reporting systemic reactions grade 1 and above (numerator; n) versus the total number of infants reporting on systemic reactions (none, grade 1, grade 2, or grade 3; denominator; N).

6.3.2 Vomiting

Figure 15 presents the distribution of daily grades for vomiting following each vaccination occasion by assigned treatment.

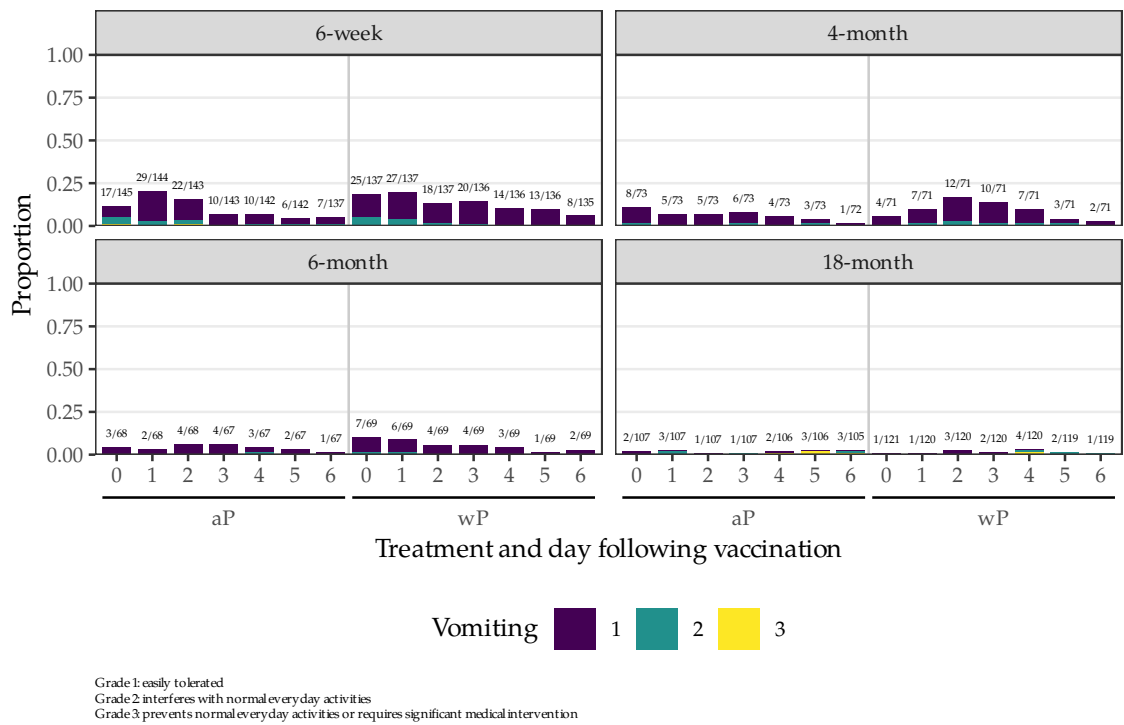


Figure 15: Vomiting in the 7 days following each pertussis-containing vaccine dose. *aP*, acellular pertussis vaccine; *wP*, whole-cell pertussis vaccine; grade 1: easily tolerated; grade 2, interferes with normal everyday activities; grade 3, prevents normal everyday activities or requires significant medical intervention. Above each bar (n/N), we indicate the number of infants reporting systemic reactions grade 1 and above (numerator; n) versus the total number of infants reporting on systemic reactions (none, grade 1, grade 2, or grade 3; denominator; N).

6.3.3 Diarrhoea

Figure 16 presents the distribution of daily grades for diarrhoea following each vaccination occasion by assigned treatment.

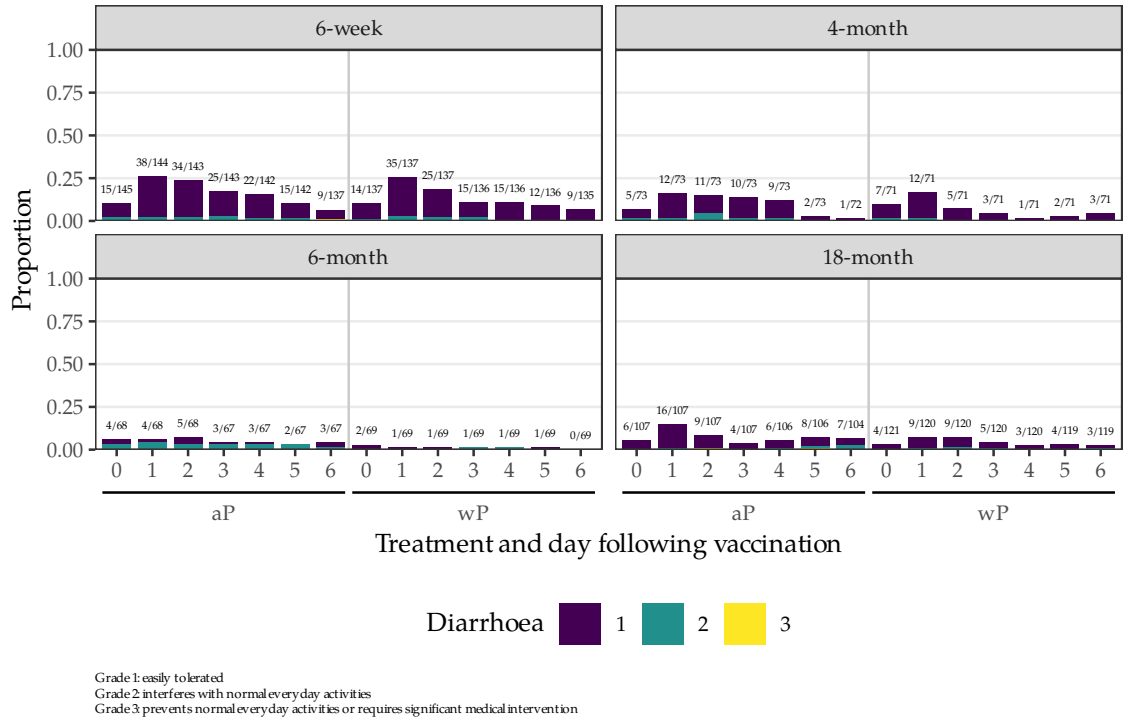


Figure 16: Diarrhoea in the 7 days following each pertussis-containing vaccine dose. aP, acellular pertussis vaccine; wP, whole-cell pertussis vaccine; grade 1: easily tolerated; grade 2, interferes with normal everyday activities; grade 3, prevents normal everyday activities or requires significant medical intervention. Above each bar (n/N), we indicate the number of infants reporting systemic reactions grade 1 and above (numerator; n) versus the total number of infants reporting on systemic reactions (none, grade 1, grade 2, or grade 3; denominator; N).

6.3.4 Decreased Feeding

Figure 17 presents the distribution of daily intensity grades for decreased feeding following each vaccination occasion by assigned treatment.

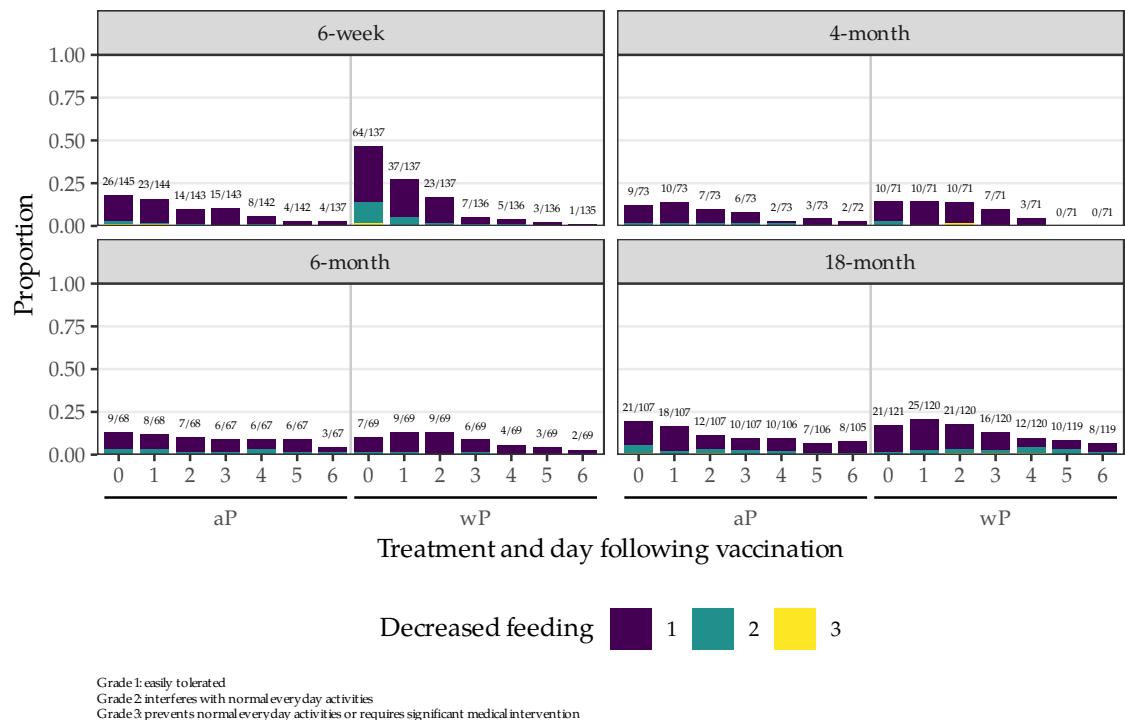


Figure 17: Decreased feeding in the 7 days following each pertussis-containing vaccine dose. *aP*, acellular pertussis vaccine; *wP*, whole-cell pertussis vaccine; grade 1: easily tolerated; grade 2, interferes with normal everyday activities; grade 3, prevents normal everyday activities or requires significant medical intervention. Above each bar (n/N), we indicate the number of infants reporting systemic reactions grade 1 and above (numerator; n) versus the total number of infants reporting on systemic reactions (none, grade 1, grade 2, or grade 3; denominator; N).

6.3.5 Drowsiness

Figure 18 presents the distribution of daily intensity grades for drowsiness following each vaccination occasion by assigned treatment.

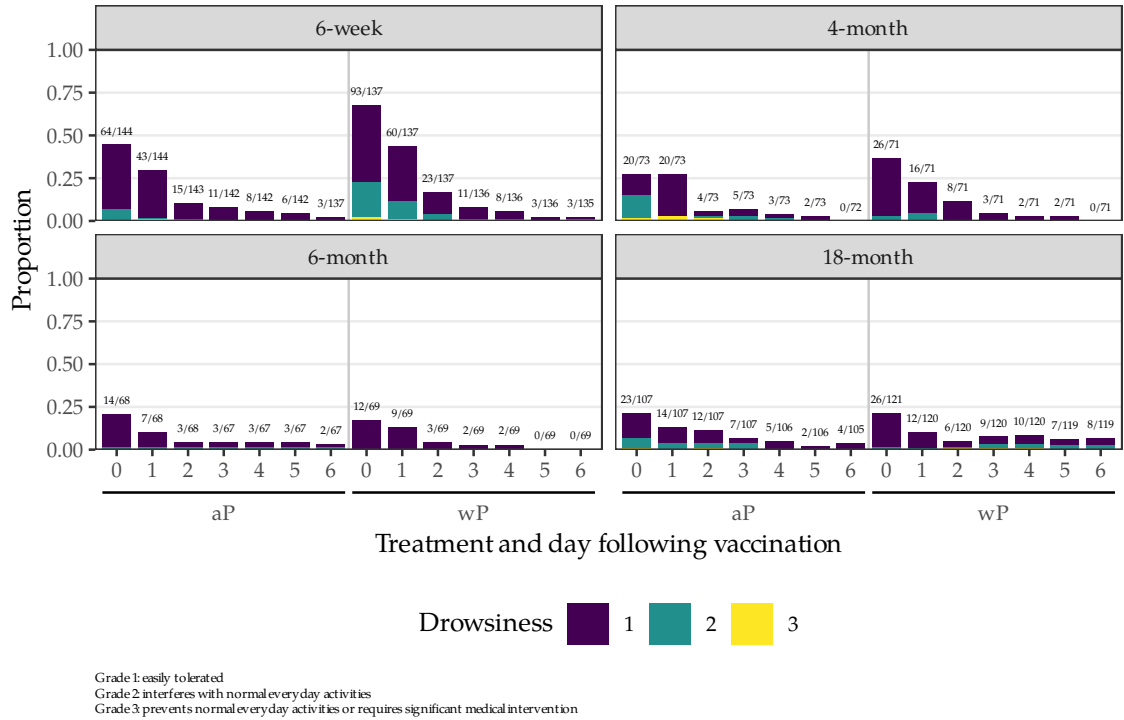


Figure 18: Drowsiness in the 7 days following each pertussis-containing vaccine dose. *aP*, acellular pertussis vaccine; *wP*, whole-cell pertussis vaccine; grade 1: easily tolerated; grade 2, interferes with normal everyday activities; grade 3, prevents normal everyday activities or requires significant medical intervention. Above each bar (n/N), we indicate the number of infants reporting systemic reactions grade 1 and above (numerator; n) versus the total number of infants reporting on systemic reactions (none, grade 1, grade 2, or grade 3; denominator; N).

6.3.6 Restlessness

Figure 19 presents the distribution of daily intensity grades for restlessness following each vaccination occasion by assigned treatment.

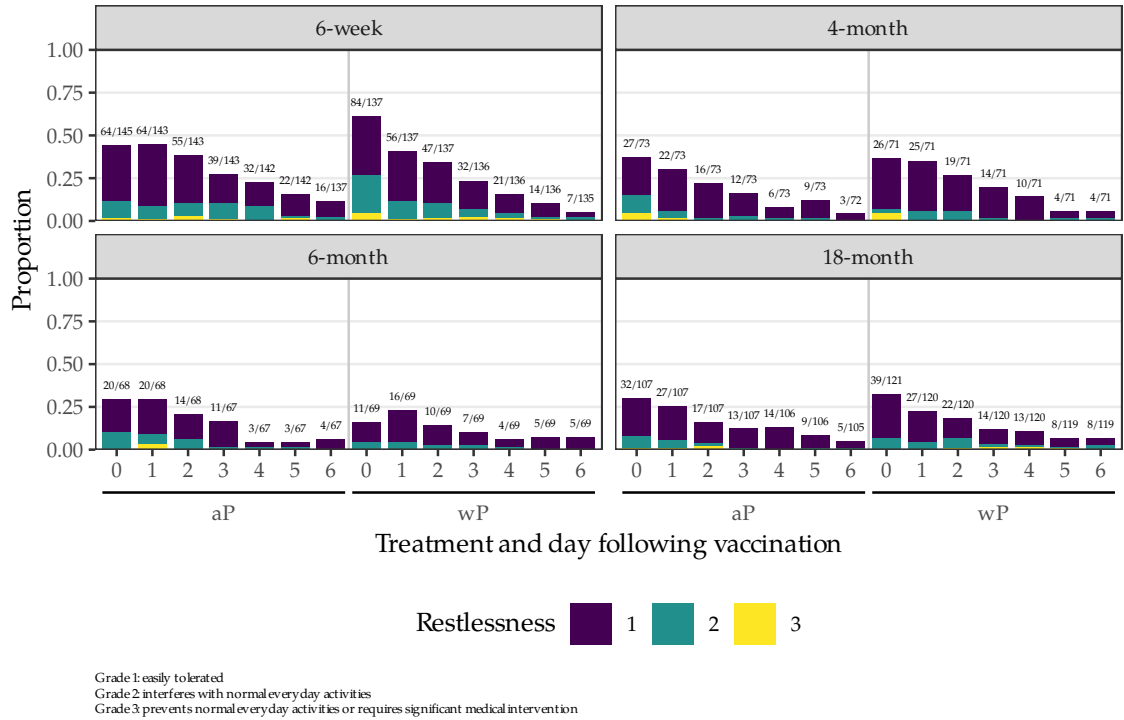


Figure 19: Restlessness in the 7 days following each pertussis-containing vaccine dose. *aP*, acellular pertussis vaccine; *wP*, whole-cell pertussis vaccine; grade 1: easily tolerated; grade 2, interferes with normal everyday activities; grade 3, prevents normal everyday activities or requires significant medical intervention. Above each bar (n/N), we indicate the number of infants reporting systemic reactions grade 1 and above (numerator; n) versus the total number of infants reporting on systemic reactions (none, grade 1, grade 2, or grade 3; denominator; N).

6.3.7 Fever

Table 56 presents the highest reported temperature following each vaccination occasion by assigned treatment. Figure 20 summarises daily temperature categories in the days following vaccination.

Table 56: Highest fever following each vaccination by assigned treatment.

	aP N = 152 (6-week, 18-month) N = 75 (4-month, 6-month)					wP N = 151 (6-week, 18-month) N = 75 (4-month, 6-month)				
	<38	38.0-38.9	39.0-39.9	≥40.0	Missing	<38	38.0-38.9	39.0-39.9	≥40.0	Missing
6-week	140 (98)	3 (2)	0 (0)	0 (0)	9	129 (96)	6 (4)	0 (0)	0 (0)	16
4-month	68 (94)	4 (6)	0 (0)	0 (0)	3	67 (96)	3 (4)	0 (0)	0 (0)	5
6-month	67 (99)	1 (1)	0 (0)	0 (0)	7	67 (99)	1 (1)	0 (0)	0 (0)	7
18-month	100 (95)	2 (2)	2 (2)	1 (1)	47	104 (87)	12 (10)	3 (3)	0 (0)	32

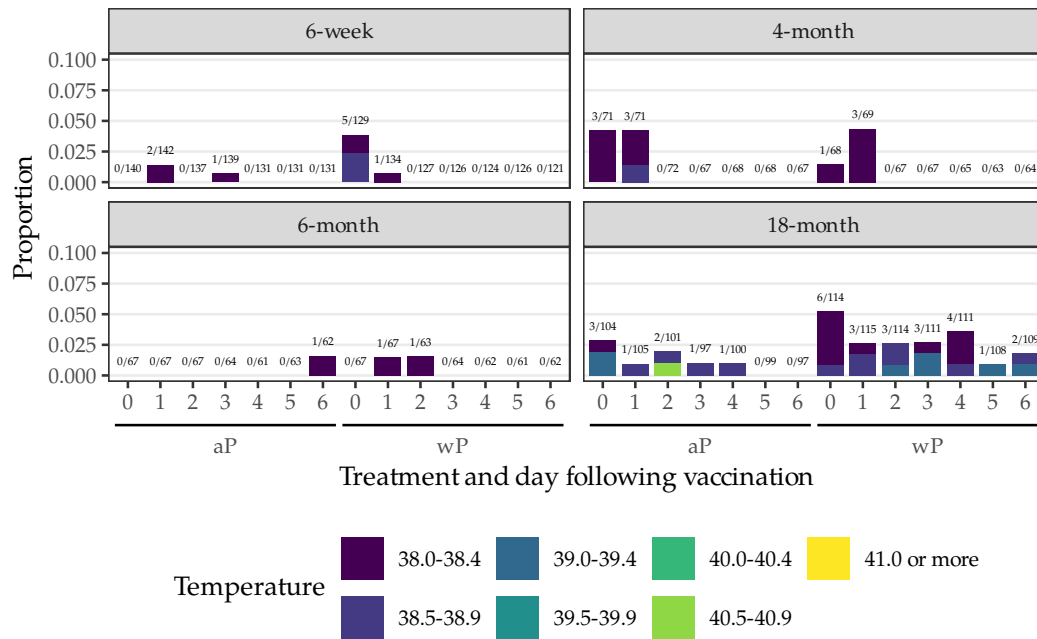


Figure 20: Fever by vaccination age and assigned treatment, stage 1 and 2 combined.

6.4 Solicited injection site reactions

The number and percentage of participants who experienced any severe injection site reaction over the 7 days following vaccination, and the highest reaction size by type over those 7 days are reported in Table 54, Table 57, and Figure 21.

The subsequent sections report on the daily injection site reaction sizes following each vaccination occasion by the type of reaction.

Table 57: Maximum injection site reaction size in the 7 days following vaccination, by vaccination age, vaccine, and assigned treatment.

	Size (mm)											
	aP (n = 152)						wP (n = 151)					
	None	>0-10	>10-25	>25-50	>50	Missing	None	>0-10	>10-25	>25-50	>50	Missing
6-week - aP/wP												
Erythema	99 (69)	45 (31)	0 (0)	0 (0)	0 (0)	8	59 (43)	71 (52)	2 (1)	4 (3)	1 (1)	14
Swelling	123 (85)	19 (13)	2 (1)	0 (0)	0 (0)	8	92 (67)	34 (25)	4 (3)	6 (4)	1 (1)	14
Induration	111 (77)	29 (20)	4 (3)	0 (0)	0 (0)	8	72 (53)	49 (36)	9 (7)	7 (5)	0 (0)	14
6-week - PCV-13												
Erythema	100 (69)	41 (28)	3 (2)	0 (0)	0 (0)	8	87 (64)	46 (34)	2 (1)	1 (1)	1 (1)	14
Swelling	126 (88)	17 (12)	1 (1)	0 (0)	0 (0)	8	110 (80)	20 (15)	5 (4)	1 (1)	1 (1)	14
Induration	108 (75)	34 (24)	2 (1)	0 (0)	0 (0)	8	98 (72)	31 (23)	6 (4)	2 (1)	0 (0)	14
4-month - DTaP-HepB-IPV-Hib												
Erythema	36 (49)	30 (41)	6 (8)	1 (1)	0 (0)	2	40 (56)	28 (39)	2 (3)	1 (1)	0 (0)	4
Swelling	53 (73)	15 (21)	2 (3)	3 (4)	0 (0)	2	62 (87)	8 (11)	0 (0)	1 (1)	0 (0)	4
Induration	44 (60)	23 (32)	3 (4)	3 (4)	0 (0)	2	54 (76)	14 (20)	2 (3)	1 (1)	0 (0)	4
4-month - PCV-13												
Erythema	39 (53)	28 (38)	5 (7)	1 (1)	0 (0)	2	39 (55)	29 (41)	2 (3)	1 (1)	0 (0)	4
Swelling	52 (71)	17 (23)	2 (3)	2 (3)	0 (0)	2	63 (89)	6 (8)	1 (1)	1 (1)	0 (0)	4
Induration	47 (64)	20 (27)	5 (7)	1 (1)	0 (0)	2	54 (76)	14 (20)	1 (1)	2 (3)	0 (0)	4
6-month - DTaP-HepB-IPV-Hib												
Erythema	26 (39)	36 (54)	4 (6)	1 (1)	0 (0)	8	28 (41)	30 (44)	9 (13)	1 (1)	0 (0)	7
Swelling	50 (75)	14 (21)	2 (3)	1 (1)	0 (0)	8	53 (78)	9 (13)	3 (4)	3 (4)	0 (0)	7
Induration	37 (55)	28 (42)	2 (3)	0 (0)	0 (0)	8	34 (50)	24 (35)	7 (10)	3 (4)	0 (0)	7
18-month - MMR												
Erythema	54 (52)	48 (47)	1 (1)	0 (0)	0 (0)	49	63 (55)	51 (44)	1 (1)	0 (0)	0 (0)	36
Swelling	85 (83)	17 (17)	1 (1)	0 (0)	0 (0)	49	90 (78)	24 (21)	1 (1)	0 (0)	0 (0)	36
Induration	82 (80)	21 (20)	0 (0)	0 (0)	0 (0)	49	90 (78)	24 (21)	1 (1)	0 (0)	0 (0)	36
18-month - DTaP-IPV												
Erythema	40 (37)	47 (44)	10 (9)	8 (7)	2 (2)	45	44 (36)	53 (44)	4 (3)	13 (11)	7 (6)	30
Swelling	63 (59)	28 (26)	10 (9)	6 (6)	0 (0)	45	79 (65)	33 (27)	2 (2)	6 (5)	1 (1)	30
Induration	60 (56)	34 (32)	8 (7)	5 (5)	0 (0)	45	74 (61)	38 (31)	6 (5)	2 (2)	1 (1)	30
18-month - Hib												
Erythema	50 (47)	52 (49)	2 (2)	2 (2)	0 (0)	46	59 (49)	51 (42)	3 (2)	7 (6)	0 (0)	31
Swelling	79 (75)	24 (23)	2 (2)	1 (1)	0 (0)	46	101 (84)	18 (15)	1 (1)	0 (0)	0 (0)	31
Induration	75 (71)	30 (28)	1 (1)	0 (0)	0 (0)	46	86 (72)	30 (25)	4 (3)	0 (0)	0 (0)	31

For participants with non-responses for some days, the maximum size is taken as the largest size reported on days with responses available. Percentages exclude participants with a missing highest grade (diary card not returned, or non-response for all days).

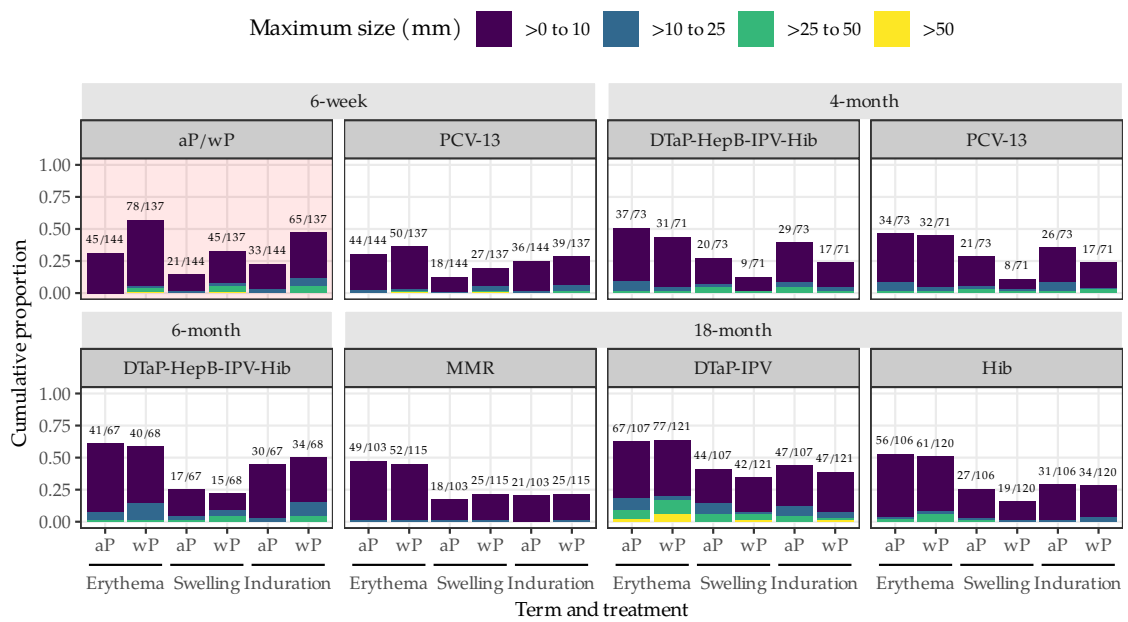


Figure 21: Maximum injection site reaction size in 7 days following each vaccination, by vaccination age, vaccine, and assigned treatment. Stage 1: 6-week, 4-month, 6-month, 18-month. Stage 2: 6-week, 18-month.

6.4.1 Erythema

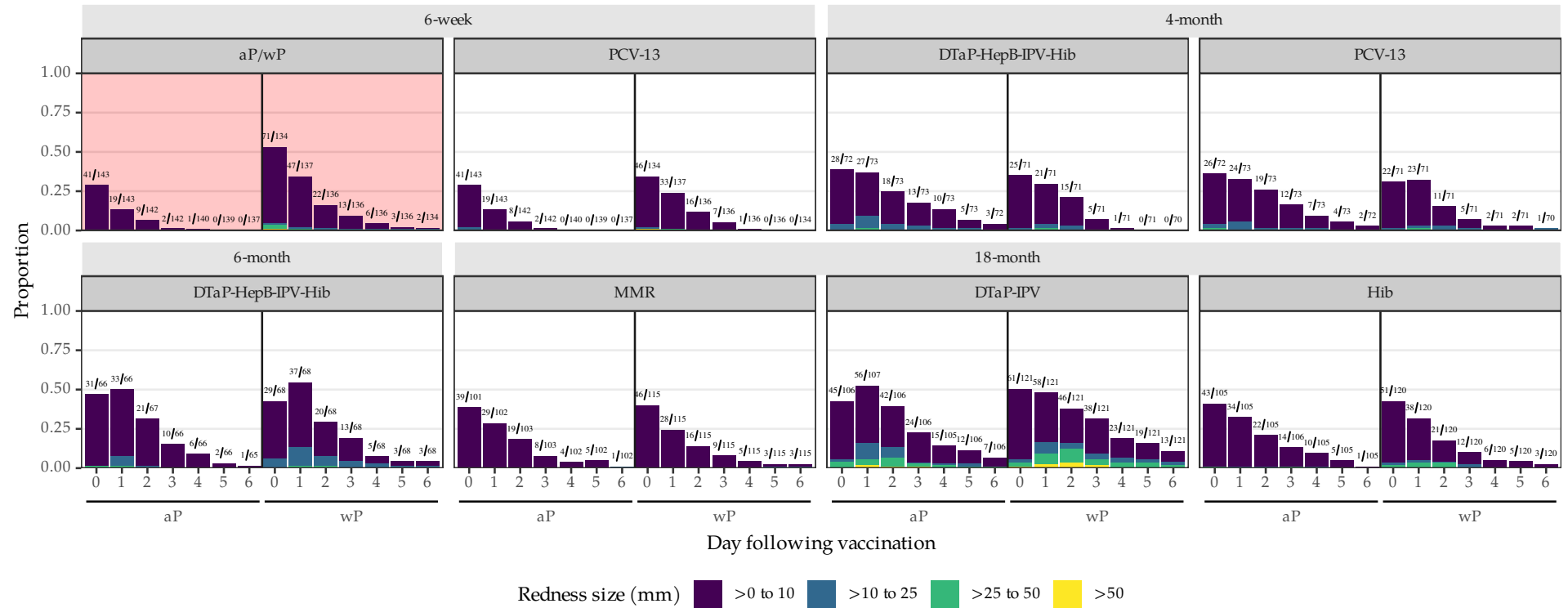


Figure 22: Erythema in the 7 days following the 6-week, 4-month, 6-month, and 18-month vaccinations. *aP*, acellular pertussis vaccine; *wP*, whole-cell pertussis vaccine. Above each bar (n/N), we indicate the number of infants reporting any injection site reaction (numerator; n) versus the total number of infants reporting on injection site reactions (denominator; N).

6.4.2 Swelling

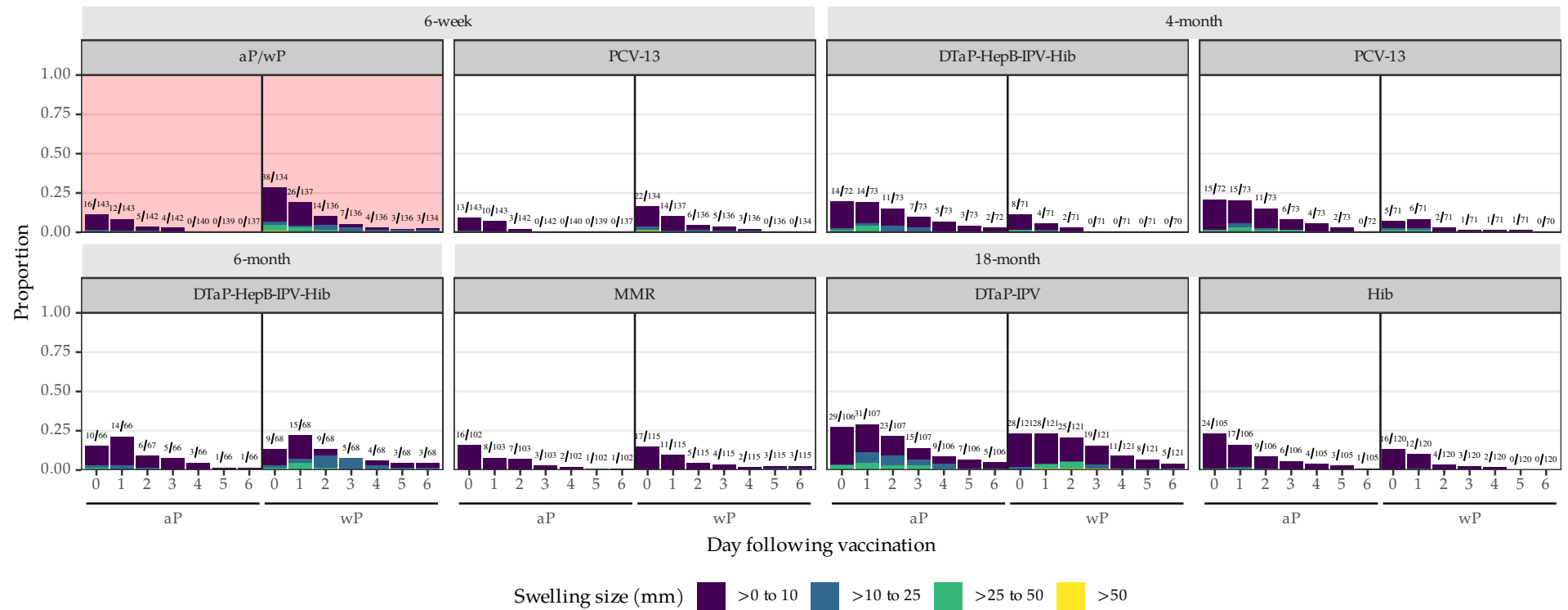


Figure 23: Swelling in the 7 days following the 6-week, 4-month, 6-month, and 18-month vaccinations. *aP*, acellular pertussis vaccine; *wP*, whole-cell pertussis vaccine. Above each bar (n/N), we indicate the number of infants reporting any injection site reaction (numerator; n) versus the total number of infants reporting on injection site reactions (denominator; N).

6.4.3 Induration

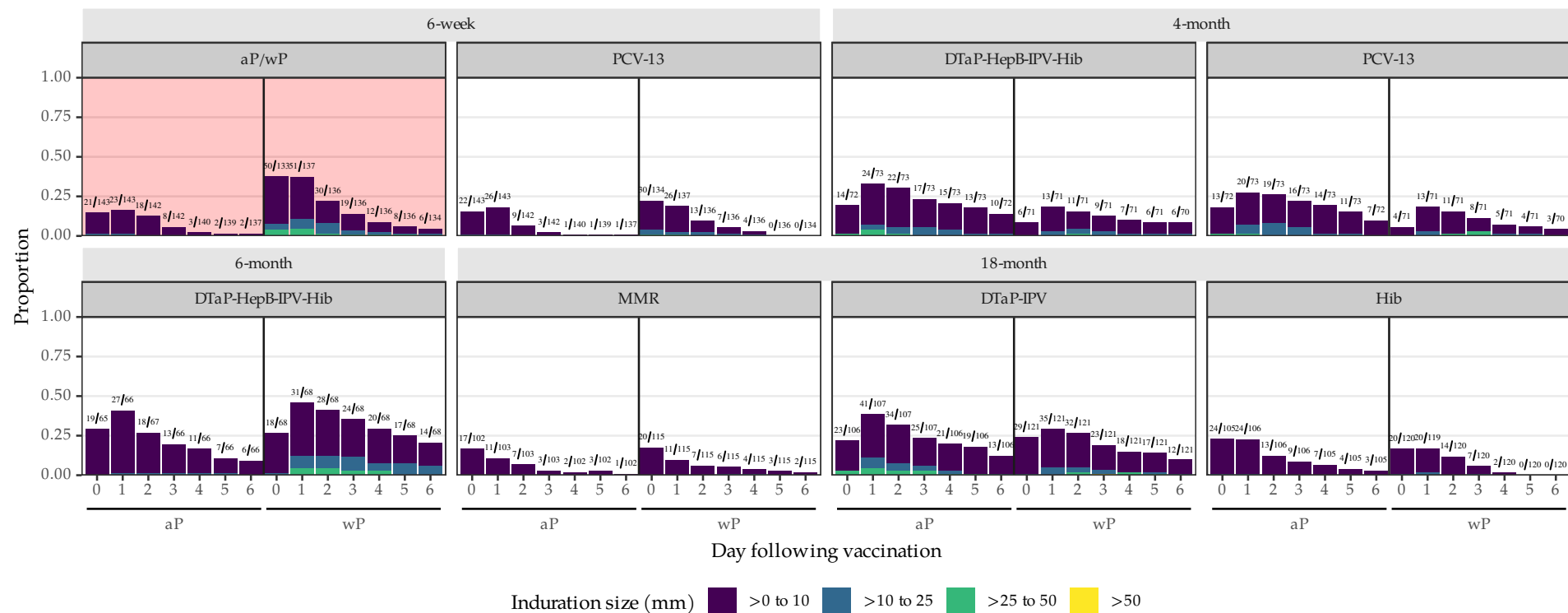


Figure 24: Induration in the 7 days following the 6-week, 4-month, 6-month, and 18-month vaccinations. *aP*, acellular pertussis vaccine; *wP*, whole-cell pertussis vaccine. Above each bar (n/N), we indicate the number of infants reporting any injection site reaction (numerator; n) versus the total number of infants reporting on injection site reactions (denominator; N).

6.4.4 Pain

Table 58: Maximum injection site pain in the 7 days following vaccination, by vaccination age, vaccine, and assigned treatment.

	Pain									
	aP (n = 152)					wP (n = 151)				
	None	Mild	Moderate	Severe	Missing	None	Mild	Moderate	Severe	Missing
6-week										
aP/wP	102 (70)	36 (25)	6 (4)	1 (1)	3	55 (40)	40 (29)	33 (24)	9 (7)	1
PCV-13	102 (70)	38 (26)	3 (2)	2 (1)	3	70 (51)	39 (28)	26 (19)	2 (1)	1
4-month										
DTaP-HepB-IPV-Hib	55 (75)	11 (15)	6 (8)	1 (1)	2	52 (73)	14 (20)	3 (4)	2 (3)	4
PCV-13	52 (71)	12 (16)	8 (11)	1 (1)	2	52 (73)	15 (21)	2 (3)	2 (3)	4
6-month										
DTaP-HepB-IPV-Hib	49 (73)	13 (19)	3 (4)	2 (3)	8	42 (62)	23 (34)	3 (4)	0 (0)	7
18-month										
MMR	67 (65)	32 (31)	4 (4)	0 (0)	14	70 (61)	27 (23)	16 (14)	2 (2)	15
DTaP-IPV	61 (57)	30 (28)	15 (14)	1 (1)	15	60 (50)	37 (31)	23 (19)	1 (1)	15
Hib	64 (60)	33 (31)	9 (8)	0 (0)	15	72 (60)	38 (32)	9 (8)	1 (1)	15

For participants with non-responses for some days, the maximum pain is taken as the greatest pain grade reported on days with responses available. Percentages exclude participants with a missing highest grade (diary card not returned, or non-response for all days).

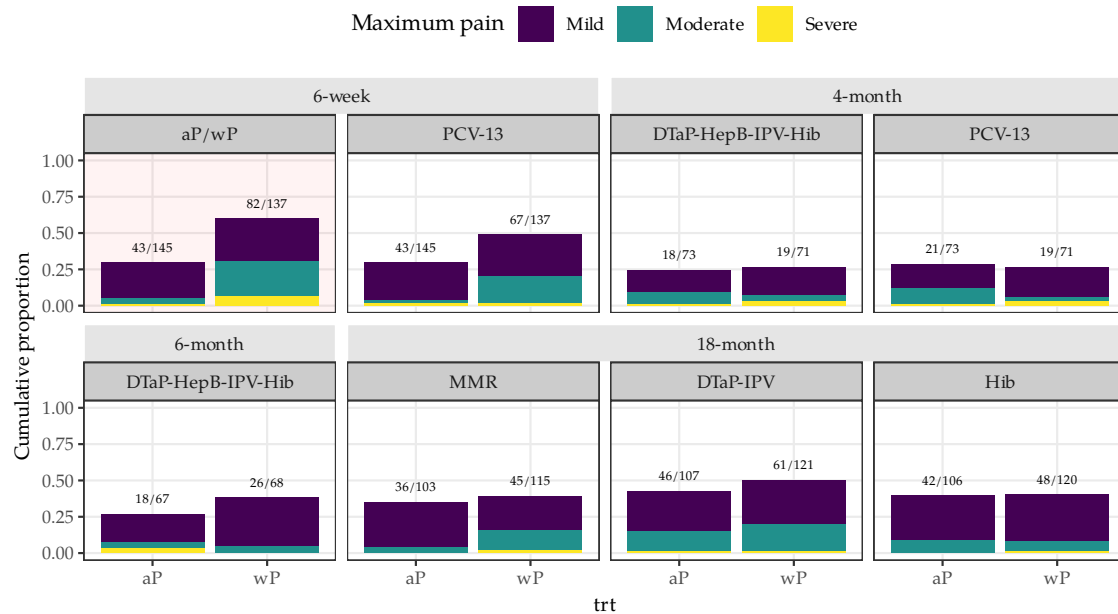


Figure 25: Maximum injection site pain in the 7 days following vaccination, by vaccination age, vaccine, and assigned treatment.

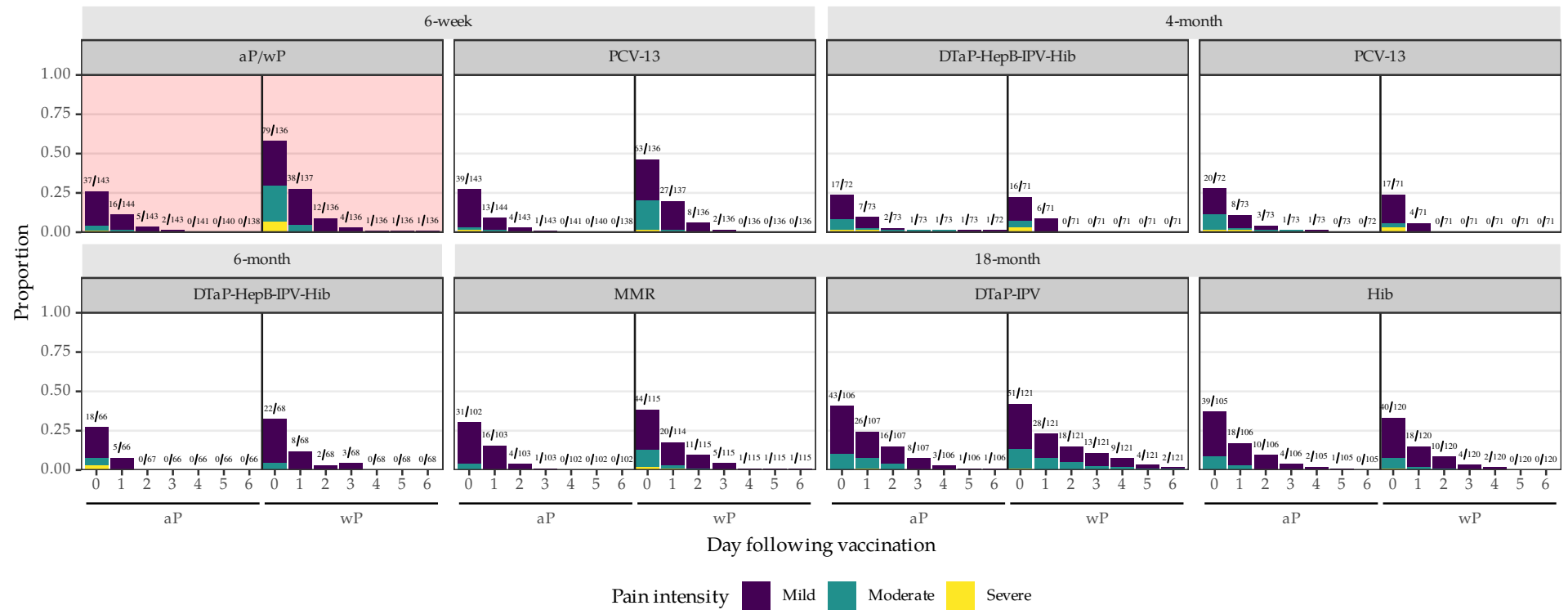


Figure 26: Pain in the 7 days following the 6-week, 4-month, 6-month, and 18-month vaccinations. *aP*, acellular pertussis vaccine; *wP*, whole-cell pertussis vaccine. Above each bar (n/N), we indicate the number of infants reporting any injection site reaction (numerator; n) versus the total number of infants reporting on injection site reactions (denominator; N).

7 Immunological Outcomes

This section reports on data relating to the immunological outcomes.

7.1 Blood collection

For participants enrolled under stage 1, blood collection was to occur at visit 3 (6 to <7-months), visit 5 (21 to 35 days after visit 3, i.e. at 7-months), visit 7 (18 to <19-months), and visit 8 (21 to 35 days after visit 7, i.e. at 19-months). For those enrolled under stage 2, blood collection was at 18 to <19-months and 21 to 35 days after that collection.

The number and percentage of infants with blood collected at each visit is reported in Table 59. Most collections occurred at the scheduled ages (Figure 27) and within the protocolised window (Figure 28) but some collections did occur outside the windows.

Table 59: Number and proportion of infants who had blood collected at the specified ages.

Visit age	aP N = 75	wP N = 75
Stage 1		
6-month	73/75 (97%)	68/75 (91%)
7-month	69/75 (92%)	68/75 (91%)
18-month	66/75 (88%)	59/75 (79%)
19-month	65/75 (87%)	55/75 (73%)
Stage 2		
18-month	55/75 (73%)	66/75 (88%)
19-month	41/75 (55%)	53/75 (71%)

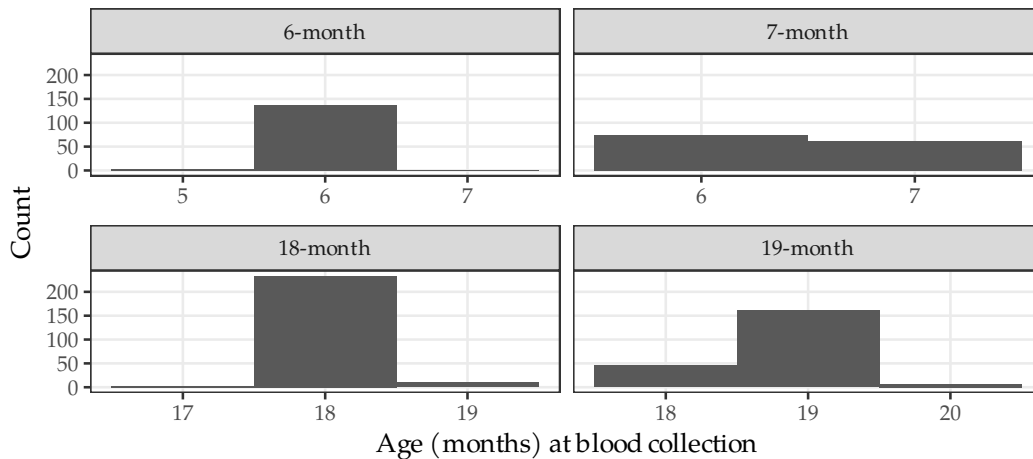


Figure 27: Age (months) at blood collection by scheduled visit based on date of birth and documented visit dates.

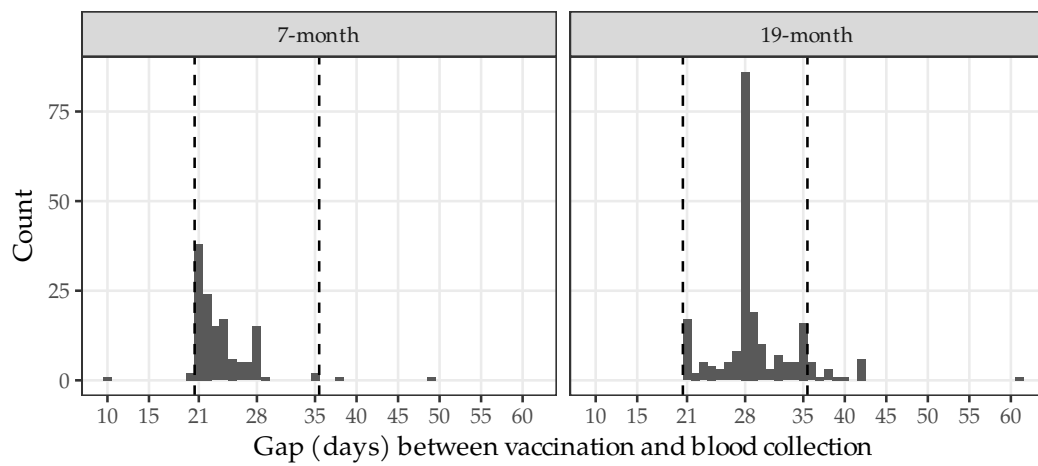


Figure 28: Gap times (days) between blood collection and previous visit for 7-month (21-35 days after 6-month vaccination) and 19-month (21-35 days after 18-month vaccination) based on documented visit dates. The protocolised windows of 21 to 35 days are indicated by dotted lines.

7.2 IgE concentrations

7.2.1 Missingness

Even if blood collection occurred, IgE concentrations may have been missing (e.g. insufficient sample collected). The majority of participants had at least one IgE measurement and a small majority had all planned measurements (Table 60). Missingness varied by the specific allergen being tested (Table 61).

Table 60: Number and proportion of participants with at least one observed IgE response (from any visit or allergen), and with all planned IgE responses observed (all visits and allergens).

Characteristic	aP N = 75	wP N = 75	Overall N = 150
At least one IgE response available	75 (100%)	69 (92%)	144 (96%)
All planned IgE responses available	47 (63%)	35 (47%)	82 (55%)

Table 61: Missingness pattern of IgE concentrations by allergen, “X” indicates a measurement was available and “-” means it was missing.

Count (%)			aP N = 75	wP N = 75
6-month	7-month	19-month		
Tetanus-toxoid				
X	X	X	53 (71%)	41 (55%)
X	X	-	7 (9%)	11 (15%)
X	-	X	5 (7%)	7 (9%)
X	-	-	2 (3%)	2 (3%)
-	X	X	6 (8%)	4 (5%)
-	X	-	0 (0%)	1 (1%)
-	-	X	1 (1%)	3 (4%)
-	-	-	1 (1%)	6 (8%)
Whole egg				
X	X	X	53 (71%)	41 (55%)
X	X	-	7 (9%)	10 (13%)
X	-	X	5 (7%)	7 (9%)
X	-	-	2 (3%)	2 (3%)
-	X	X	6 (8%)	4 (5%)
-	X	-	1 (1%)	2 (3%)
-	-	X	1 (1%)	3 (4%)
-	-	-	0 (0%)	6 (8%)
Egg white				
X	X	X	52 (69%)	40 (53%)
X	X	-	8 (11%)	12 (16%)
X	-	X	5 (7%)	7 (9%)
X	-	-	2 (3%)	2 (3%)
-	X	X	6 (8%)	4 (5%)
-	X	-	1 (1%)	1 (1%)
-	-	X	1 (1%)	3 (4%)
-	-	-	0 (0%)	6 (8%)
Total IgE				
X	X	X	50 (67%)	40 (53%)
X	X	-	10 (13%)	12 (16%)
X	-	X	5 (7%)	7 (9%)
X	-	-	2 (3%)	2 (3%)
-	X	X	6 (8%)	3 (4%)
-	X	-	0 (0%)	2 (3%)
-	-	X	1 (1%)	3 (4%)
-	-	-	1 (1%)	6 (8%)
Peanut				
-	-	X	64 (85%)	55 (73%)
-	-	-	11 (15%)	20 (27%)
Cashew				
-	-	X	64 (85%)	53 (71%)
-	-	-	11 (15%)	22 (29%)
Cow's milk				
-	-	X	62 (83%)	53 (71%)
-	-	-	13 (17%)	22 (29%)
<i>D. pteronyssinus</i>				
-	-	X	60 (80%)	47 (63%)
-	-	-	15 (20%)	28 (37%)
Cat				
-	-	X	60 (80%)	47 (63%)
-	-	-	15 (20%)	28 (37%)
Rye grass				
-	-	X	58 (77%)	47 (63%)
-	-	-	17 (23%)	28 (37%)

7.2.2 Descriptive

Descriptive summaries of IgE concentrations (kU/L) are reported in Table 62 including the number of observations (n), the geometric mean concentration (GMC), the range of concentrations (min-max), and the proportion of concentrations exceeding either at least 0.01 or 0.1 kU/L.

Table 62: Descriptive summary of IgE concentrations (kU/L) by allergen type, visit age, and assigned vaccination schedule.

	aP N = 75					wP N = 75				
	n	GMC [†]	Min-Max	≥0.01	≥0.1	n	GMC [†]	Min-Max	≥0.01	≥0.1
Total IgE										
6-month	67	4.76	0.40–121.43	100%	100%	61	6.11	0.19–141.38	100%	100%
7-month	66	6.36	0.52–241.99	100%	100%	57	7.71	0.52–148.70	100%	100%
19-month	62	17.64	0.92–254.03	100%	100%	53	20.43	1.87–951.55	100%	100%
Tetanus-toxoid										
6-month	67	0.00	0.00–0.22	12%	1%	61	0.00	0.00–1.71	21%	7%
7-month	66	0.00	0.00–0.20	21%	3%	57	0.00	0.00–0.82	21%	7%
19-month	65	0.02	0.00–4.39	65%	26%	55	0.02	0.00–5.05	64%	31%
Egg white										
6-month	67	0.00	0.00–14.91	36%	18%	61	0.00	0.00–3.00	26%	20%
7-month	67	0.01	0.00–39.50	54%	24%	57	0.00	0.00–4.51	42%	19%
19-month	64	0.03	0.00–59.70	75%	22%	54	0.04	0.00–6.67	74%	28%
Whole egg										
6-month	67	0.00	0.00–14.85	40%	18%	60	0.00	0.00–3.51	42%	22%
7-month	67	0.01	0.00–33.70	52%	22%	57	0.00	0.00–5.16	40%	21%
19-month	65	0.03	0.00–64.03	58%	26%	55	0.03	0.00–6.95	71%	27%
Peanut										
19-month	64	0.03	0.01–79.32	89%	12%	55	0.04	0.01–3.51	93%	16%
Cashew										
19-month	64	0.01	0.00–3.99	45%	6%	53	0.01	0.00–2.65	45%	9%
Cow's milk										
19-month	62	0.05	0.01–6.32	98%	21%	53	0.06	0.01–159.54	100%	21%
<i>D. pteronyssinus</i>										
19-month	60	0.02	0.01–0.20	97%	3%	47	0.02	0.01–0.43	98%	2%
Cat										
19-month	60	0.01	0.00–3.02	37%	2%	47	0.01	0.00–1.29	34%	2%
Rye grass										
19-month	58	0.01	0.00–0.12	52%	2%	47	0.01	0.00–1.68	45%	4%

[†]Values of 0.00 kU/L and 0.000 kU/L set to 0.005 and 0.0005 respectively for calculation of GMC.

Table 63: Descriptive summary of geometric mean fold-rise (GMFR), geometric mean ratio (GMR), and geometric mean fold-rise ratio (GMFRR) of wP compared to aP. Number in brackets (n) indicates the number of infants contributing to the calculation.

	aP N = 75		wP N = 75			
	GMC (n) [†]	GMFR (n)	GMC (n) [†]	GMFR (n)	GMR	GMFRR
Total IgE						
6-month	4.76 (67)	— (—)	6.11 (61)	— (—)	1.28	—
7-month	6.36 (66)	1.36 (60)	7.71 (57)	1.23 (52)	1.21	0.9
19-month	17.64 (62)	3.01 (56)	20.43 (53)	2.62 (43)	1.16	0.87
Tetanus-toxoid						
6-month	0.00 (67)	— (—)	0.00 (61)	— (—)	1.61	—
7-month	0.00 (66)	1.41 (60)	0.00 (57)	1.01 (52)	1.10	0.72
19-month	0.02 (65)	15.71 (59)	0.02 (55)	19.43 (45)	1.04	1.24
Egg white						
6-month	0.00 (67)	— (—)	0.00 (61)	— (—)	0.79	—
7-month	0.01 (67)	2.19 (60)	0.00 (57)	2.19 (52)	0.65	1.00
19-month	0.03 (64)	4.85 (58)	0.04 (54)	7.99 (44)	1.16	1.65
Whole egg						
6-month	0.00 (67)	— (—)	0.00 (60)	— (—)	1.09	—
7-month	0.01 (67)	1.76 (60)	0.00 (57)	1.30 (51)	0.68	0.74
19-month	0.03 (65)	4.05 (59)	0.03 (55)	6.30 (45)	1.21	1.56
Peanut						
19-month	0.03 (64)	— (—)	0.04 (55)	— (—)	1.12	—
Cashew						
19-month	0.01 (64)	— (—)	0.01 (53)	— (—)	0.90	—
Cow's milk						
19-month	0.05 (62)	— (—)	0.06 (53)	— (—)	1.08	—
<i>D. pteromyssinus</i>						
19-month	0.02 (60)	— (—)	0.02 (47)	— (—)	0.99	—
Cat						
19-month	0.01 (60)	— (—)	0.01 (47)	— (—)	0.81	—
Rye grass						
19-month	0.01 (58)	— (—)	0.01 (47)	— (—)	0.93	—

¹Values of 0.00 kU/L and 0.000 kU/L set to 0.005 and 0.0005 respectively for calculation of GMC.

7.2.3 Geometric mean concentration and fold-rise ratio

Model estimated geometric mean ratios (GMRs) and geometric mean fold-rise ratios (GMFRRs) are reported in Table 64. For completeness, all IgE types are reported, however, in some cases the GMR and GMFRR may not be meaningful due to the small number of participants with measurable quantities (e.g. tetanus toxoid at 6-months and 7-months) resulting in GMC's close to zero and GMR's which are ratios of two near zero values.

For total IgE, the GMFRR from 6 to 7 months was 0.94 (0.72, 1.22) and the probability of a reduction in the GMFR was 0.69.

Table 64: Geometric mean ratio (GMR) and geometric mean fold rise ratio (GMFRR) for wP compared to aP by age.

	Mean $\pm$ std	Median	95% CrI	Pr(< 1)
Total IgE				
6-month	1.01 $\pm$ 0.22	0.99	(0.64, 1.51)	0.52
7-month	0.94 $\pm$ 0.20	0.92	(0.61, 1.40)	0.64
19-month	0.97 $\pm$ 0.24	0.94	(0.57, 1.51)	0.60
7/6-month (GMFR)	0.94 $\pm$ 0.13	0.94	(0.72, 1.22)	0.69
19/7-month (GMFR)	1.03 $\pm$ 0.20	1.02	(0.69, 1.48)	0.46
Tetanus toxoid				
6-month	25.71 $\pm$ 178.98	6.15	(0.55, 144.22)	0.07
7-month	3.44 $\pm$ 5.85	1.96	(0.31, 15.78)	0.23
19-month	1.42 $\pm$ 0.68	1.27	(0.54, 3.12)	0.29
7/6-month (GMFR)	0.43 $\pm$ 0.42	0.32	(0.04, 1.55)	0.93
19/7-month (GMFR)	0.88 $\pm$ 1.11	0.65	(0.11, 2.96)	0.71
Egg white				
6-month	0.75 $\pm$ 0.90	0.49	(0.07, 2.98)	0.78
7-month	0.81 $\pm$ 0.64	0.65	(0.16, 2.46)	0.74
19-month	1.10 $\pm$ 0.39	1.04	(0.53, 2.02)	0.46
7/6-month (GMFR)	1.65 $\pm$ 1.30	1.32	(0.41, 4.78)	0.32
19/7-month (GMFR)	2.00 $\pm$ 1.48	1.62	(0.47, 5.80)	0.22
Whole egg				
6-month	1.31 $\pm$ 1.16	0.98	(0.21, 4.28)	0.51
7-month	0.95 $\pm$ 0.80	0.73	(0.17, 2.94)	0.67
19-month	1.29 $\pm$ 0.57	1.18	(0.52, 2.69)	0.35
7/6-month (GMFR)	0.83 $\pm$ 0.39	0.75	(0.31, 1.80)	0.74
19/7-month (GMFR)	2.08 $\pm$ 1.68	1.61	(0.43, 6.43)	0.25

All models adjusted for sex, birth order, breastfeeding status, delivery method, family history of atopic disease, and parental income at baseline.

7.2.4 Quantitation

Model estimate difference in probability of concentrations ≥ 0.01 kU/L, and conditional odds ratios, comparing wP with aP, are reported in Table 65. Estimates had a large magnitude of uncertainty, but generally, there was only weak support for any direction of effect on probability of IgE concentration exceeding 0.01 kU/L.

Table 65: The model estimated standardised risk, risk difference, and conditional odds ratio for IgE ≥ 0.01 kU/L comparing wP with aP.

	aP N = 75		wP N = 75		Difference	Conditional odds ratio
	y/n (%)	Median (95% CrI)	y/n (%)	Median (95% CrI)	Median (95% CrI), Pr(<0)	Median (95% CrI)
Tetanus-toxoid						
6-month	8/67 (12%)	0.17 (0.09, 0.28)	13/61 (21%)	0.20 (0.11, 0.31)	0.03 (-0.07, 0.13), 0.29	1.81 (0.25, 14.66)
7-month	14/66 (21%)	0.23 (0.13, 0.35)	12/57 (21%)	0.22 (0.12, 0.34)	-0.02 (-0.14, 0.11), 0.60	0.85 (0.08, 8.95)
19-month	42/65 (65%)	0.62 (0.50, 0.74)	35/55 (64%)	0.64 (0.51, 0.76)	0.02 (-0.12, 0.15), 0.40	1.55 (0.18, 14.94)
Egg white						
6-month	24/67 (36%)	0.37 (0.26, 0.49)	16/61 (26%)	0.27 (0.17, 0.40)	-0.09 (-0.23, 0.04), 0.91	0.48 (0.13, 1.64)
7-month	36/67 (54%)	0.53 (0.41, 0.65)	24/57 (42%)	0.43 (0.30, 0.56)	-0.10 (-0.26, 0.05), 0.91	0.53 (0.13, 1.99)
19-month	48/64 (75%)	0.72 (0.60, 0.82)	40/54 (74%)	0.74 (0.61, 0.84)	0.02 (-0.13, 0.15), 0.41	1.23 (0.28, 5.46)
Whole egg						
6-month	27/67 (40%)	0.42 (0.30, 0.54)	25/60 (42%)	0.41 (0.29, 0.54)	-0.01 (-0.15, 0.13), 0.54	1.06 (0.27, 4.13)
7-month	35/67 (52%)	0.51 (0.39, 0.64)	23/57 (40%)	0.42 (0.29, 0.55)	-0.10 (-0.25, 0.06), 0.89	0.45 (0.09, 2.02)
19-month	38/65 (58%)	0.58 (0.46, 0.69)	39/55 (71%)	0.68 (0.56, 0.79)	0.10 (-0.04, 0.25), 0.07	3.49 (0.73, 17.96)
Peanut						
19-month	57/64 (89%)	0.89 (0.81, 0.95)	51/55 (93%)	0.92 (0.83, 0.97)	0.02 (-0.07, 0.12), 0.31	1.63 (0.45, 6.27)
Cashew						
19-month	29/64 (45%)	0.45 (0.35, 0.57)	24/53 (45%)	0.45 (0.33, 0.58)	0.00 (-0.17, 0.16), 0.51	0.97 (0.44, 2.11)
Cow's milk						
19-month	61/62 (98%)	0.99 (0.94, 1.00)	53/53 (100%)	0.99 (0.96, 1.00)	0.01 (-0.02, 0.05), 0.24	2.12 (0.17, 84.11)
<i>D. pteronyssinus</i>						
19-month	58/60 (97%)	0.97 (0.91, 0.99)	46/47 (98%)	0.98 (0.92, 1.00)	0.01 (-0.04, 0.07), 0.32	1.41 (0.20, 13.30)
Cat						
19-month	22/60 (37%)	0.37 (0.26, 0.48)	16/47 (34%)	0.34 (0.22, 0.46)	-0.03 (-0.19, 0.13), 0.65	0.83 (0.35, 1.98)
Rye grass						
19-month	30/58 (52%)	0.51 (0.39, 0.63)	21/47 (45%)	0.45 (0.33, 0.58)	-0.06 (-0.23, 0.11), 0.75	0.84 (0.36, 1.99)

All models adjust for and standardise over sex, birth order, breastfeeding status, delivery method, family history of atopic disease, and parental income at baseline.

7.2.5 Sensitisation

Model estimate difference in probability of concentrations ≥ 0.1 kU/L, and conditional odds ratios, comparing wP with aP, are reported in Table 66. Estimates had a large magnitude of uncertainty, but generally, there was only weak support for any direction of effect on probability of IgE concentration exceeding 0.1 kU/L.

Table 66: The model estimated standardised risk, risk difference, and conditional odds ratio for IgE ≥ 0.1 kU/L comparing wP with aP.

	aP N = 75		wP N = 75		Difference	Conditional odds ratio
	y/n (%)	Median (95% CrI)	y/n (%)	Median (95% CrI)	Median (95% CrI), Pr(<0)	Median (95% CrI)
Tetanus-toxoid						
6-month	1/67 (1%)	0.04 (0.01, 0.10)	4/61 (7%)	0.05 (0.02, 0.12)	0.01 (-0.03, 0.07), 0.26	1.85 (0.28, 14.61)
7-month	2/66 (3%)	0.04 (0.01, 0.11)	4/57 (7%)	0.06 (0.02, 0.14)	0.01 (-0.05, 0.09), 0.31	1.83 (0.14, 23.11)
19-month	17/65 (26%)	0.26 (0.16, 0.38)	17/55 (31%)	0.32 (0.21, 0.45)	0.06 (-0.08, 0.20), 0.21	2.13 (0.39, 13.60)
Egg white						
6-month	12/67 (18%)	0.19 (0.11, 0.30)	12/61 (20%)	0.20 (0.11, 0.31)	0.00 (-0.10, 0.11), 0.47	1.23 (0.18, 8.61)
7-month	16/67 (24%)	0.25 (0.15, 0.37)	11/57 (19%)	0.21 (0.11, 0.34)	-0.04 (-0.16, 0.09), 0.72	0.60 (0.07, 5.50)
19-month	14/64 (22%)	0.25 (0.15, 0.37)	15/54 (28%)	0.26 (0.15, 0.38)	0.01 (-0.12, 0.14), 0.44	1.41 (0.16, 12.95)
Whole egg						
6-month	12/67 (18%)	0.20 (0.11, 0.32)	13/60 (22%)	0.22 (0.12, 0.33)	0.01 (-0.09, 0.12), 0.39	1.54 (0.21, 12.37)
7-month	15/67 (22%)	0.23 (0.13, 0.35)	12/57 (21%)	0.23 (0.13, 0.36)	0.00 (-0.12, 0.13), 0.48	1.23 (0.13, 11.70)
19-month	17/65 (26%)	0.28 (0.18, 0.40)	15/55 (27%)	0.26 (0.16, 0.38)	-0.02 (-0.15, 0.10), 0.64	0.84 (0.08, 7.92)
Peanut						
19-month	8/64 (12%)	0.13 (0.06, 0.21)	9/55 (16%)	0.16 (0.08, 0.26)	0.03 (-0.08, 0.15), 0.29	1.47 (0.52, 4.31)
Cashew						
19-month	4/64 (6%)	0.07 (0.02, 0.14)	5/53 (9%)	0.10 (0.04, 0.19)	0.03 (-0.06, 0.13), 0.25	1.61 (0.44, 6.05)
Cow's milk						
19-month	13/62 (21%)	0.21 (0.13, 0.31)	11/53 (21%)	0.21 (0.12, 0.33)	0.00 (-0.13, 0.14), 0.48	1.04 (0.41, 2.69)
<i>D. pteromyssinus</i>						
19-month	2/60 (3%)	0.04 (0.01, 0.09)	1/47 (2%)	0.02 (0.00, 0.08)	-0.01 (-0.07, 0.04), 0.68	0.82 (0.09, 6.02)
Cat						
19-month	1/60 (2%)	0.02 (0.00, 0.07)	1/47 (2%)	0.02 (0.00, 0.07)	0.00 (-0.04, 0.05), 0.46	1.06 (0.11, 10.99)
Rye grass						
19-month	1/58 (2%)	0.02 (0.00, 0.07)	2/47 (4%)	0.04 (0.01, 0.10)	0.01 (-0.04, 0.07), 0.28	1.51 (0.20, 13.30)

All models adjust for and standardise over sex, birth order, breastfeeding status, delivery method, family history of atopic disease, and parental income at baseline.

7.2.6 19-month only

In the following, analysis is restricted to only the 19-month IgE measurements.

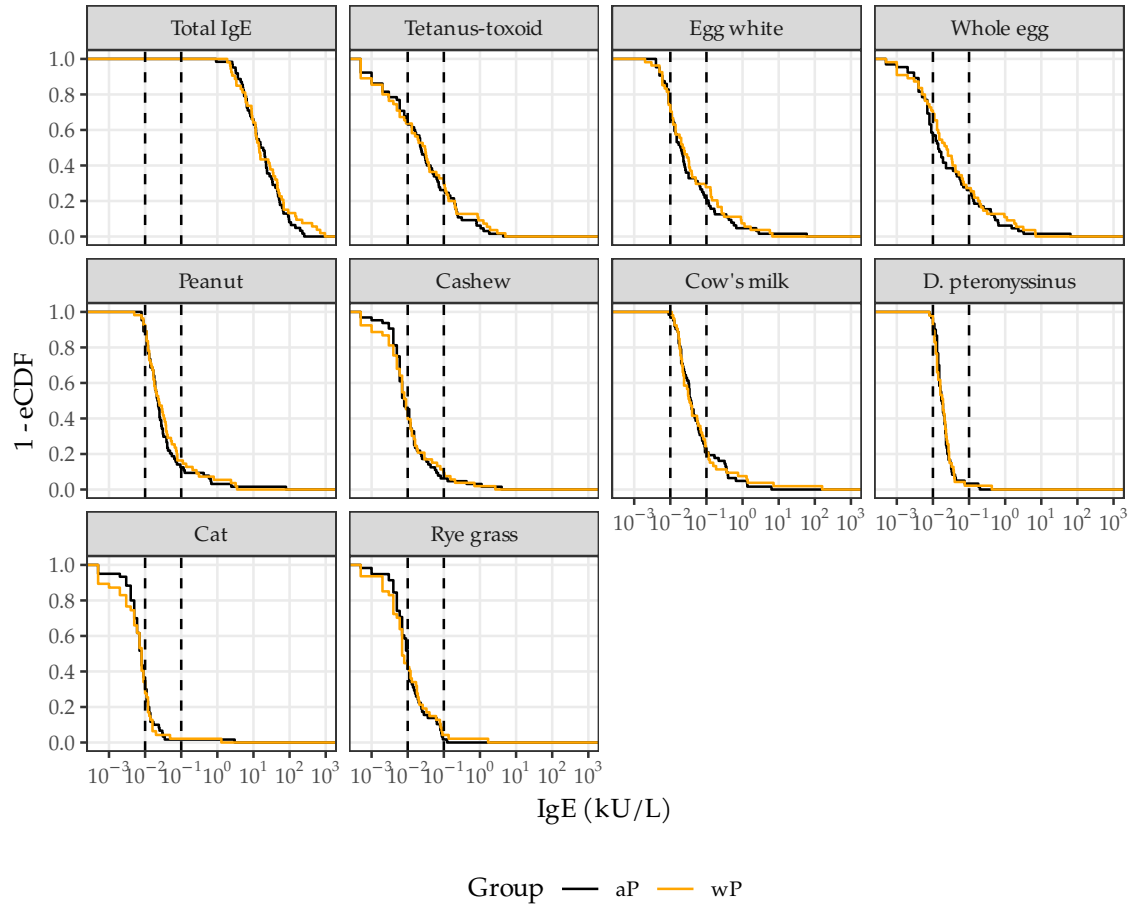


Figure 29: Distribution of IgE (kU/L) concentrations at 19-months of age by IgE type and assigned treatment group.

Table 67: Estimated geometric mean ratio (GMR) and %pt difference in quantitation and sensitisation at 19-months of age, wP versus aP, by allergen.

IgE Type	aP				wP				Median (95% CrI)		
	N	GMC	≥0.01	≥0.1	N	GMC	≥0.01	≥0.1	GMR	Diff. ≥0.01	Diff. ≥0.1
Total IgE	62	17.64	100%	100%	53	20.43	100%	100%	1.15 (0.70, 1.95)	0 (-2, 2)	0 (-2, 2)
Tetanus-toxoid	65	0.02	65%	26%	55	0.02	64%	31%	1.04 (0.42, 2.54)	-1 (-18, 16)	4 (-11, 21)
Egg white	64	0.03	75%	22%	54	0.04	74%	28%	1.16 (0.57, 2.31)	-1 (-17, 14)	6 (-9, 21)
Whole egg	65	0.03	58%	26%	55	0.03	71%	27%	1.22 (0.53, 2.74)	12 (-5, 29)	1 (-15, 17)
Peanut	64	0.03	89%	12%	55	0.04	93%	16%	1.12 (0.64, 1.90)	3 (-7, 13)	3 (-8, 17)
Cashew	64	0.01	45%	6%	53	0.01	45%	9%	0.90 (0.50, 1.61)	0 (-17, 17)	3 (-6, 13)
Cow's milk	62	0.05	98%	21%	53	0.06	100%	21%	1.08 (0.61, 1.95)	1 (-3, 5)	0 (-14, 14)
D. pteronyssinus	60	0.02	97%	3%	47	0.02	98%	2%	1.00 (0.77, 1.27)	1 (-5, 7)	-1 (-7, 5)
Cat	60	0.01	37%	2%	47	0.01	34%	2%	0.82 (0.50, 1.31)	-3 (-19, 15)	0 (-5, 6)
Rye grass	58	0.01	52%	2%	47	0.01	45%	4%	0.93 (0.55, 1.56)	-6 (-25, 12)	2 (-4, 9)

7.3 IgG concentrations

7.3.1 Missingness

Table 68: Number and proportion of participants with at least one observed response, and with all planned responses observed.

Characteristic	aP N = 150	wP N = 150	Overall N = 300
At least one IgG response available	131 (87%)	137 (91%)	268 (89%)
All planned IgG responses available	98 (65%)	102 (68%)	200 (67%)

Table 69: Missingness pattern of IgG concentrations (anti-PT) by study stage.

Count (%)				aP N = 150	wP N = 150
6-month	7-month	18-month	19-month		
Stage 1					
X	X	X	X	58 (39%)	51 (34%)
X	X	X	-	1 (1%)	4 (3%)
X	X	-	-	7 (5%)	10 (7%)
X	-	X	X	4 (3%)	1 (1%)
X	-	X	-	0 (0%)	1 (1%)
X	-	-	-	2 (1%)	1 (1%)
-	X	X	X	2 (1%)	1 (1%)
-	X	-	-	1 (1%)	1 (1%)
-	-	X	X	0 (0%)	1 (1%)
-	-	-	-	0 (0%)	4 (3%)
Stage 2					
-	-	X	X	40 (27%)	52 (35%)
-	-	X	-	15 (10%)	14 (9%)
-	-	-	X	1 (1%)	0 (0%)
-	-	-	-	19 (13%)	9 (6%)

7.3.2 Descriptive summaries

Descriptive summaries of 13vPCV serotype-specific IgG concentrations are reported in Table 72. Summaries of fold-rises and ratios are reported in Table 71. The empirical distribution of IgG concentrations in each treatment group are in the appendix.

Table 70: Descriptive summaries for IgG concentrations (IU/mL) by assigned treatment group.

	aP N = 150						wP N = 150					
	n	GMC ¹	GSD ²	Q1-Q3 ³	Min-Max	S+ ⁴	n	GMC ¹	GSD ²	Q1-Q3 ³	Min-Max	S+ ⁴
HBsAg												
6-month	72	3.81	3.78	1.91–9.00	0.04–33.7	100%	68	4.80	3.66	2.32–12.2	0.06–51.4	100%
7-month	69	7.53	2.90	3.97–14.8	0.24–42.3	100%	66	8.70	3.10	3.79–20.6	0.53–51.8	100%
18-month	120	0.41	4.18	0.16–1.13	0.02–8.03	100%	124	0.45	4.40	0.17–1.43	0.00–14.2	98%
19-month	64	0.46	4.33	0.20–1.27	0.02–7.06	100%	53	0.40	3.86	0.21–1.15	0.02–8.35	100%
Hib-PRP												
6-month	72	0.04	3.71	0.02–0.06	0.00–1.98	12%	68	0.11	4.85	0.03–0.28	0.01–9.03	38%
7-month	69	0.20	5.54	0.07–0.67	0.00–8.28	54%	66	0.64	5.19	0.27–2.46	0.02–28.3	82%
18-month	120	0.09	3.25	0.04–0.19	0.01–2.74	4%	124	0.17	4.20	0.06–0.46	0.01–21.4	9%
19-month	105	11.36	5.60	6.25–35.9	0.06–232	87%	105	19.12	6.46	7.95–56.8	0.09–984	90%
DT												
6-month	72	0.14	2.17	0.08–0.23	0.03–1.01	62%	68	0.15	2.04	0.09–0.23	0.03–2.16	71%
7-month	69	0.61	2.01	0.42–0.98	0.08–3.20	99%	67	0.86	1.98	0.51–1.29	0.24–4.84	100%
18-month	120	0.09	2.67	0.04–0.17	0.01–0.89	42%	125	0.12	2.75	0.06–0.24	0.01–1.35	50%
19-month	105	2.73	2.02	1.78–4.52	0.51–13.9	100%	106	3.70	2.13	2.47–5.73	0.25–20.4	100%
FHA												
6-month	72	45.24	2.11	27.8–77.8	8.92–193	100%	68	15.02	2.16	8.86–24.6	4.47–180	99%
7-month	69	82.39	1.88	52.9–123	19.8–563	100%	67	56.12	2.08	33.9–99.9	9.68–240	100%
18-month	120	9.20	2.58	5.19–16.4	0.38–153	75%	125	8.69	2.68	5.19–14.7	0.65–164	78%
19-month	105	142.62	2.83	79.3–299	2.65–1223	99%	106	162.30	2.27	92.8–257	27.6–1120	100%
FIM2/3												
6-month	72	5.40	7.35	1.21–20.2	0.02–284	51%	68	9.77	4.06	4.49–24.9	0.24–133	72%
7-month	69	3.89	6.62	0.86–12.3	0.09–177	43%	67	7.12	3.99	3.06–19.5	0.12–79.9	58%
18-month	65	0.18	2.44	0.10–0.27	0.02–1.82	0%	59	0.25	3.12	0.11–0.41	0.06–8.33	5%
19-month	64	0.46	3.52	0.21–0.56	0.08–34.5	6%	54	0.71	4.47	0.21–1.88	0.07–49.6	9%
PRN												
6-month	72	15.21	2.71	7.28–32.2	2.15–144	86%	68	14.87	2.38	7.65–28.8	2.33–190	93%
7-month	69	51.19	2.31	30.6–81.7	7.77–378	100%	67	38.36	2.48	20.3–68.2	7.58–610	100%
18-month	120	3.30	4.03	1.23–8.94	0.12–115	38%	125	3.39	4.27	1.26–8.35	0.17–246	42%
19-month	105	239.08	3.16	125–554	10.4–2139	100%	106	226.41	3.04	122–515	5.56–3372	100%
PT												
6-month	72	13.66	2.22	7.22–22.5	2.31–75.1	90%	68	4.94	1.70	3.42–6.90	1.83–18.3	49%
7-month	69	27.69	2.14	17.5–45.8	5.14–145	100%	67	26.38	2.08	15.2–39.9	3.59–152	99%
18-month	120	2.36	2.50	1.24–4.74	0.27–18.4	23%	125	2.92	2.63	1.52–6.03	0.29–25.5	32%
19-month	105	41.03	2.59	22.6–74.3	4.53–340	98%	106	73.04	2.33	46.5–121	11.6–546	100%
TT												
6-month	72	0.70	2.03	0.44–1.19	0.16–3.60	100%	68	0.78	1.98	0.49–1.17	0.19–2.84	100%
7-month	69	1.41	1.80	0.95–2.02	0.44–6.86	100%	67	1.63	1.96	1.14–2.47	0.38–9.84	100%
18-month	120	0.52	3.59	0.23–1.16	0.02–14.0	92%	125	0.62	3.19	0.30–1.27	0.02–6.49	93%
19-month	105	5.29	2.97	3.01–11.9	0.12–47.3	100%	106	7.16	2.52	3.70–14.7	0.59–41.2	100%

¹Geometric mean concentration

²Geometric standard deviation

³Q1 - 25th sample percentile, Q3 - 75th sample percentile

⁴Seroprotection at specified levels

Table 71: Descriptive summary of geometric mean fold-rise (GMFR), percentage with 4-fold rise, geometric mean ratio (GMR), and geometric mean fold-rise ratio (GMFRR) of wP compared to aP. Number in brackets (n) indicates the number of infants contributing to the calculation of the quantity (GMC and GMFR).

	aP N = 150			wP N = 150			GMR	GMFRR
	GMC (n)	GMFR (n)	4-fold rise	GMC (n)	GMFR (n)	4-fold rise		
HBsAg								
6-month	3.81 (72)			4.80 (68)			1.26	
7-month	7.53 (69)	1.98 (66)	12%	8.70 (66)	1.86 (64)	9%	1.16	0.94
18-month	0.41 (120)			0.45 (124)			1.10	
19-month	0.46 (64)	0.90 (64)	0%	0.40 (53)	0.86 (53)	0%	0.87	0.96
Hib-PRP								
6-month	0.04 (72)			0.11 (68)			2.85	
7-month	0.20 (69)	5.20 (66)	52%	0.64 (66)	5.71 (64)	59%	3.21	1.10
18-month	0.09 (120)			0.17 (124)			1.80	
19-month	11.36 (105)	121.11 (104)	97%	19.12 (105)	112.87 (105)	96%	1.68	0.93
DT								
6-month	0.14 (72)			0.15 (68)			1.06	
7-month	0.61 (69)	4.79 (66)	56%	0.86 (67)	5.79 (65)	72%	1.40	1.21
18-month	0.09 (120)			0.12 (125)			1.35	
19-month	2.73 (105)	32.48 (104)	100%	3.70 (106)	31.88 (106)	100%	1.35	0.98
FHA								
6-month	45.24 (72)			15.02 (68)			0.33	
7-month	82.39 (69)	1.90 (66)	8%	56.12 (67)	3.58 (65)	54%	0.68	1.88
18-month	9.20 (120)			8.69 (125)			0.94	
19-month	142.62 (105)	15.58 (104)	89%	162.30 (106)	18.51 (106)	94%	1.14	1.19
FIM2/3								
6-month	5.40 (72)			9.77 (68)			1.81	
7-month	3.89 (69)	0.77 (66)	2%	7.12 (67)	0.73 (65)	0%	1.83	0.94
18-month	0.18 (65)			0.25 (59)			1.42	
19-month	0.46 (64)	2.55 (64)	28%	0.71 (54)	2.90 (54)	31%	1.55	1.14
PRN								
6-month	15.21 (72)			14.87 (68)			0.98	
7-month	51.19 (69)	3.46 (66)	42%	38.36 (67)	2.52 (65)	29%	0.75	0.73
18-month	3.30 (120)			3.39 (125)			1.03	
19-month	239.08 (105)	67.18 (104)	99%	226.41 (106)	65.15 (106)	99%	0.95	0.97
PT								
6-month	13.66 (72)			4.94 (68)			0.36	
7-month	27.69 (69)	2.04 (66)	11%	26.38 (67)	5.31 (65)	74%	0.95	2.60
18-month	2.36 (120)			2.92 (125)			1.24	
19-month	41.03 (105)	17.02 (104)	96%	73.04 (106)	25.49 (106)	98%	1.78	1.50
TT								
6-month	0.70 (72)			0.78 (68)			1.12	
7-month	1.41 (69)	2.04 (66)	9%	1.63 (67)	2.03 (65)	12%	1.16	0.99
18-month	0.52 (120)			0.62 (125)			1.19	
19-month	5.29 (105)	9.42 (104)	84%	7.16 (106)	12.58 (106)	97%	1.35	1.33

Table 72: Descriptive summaries for IgG concentrations ($\mu\text{g/mL}$) for 13-valent pneumococcal vaccine serotypes by assigned treatment group.

	aP N = 150						wP N = 150					
	n	GMC ¹	GSD ²	Q1-Q3 ³	Min-Max	S+ ⁴	n	GMC ¹	GSD ²	Q1-Q3 ³	Min-Max	S+ ⁴
PnPs 1												
6-month	72	0.63	3.02	0.29–1.56	0.04–10.5	68%	68	0.86	2.83	0.35–1.93	0.16–8.85	74%
7-month	69	0.51	2.87	0.24–1.11	0.04–7.26	55%	67	0.68	2.74	0.31–1.50	0.10–4.87	72%
18-month	120	0.53	2.67	0.27–0.89	0.07–7.37	68%	125	0.60	2.49	0.31–1.11	0.09–8.23	71%
19-month	64	0.48	2.89	0.22–0.89	0.06–8.77	61%	54	0.65	2.54	0.31–1.16	0.11–5.14	72%
PnPs 3												
6-month	72	0.28	2.39	0.19–0.47	0.02–1.89	39%	68	0.43	2.22	0.29–0.70	0.05–2.06	63%
7-month	69	0.22	2.32	0.12–0.33	0.02–1.16	23%	67	0.33	2.30	0.20–0.52	0.05–2.58	45%
18-month	120	0.25	2.92	0.13–0.48	0.02–13.0	35%	125	0.26	2.31	0.16–0.41	0.01–4.52	36%
19-month	64	0.22	2.60	0.12–0.40	0.03–3.33	30%	54	0.30	2.19	0.15–0.43	0.09–2.25	41%
PnPs 4												
6-month	72	0.66	3.60	0.29–1.64	0.02–10.5	69%	68	1.03	3.13	0.54–2.09	0.04–7.64	85%
7-month	69	0.54	3.26	0.25–1.21	0.02–9.18	62%	67	0.82	2.68	0.44–1.54	0.04–4.72	81%
18-month	120	0.75	2.43	0.45–1.28	0.09–8.31	85%	125	0.73	2.37	0.44–1.29	0.09–14.0	80%
19-month	64	0.61	2.45	0.40–1.22	0.06–10.6	80%	54	0.80	2.43	0.50–1.39	0.14–13.2	85%
PnPs 5												
6-month	72	0.53	3.69	0.17–1.40	0.02–8.19	72%	68	0.82	2.96	0.44–1.77	0.03–5.37	81%
7-month	69	0.50	3.35	0.22–1.20	0.03–4.89	70%	67	0.72	2.81	0.34–1.50	0.03–4.75	73%
18-month	120	0.61	2.28	0.41–1.00	0.03–5.66	81%	125	0.72	2.19	0.43–1.21	0.09–9.46	83%
19-month	64	0.67	2.23	0.38–1.08	0.14–5.16	77%	54	0.78	2.20	0.50–1.27	0.11–6.22	89%
PnPs 6A												
6-month	72	0.87	4.96	0.31–2.83	0.01–30.5	72%	68	1.60	3.87	0.81–4.22	0.02–14.4	85%
7-month	69	0.75	4.41	0.25–2.21	0.03–18.1	70%	67	1.39	3.38	0.75–3.45	0.03–9.41	88%
18-month	120	2.24	2.37	1.29–3.78	0.34–15.1	99%	125	2.41	2.25	1.46–3.89	0.35–45.5	100%
19-month	64	2.26	2.60	1.11–4.26	0.38–16.4	100%	54	2.57	2.36	1.48–3.40	0.42–32.2	100%
PnPs 6B												
6-month	72	0.06	5.60	0.02–0.21	0.00–5.49	14%	68	0.14	5.71	0.03–0.40	0.01–14.2	29%
7-month	69	0.07	4.95	0.02–0.25	0.00–4.25	17%	67	0.14	5.27	0.04–0.46	0.01–8.96	31%
18-month	120	0.94	3.33	0.41–2.24	0.03–20.4	79%	125	1.28	2.87	0.65–2.40	0.05–100	91%
19-month	64	0.81	4.07	0.29–2.62	0.03–15.5	73%	54	1.44	2.93	0.68–2.80	0.18–62.4	94%
PnPs 7F												
6-month	72	1.31	2.60	0.84–2.60	0.06–10.6	94%	68	1.79	2.38	1.06–3.26	0.07–8.35	99%
7-month	69	1.12	2.61	0.73–2.23	0.04–6.85	91%	67	1.47	2.36	0.84–2.94	0.07–6.32	94%
18-month	120	1.27	2.06	0.81–2.09	0.24–6.53	95%	125	1.45	1.89	1.01–2.20	0.29–18.7	98%
19-month	64	1.34	2.13	0.94–2.08	0.23–8.76	94%	54	1.79	1.86	1.21–2.51	0.47–13.5	100%
PnPs 9V												
6-month	72	0.49	4.39	0.26–1.50	0.02–14.8	68%	68	0.87	3.14	0.45–1.96	0.04–6.62	79%
7-month	69	0.43	3.56	0.20–1.07	0.02–6.72	59%	67	0.73	2.74	0.35–1.52	0.04–5.18	75%
18-month	120	0.68	2.41	0.39–1.04	0.11–10.3	79%	125	0.79	2.25	0.45–1.33	0.08–8.86	84%
19-month	64	0.64	2.72	0.29–1.22	0.11–9.05	69%	54	0.83	2.38	0.40–1.41	0.13–5.66	83%
PnPs 14												
6-month	72	1.06	4.21	0.26–3.71	0.07–15.0	69%	68	2.38	3.23	1.20–4.76	0.05–17.1	96%
7-month	69	1.02	4.20	0.39–2.86	0.04–29.3	75%	67	2.11	2.94	1.23–4.68	0.06–15.8	94%
18-month	120	1.57	2.91	0.78–3.40	0.09–15.9	92%	125	1.66	2.64	1.05–3.38	0.05–14.8	92%
19-month	64	1.33	3.17	0.57–2.73	0.09–17.6	88%	54	1.65	2.49	1.14–3.12	0.04–10.4	96%
PnPs 18C												
6-month	72	0.76	2.68	0.44–1.53	0.06–7.64	78%	68	1.03	2.63	0.60–2.04	0.05–6.08	90%
7-month	69	0.58	2.43	0.31–1.14	0.08–5.22	70%	67	0.82	2.36	0.48–1.49	0.06–4.97	87%
18-month	120	0.45	2.30	0.28–0.78	0.05–4.06	63%	125	0.52	2.14	0.33–0.88	0.07–9.65	73%
19-month	64	0.45	2.32	0.24–0.80	0.09–4.52	58%	54	0.55	2.27	0.32–0.98	0.12–9.61	70%

PnPs 19A

6-month	72	0.36	3.45	0.12-0.90	0.04-5.78	54%	68	0.47	3.71	0.21-1.15	0.01-6.13	66%
7-month	69	0.27	3.08	0.09-0.63	0.04-4.31	46%	67	0.33	3.35	0.15-0.79	0.01-4.59	51%
18-month	120	0.59	4.04	0.22-1.30	0.04-164	62%	125	0.59	3.34	0.29-1.02	0.02-44.0	66%
19-month	64	0.60	4.61	0.21-1.19	0.07-158	58%	54	0.73	3.52	0.38-1.40	0.05-37.7	76%

PnPs 19F

6-month	72	2.20	3.65	1.15-5.64	0.02-27.2	93%	68	3.33	3.42	1.96-7.16	0.01-36.1	97%
7-month	69	1.55	3.53	0.91-3.84	0.01-13.9	91%	67	2.49	3.11	1.50-4.30	0.01-30.5	97%
18-month	120	1.95	3.23	0.95-4.47	0.07-112	95%	125	2.57	2.59	1.49-4.24	0.25-57.7	99%
19-month	64	1.24	3.61	0.53-3.30	0.08-101	86%	54	2.71	3.24	1.49-4.91	0.30-145	98%

PnPs 23F

6-month	72	0.24	4.60	0.11-0.59	0.00-4.91	44%	68	0.49	4.03	0.19-1.66	0.01-6.58	56%
7-month	69	0.19	4.17	0.09-0.49	0.00-2.83	38%	67	0.39	3.64	0.17-1.04	0.01-4.97	51%
18-month	120	0.64	3.05	0.33-1.52	0.04-5.73	72%	125	1.10	3.46	0.51-1.89	0.05-72.3	84%
19-month	64	0.62	3.41	0.29-1.48	0.05-10.8	64%	54	1.37	3.57	0.56-3.32	0.17-44.1	83%

¹Geometric mean concentration

²Geometric standard deviation

³Q1 - 25th sample percentile, Q3 - 75th sample percentile

⁴Seroprotection at specified levels

7.3.3 Geometric mean ratio and fold-rise ratio

Model estimated geometric mean ratios and geometric mean fold-rise ratios are reported for all antigens in Table 73 and geometric mean ratios are displayed in Figure 30. At 6-months, geometric mean concentrations were lower under wP compared to aP for FHA and PT. For other ages and antigens, concentrations were generally similar (GMR of at least 2/3) or higher under wP compared to aP.

Table 73: Adjusted IgG GMR (wP versus aP) posterior summaries by serotype/antigen and age.

	Mean $\pm$ std	Median	95% CrI	Pr(> 1)	Pr(> 2/3)
HBsAg					
6-month	1.58 $\pm$ 0.30	1.56	(1.07, 2.27)	0.99	1.00
7-month	1.43 $\pm$ 0.22	1.41	(1.04, 1.91)	0.99	1.00
18-month	1.12 $\pm$ 0.21	1.11	(0.77, 1.58)	0.71	1.00
19-month	1.08 $\pm$ 0.20	1.06	(0.74, 1.50)	0.62	0.99
7/6-month (GMFR)	0.92 $\pm$ 0.09	0.91	(0.76, 1.10)	0.16	—
19/18-month (GMFR)	0.96 $\pm$ 0.05	0.96	(0.87, 1.06)	0.18	—
Hib-PRP					
6-month	2.76 $\pm$ 0.68	2.68	(1.64, 4.31)	1.00	1.00
7-month	2.65 $\pm$ 0.71	2.56	(1.50, 4.33)	1.00	1.00
18-month	1.76 $\pm$ 0.29	1.74	(1.25, 2.40)	1.00	1.00
19-month	1.66 $\pm$ 0.42	1.61	(0.98, 2.62)	0.97	1.00
7/6-month (GMFR)	0.98 $\pm$ 0.21	0.96	(0.63, 1.46)	0.41	—
19/18-month (GMFR)	0.95 $\pm$ 0.22	0.93	(0.59, 1.45)	0.37	—
DT					
6-month	1.15 $\pm$ 0.14	1.14	(0.89, 1.47)	0.86	1.00
7-month	1.40 $\pm$ 0.15	1.39	(1.11, 1.72)	1.00	1.00
18-month	1.35 $\pm$ 0.17	1.34	(1.04, 1.72)	0.99	1.00
19-month	1.31 $\pm$ 0.13	1.31	(1.06, 1.59)	1.00	1.00
7/6-month (GMFR)	1.23 $\pm$ 0.15	1.22	(0.96, 1.53)	0.95	—
19/18-month (GMFR)	0.98 $\pm$ 0.11	0.98	(0.79, 1.22)	0.42	—
FHA					
6-month	0.37 $\pm$ 0.05	0.37	(0.29, 0.47)	0.00	0.00
7-month	0.74 $\pm$ 0.08	0.73	(0.59, 0.91)	0.00	0.79
18-month	0.89 $\pm$ 0.11	0.89	(0.69, 1.13)	0.16	0.99
19-month	1.07 $\pm$ 0.14	1.06	(0.81, 1.37)	0.68	1.00
7/6-month (GMFR)	2.02 $\pm$ 0.24	2.00	(1.60, 2.53)	1.00	—
19/18-month (GMFR)	1.21 $\pm$ 0.17	1.20	(0.92, 1.59)	0.90	—
FIM2/3					
6-month	2.14 $\pm$ 0.66	2.05	(1.14, 3.70)	0.99	1.00
7-month	2.04 $\pm$ 0.59	1.95	(1.12, 3.42)	0.99	1.00
18-month	1.40 $\pm$ 0.27	1.38	(0.95, 2.03)	0.95	1.00
19-month	1.67 $\pm$ 0.44	1.62	(0.97, 2.70)	0.97	1.00
7/6-month (GMFR)	0.96 $\pm$ 0.07	0.95	(0.83, 1.10)	0.25	—
19/18-month (GMFR)	1.19 $\pm$ 0.25	1.17	(0.78, 1.73)	0.78	—
PRN					
6-month	1.11 $\pm$ 0.19	1.09	(0.79, 1.53)	0.71	1.00
7-month	1.02 $\pm$ 0.14	1.01	(0.78, 1.31)	0.54	1.00
18-month	1.01 $\pm$ 0.18	0.99	(0.70, 1.40)	0.48	0.99
19-month	0.93 $\pm$ 0.14	0.92	(0.68, 1.24)	0.30	0.98
7/6-month (GMFR)	0.94 $\pm$ 0.14	0.92	(0.69, 1.25)	0.30	—
19/18-month (GMFR)	0.94 $\pm$ 0.16	0.93	(0.66, 1.30)	0.32	—
PT					

6-month	0.41 ± 0.05	0.40	(0.32, 0.50)	0.00	0.00
7-month	1.09 ± 0.13	1.09	(0.86, 1.36)	0.75	1.00
18-month	1.20 ± 0.15	1.19	(0.94, 1.51)	0.92	1.00
19-month	1.70 ± 0.21	1.69	(1.33, 2.14)	1.00	1.00
7/6-month (GMFR)	2.71 ± 0.28	2.69	(2.20, 3.28)	1.00	—
19/18-month (GMFR)	1.43 ± 0.19	1.42	(1.09, 1.84)	0.99	—
TT					
6-month	1.21 ± 0.14	1.20	(0.96, 1.52)	0.94	1.00
7-month	1.24 ± 0.12	1.23	(1.02, 1.50)	0.98	1.00
18-month	1.22 ± 0.19	1.21	(0.88, 1.66)	0.89	1.00
19-month	1.50 ± 0.20	1.49	(1.14, 1.93)	1.00	1.00
7/6-month (GMFR)	1.03 ± 0.11	1.03	(0.84, 1.25)	0.60	—
19/18-month (GMFR)	1.24 ± 0.14	1.23	(0.99, 1.53)	0.97	—
PnPs 1					
6-month	1.35 ± 0.22	1.33	(0.97, 1.84)	0.96	1.00
7-month	1.22 ± 0.19	1.20	(0.89, 1.63)	0.89	1.00
18-month	1.11 ± 0.13	1.10	(0.87, 1.38)	0.79	1.00
19-month	1.10 ± 0.13	1.09	(0.86, 1.39)	0.77	1.00
7/6-month (GMFR)	0.90 ± 0.04	0.90	(0.83, 0.99)	0.01	—
19/18-month (GMFR)	1.00 ± 0.04	1.00	(0.92, 1.08)	0.46	—
PnPs 3					
6-month	1.41 ± 0.19	1.40	(1.07, 1.81)	0.99	1.00
7-month	1.35 ± 0.18	1.34	(1.03, 1.72)	0.98	1.00
18-month	1.03 ± 0.12	1.03	(0.82, 1.29)	0.58	1.00
19-month	1.07 ± 0.13	1.07	(0.84, 1.34)	0.70	1.00
7/6-month (GMFR)	0.96 ± 0.04	0.96	(0.88, 1.04)	0.16	—
19/18-month (GMFR)	1.04 ± 0.06	1.04	(0.92, 1.17)	0.73	—
PnPs 4					
6-month	1.56 ± 0.30	1.53	(1.07, 2.25)	0.99	1.00
7-month	1.35 ± 0.24	1.32	(0.95, 1.87)	0.95	1.00
18-month	0.95 ± 0.11	0.95	(0.77, 1.19)	0.32	1.00
19-month	1.03 ± 0.12	1.02	(0.82, 1.30)	0.57	1.00
7/6-month (GMFR)	0.87 ± 0.05	0.86	(0.78, 0.97)	0.00	—
19/18-month (GMFR)	1.08 ± 0.05	1.08	(0.99, 1.18)	0.96	—
PnPs 5					
6-month	1.63 ± 0.30	1.61	(1.11, 2.30)	1.00	1.00
7-month	1.48 ± 0.26	1.45	(1.03, 2.05)	0.98	1.00
18-month	1.11 ± 0.11	1.10	(0.90, 1.36)	0.84	1.00
19-month	1.08 ± 0.11	1.07	(0.87, 1.32)	0.76	1.00
7/6-month (GMFR)	0.91 ± 0.04	0.91	(0.83, 0.99)	0.01	—
19/18-month (GMFR)	0.97 ± 0.04	0.97	(0.90, 1.05)	0.22	—
PnPs 6A					
6-month	1.98 ± 0.47	1.92	(1.23, 3.07)	1.00	1.00
7-month	1.74 ± 0.39	1.70	(1.12, 2.62)	0.99	1.00
18-month	1.07 ± 0.12	1.06	(0.86, 1.32)	0.72	1.00
19-month	1.05 ± 0.12	1.05	(0.84, 1.31)	0.65	1.00
7/6-month (GMFR)	0.89 ± 0.04	0.88	(0.80, 0.98)	0.01	—
19/18-month (GMFR)	0.98 ± 0.04	0.98	(0.90, 1.07)	0.33	—
PnPs 6B					
6-month	2.01 ± 0.54	1.94	(1.15, 3.27)	1.00	1.00
7-month	1.85 ± 0.45	1.79	(1.11, 2.87)	0.99	1.00
18-month	1.34 ± 0.19	1.33	(1.01, 1.75)	0.98	1.00
19-month	1.30 ± 0.19	1.29	(0.97, 1.71)	0.96	1.00
7/6-month (GMFR)	0.93 ± 0.06	0.92	(0.81, 1.05)	0.11	—
19/18-month (GMFR)	0.97 ± 0.04	0.97	(0.89, 1.06)	0.24	—

PnPs 7F					
6-month	1.29 ± 0.19	1.28	(0.96, 1.69)	0.95	1.00
7-month	1.24 ± 0.18	1.23	(0.93, 1.61)	0.92	1.00
18-month	1.15 ± 0.10	1.14	(0.96, 1.35)	0.94	1.00
19-month	1.14 ± 0.10	1.14	(0.95, 1.36)	0.92	1.00
7/6-month (GMFR)	0.96 ± 0.04	0.96	(0.89, 1.03)	0.13	—
19/18-month (GMFR)	1.00 ± 0.04	1.00	(0.92, 1.08)	0.47	—
PnPs 9V					
6-month	1.78 ± 0.38	1.74	(1.16, 2.64)	1.00	1.00
7-month	1.62 ± 0.30	1.59	(1.11, 2.29)	0.99	1.00
18-month	1.16 ± 0.12	1.15	(0.93, 1.42)	0.90	1.00
19-month	1.13 ± 0.13	1.12	(0.90, 1.41)	0.85	1.00
7/6-month (GMFR)	0.92 ± 0.05	0.92	(0.82, 1.02)	0.06	—
19/18-month (GMFR)	0.98 ± 0.04	0.98	(0.90, 1.05)	0.28	—
PnPs 14					
6-month	2.22 ± 0.48	2.17	(1.43, 3.28)	1.00	1.00
7-month	2.12 ± 0.44	2.08	(1.39, 3.11)	1.00	1.00
18-month	1.07 ± 0.15	1.06	(0.81, 1.37)	0.65	1.00
19-month	1.02 ± 0.14	1.01	(0.76, 1.32)	0.52	1.00
7/6-month (GMFR)	0.96 ± 0.06	0.96	(0.84, 1.09)	0.26	—
19/18-month (GMFR)	0.95 ± 0.04	0.95	(0.88, 1.04)	0.13	—
PnPs 18C					
6-month	1.41 ± 0.22	1.39	(1.03, 1.87)	0.98	1.00
7-month	1.34 ± 0.19	1.33	(1.01, 1.73)	0.98	1.00
18-month	1.13 ± 0.12	1.13	(0.92, 1.38)	0.88	1.00
19-month	1.12 ± 0.12	1.12	(0.91, 1.39)	0.84	1.00
7/6-month (GMFR)	0.96 ± 0.04	0.95	(0.88, 1.04)	0.15	—
19/18-month (GMFR)	0.99 ± 0.04	0.99	(0.92, 1.07)	0.41	—
PnPs 19A					
6-month	1.36 ± 0.29	1.33	(0.86, 2.01)	0.90	1.00
7-month	1.24 ± 0.24	1.22	(0.82, 1.76)	0.84	1.00
18-month	0.99 ± 0.16	0.98	(0.71, 1.35)	0.45	0.99
19-month	0.98 ± 0.18	0.97	(0.68, 1.37)	0.43	0.98
7/6-month (GMFR)	0.91 ± 0.04	0.91	(0.84, 0.99)	0.02	—
19/18-month (GMFR)	0.99 ± 0.06	0.99	(0.87, 1.12)	0.41	—
PnPs 19F					
6-month	1.40 ± 0.28	1.37	(0.93, 2.04)	0.94	1.00
7-month	1.35 ± 0.26	1.33	(0.91, 1.92)	0.93	1.00
18-month	1.34 ± 0.18	1.32	(1.02, 1.72)	0.98	1.00
19-month	1.38 ± 0.19	1.37	(1.04, 1.79)	0.99	1.00
7/6-month (GMFR)	0.97 ± 0.04	0.97	(0.90, 1.05)	0.19	—
19/18-month (GMFR)	1.03 ± 0.05	1.03	(0.95, 1.13)	0.77	—
PnPs 23F					
6-month	2.04 ± 0.49	1.98	(1.25, 3.14)	1.00	1.00
7-month	1.82 ± 0.40	1.78	(1.16, 2.74)	0.99	1.00
18-month	1.72 ± 0.26	1.70	(1.27, 2.28)	1.00	1.00
19-month	1.72 ± 0.28	1.70	(1.24, 2.35)	1.00	1.00
7/6-month (GMFR)	0.90 ± 0.04	0.90	(0.82, 0.99)	0.02	—
19/18-month (GMFR)	1.00 ± 0.07	1.00	(0.86, 1.15)	0.50	—

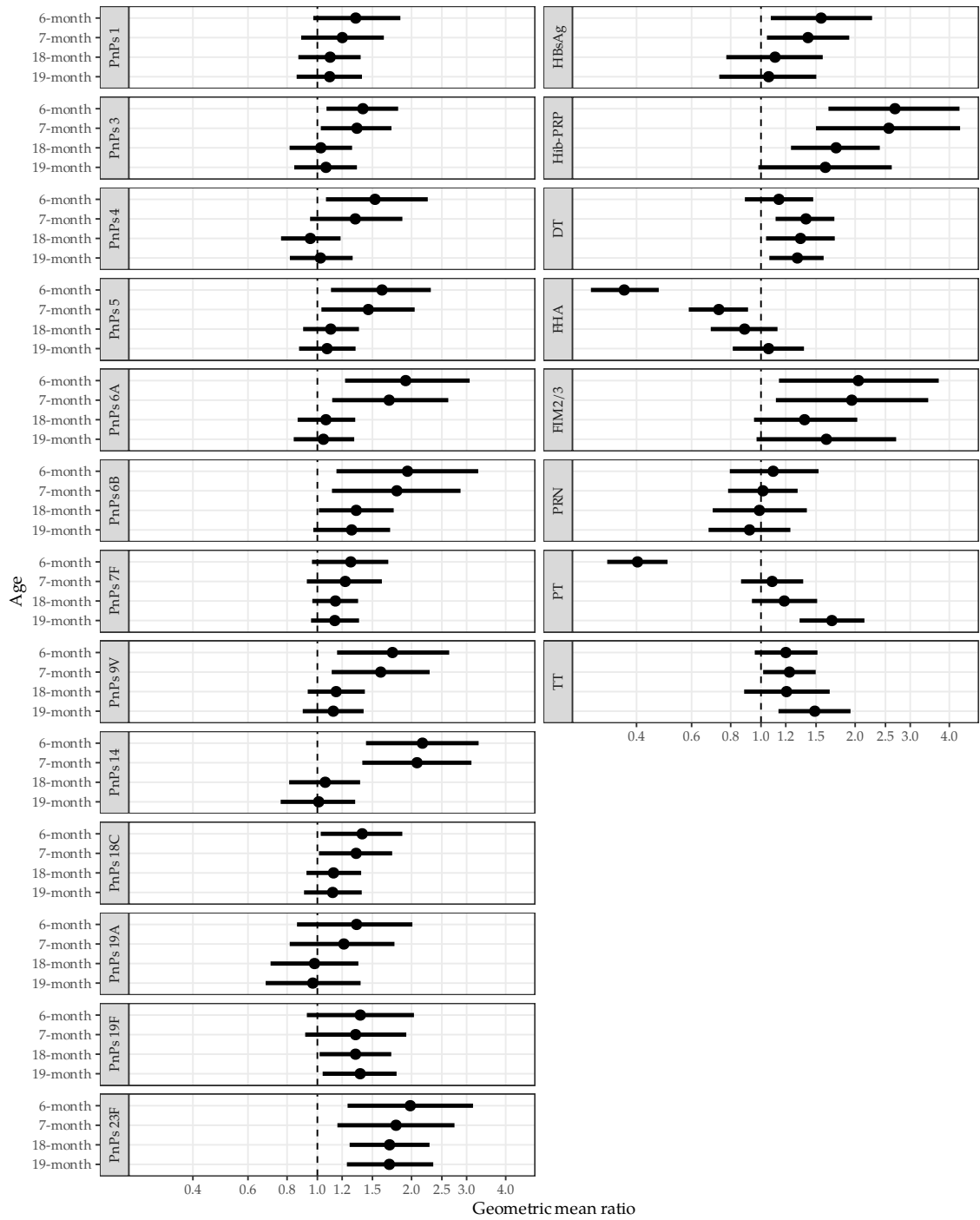


Figure 30: Posterior GMR (median and 95% CrI) for wP versus aP by age and antigen.

7.3.4 Seroresponses and seroprotection to non-pertussis antigens

Model estimated differences in seroresponse and seroprotection (where applicable) are summarised in Table 74 and Figure 31.

Table 74: The model estimated standardised risk, risk difference, and conditional odds ratio for seroresponses and seroprotection (where applicable) comparing recipients of wP as a first dose versus aP as a first dose.

	aP N = 150		wP N = 150		Difference	Conditional odds ratio
	y/n (%)	Median (95% CrI)	y/n (%)	Median (95% CrI)	Median (95% CrI), Pr(<0)	Median (95% CrI)
HBsAg						
6-month	72/72 (100%)	1.00 (1.00, 1.00)	68/68 (100%)	1.00 (0.99, 1.00)	0.00 (-0.01, 0.00), 0.74	0.40 (0.01, 8.93)
7-month	69/69 (100%)	1.00 (0.99, 1.00)	66/66 (100%)	1.00 (0.99, 1.00)	0.00 (-0.01, 0.00), 0.62	0.51 (0.00, 90.33)
18-month	120/120 (100%)	1.00 (0.99, 1.00)	122/124 (98%)	0.99 (0.96, 1.00)	-0.01 (-0.04, 0.00), 0.95	0.07 (0.00, 1.60)
19-month	64/64 (100%)	1.00 (0.99, 1.00)	53/53 (100%)	1.00 (0.98, 1.00)	0.00 (-0.01, 0.00), 0.63	0.51 (0.00, 147.17)
Hib-PRP						
6-month	9/72 (12%)	0.17 (0.09, 0.28)	26/68 (38%)	0.32 (0.21, 0.44)	0.15 (0.05, 0.26), 0.00	6.12 (1.74, 22.66)
7-month	37/69 (54%)	0.55 (0.43, 0.67)	54/66 (82%)	0.78 (0.67, 0.88)	0.23 (0.08, 0.37), 0.00	12.00 (2.53, 60.55)
18-month	5/120 (4%)	0.06 (0.02, 0.12)	11/124 (9%)	0.10 (0.05, 0.17)	0.04 (-0.01, 0.11), 0.07	3.39 (0.73, 20.71)
19-month	91/105 (87%)	0.86 (0.78, 0.92)	95/105 (90%)	0.90 (0.83, 0.95)	0.04 (-0.03, 0.13), 0.14	3.41 (0.72, 17.87)
DT						
6-month	45/72 (62%)	0.62 (0.50, 0.73)	48/68 (71%)	0.70 (0.59, 0.80)	0.08 (-0.05, 0.21), 0.12	1.82 (0.70, 4.83)
7-month	68/69 (99%)	0.98 (0.93, 1.00)	67/67 (100%)	1.00 (0.96, 1.00)	0.01 (-0.01, 0.06), 0.11	6.43 (0.35, 392.10)
18-month	51/120 (42%)	0.43 (0.34, 0.53)	63/125 (50%)	0.51 (0.41, 0.61)	0.08 (-0.04, 0.19), 0.10	1.68 (0.77, 3.84)
19-month	105/105 (100%)	1.00 (0.97, 1.00)	106/106 (100%)	1.00 (0.99, 1.00)	0.00 (-0.01, 0.02), 0.24	3.02 (0.12, 288.14)
FHA						
6-month	72/72 (100%)	0.99 (0.95, 1.00)	67/68 (99%)	0.99 (0.96, 1.00)	0.00 (-0.02, 0.03), 0.43	1.19 (0.12, 11.82)
7-month	69/69 (100%)	1.00 (0.97, 1.00)	67/67 (100%)	1.00 (0.98, 1.00)	0.00 (-0.01, 0.02), 0.38	1.87 (0.04, 230.47)
18-month	90/120 (75%)	0.76 (0.67, 0.84)	98/125 (78%)	0.78 (0.70, 0.86)	0.02 (-0.07, 0.12), 0.32	1.19 (0.46, 3.57)
19-month	104/105 (99%)	0.99 (0.96, 1.00)	106/106 (100%)	1.00 (0.98, 1.00)	0.00 (-0.01, 0.03), 0.20	3.65 (0.17, 296.74)
FIM2/3						
6-month	37/72 (51%)	0.52 (0.39, 0.64)	49/68 (72%)	0.69 (0.56, 0.81)	0.17 (0.03, 0.31), 0.01	8.73 (1.47, 58.17)
7-month	30/69 (43%)	0.46 (0.33, 0.58)	39/67 (58%)	0.58 (0.44, 0.70)	0.12 (-0.02, 0.28), 0.05	4.57 (0.75, 33.90)
18-month	0/65 (0%)	0.02 (0.00, 0.08)	3/59 (5%)	0.05 (0.01, 0.12)	0.03 (-0.02, 0.09), 0.09	6.89 (0.21, 276.38)
19-month	4/64 (6%)	0.10 (0.04, 0.20)	5/54 (9%)	0.10 (0.04, 0.21)	0.00 (-0.10, 0.10), 0.53	0.72 (0.04, 11.58)
PRN						
6-month	62/72 (86%)	0.86 (0.78, 0.93)	63/68 (93%)	0.93 (0.85, 0.97)	0.06 (-0.02, 0.15), 0.07	2.64 (0.78, 10.80)
7-month	69/69 (100%)	1.00 (0.97, 1.00)	67/67 (100%)	1.00 (0.98, 1.00)	0.00 (-0.01, 0.03), 0.19	4.66 (0.18, 408.55)
18-month	45/120 (38%)	0.37 (0.28, 0.47)	53/125 (42%)	0.42 (0.33, 0.53)	0.05 (-0.07, 0.17), 0.23	1.44 (0.67, 3.27)
19-month	105/105 (100%)	1.00 (0.97, 1.00)	106/106 (100%)	1.00 (0.98, 1.00)	0.00 (-0.01, 0.02), 0.17	5.01 (0.18, 604.89)
PT						
6-month	65/72 (90%)	0.86 (0.76, 0.93)	33/68 (49%)	0.55 (0.43, 0.66)	-0.31 (-0.44, -0.16), 1.00	0.07 (0.02, 0.24)
7-month	69/69 (100%)	1.00 (0.97, 1.00)	66/67 (99%)	0.98 (0.95, 1.00)	-0.01 (-0.05, 0.01), 0.87	0.22 (0.01, 5.12)
18-month	28/120 (23%)	0.24 (0.16, 0.33)	40/125 (32%)	0.33 (0.24, 0.42)	0.09 (-0.01, 0.19), 0.05	1.84 (0.81, 5.22)
19-month	103/105 (98%)	0.98 (0.94, 1.00)	106/106 (100%)	1.00 (0.97, 1.00)	0.01 (-0.01, 0.05), 0.15	6.29 (0.22, 5,535.78)
TT						
6-month	72/72 (100%)	1.00 (0.98, 1.00)	68/68 (100%)	1.00 (0.98, 1.00)	0.00 (-0.01, 0.01), 0.34	1.55 (0.16, 24.65)
7-month	69/69 (100%)	1.00 (0.98, 1.00)	67/67 (100%)	1.00 (0.98, 1.00)	0.00 (-0.01, 0.02), 0.41	2.01 (0.03, 376.12)
18-month	110/120 (92%)	0.91 (0.85, 0.96)	116/125 (93%)	0.93 (0.87, 0.97)	0.01 (-0.05, 0.08), 0.31	1.23 (0.24, 6.28)
19-month	105/105 (100%)	1.00 (0.98, 1.00)	106/106 (100%)	1.00 (0.99, 1.00)	0.00 (-0.01, 0.01), 0.34	2.06 (0.03, 408.99)
PnPs 1						
6-month	49/72 (68%)	0.68 (0.56, 0.77)	50/68 (74%)	0.72 (0.60, 0.82)	0.04 (-0.07, 0.15), 0.20	1.55 (0.35, 6.23)
7-month	38/69 (55%)	0.57 (0.45, 0.67)	48/67 (72%)	0.69 (0.58, 0.80)	0.12 (0.00, 0.24), 0.03	4.12 (0.73, 23.65)
18-month	82/120 (68%)	0.69 (0.59, 0.78)	89/125 (71%)	0.71 (0.60, 0.80)	0.02 (-0.08, 0.12), 0.35	1.32 (0.38, 5.13)
19-month	39/64 (61%)	0.65 (0.53, 0.75)	39/54 (72%)	0.69 (0.56, 0.80)	0.04 (-0.10, 0.16), 0.27	1.40 (0.20, 8.35)

PnPs 3						
6-month	28/72 (39%)	0.41 (0.29, 0.54)	43/68 (63%)	0.58 (0.46, 0.70)	0.16 (0.04, 0.30), 0.00	5.34 (1.53, 22.10)
7-month	16/69 (23%)	0.24 (0.15, 0.35)	30/67 (45%)	0.41 (0.30, 0.54)	0.16 (0.04, 0.29), 0.00	7.00 (1.60, 32.28)
18-month	42/120 (35%)	0.37 (0.27, 0.48)	45/125 (36%)	0.38 (0.28, 0.49)	0.01 (-0.10, 0.12), 0.46	1.14 (0.32, 4.15)
19-month	19/64 (30%)	0.32 (0.21, 0.43)	22/54 (41%)	0.39 (0.27, 0.51)	0.07 (-0.07, 0.19), 0.17	2.24 (0.48, 9.14)
PnPs 4						
6-month	50/72 (69%)	0.72 (0.61, 0.82)	58/68 (85%)	0.82 (0.72, 0.90)	0.09 (-0.01, 0.20), 0.04	3.31 (0.89, 13.85)
7-month	43/69 (62%)	0.65 (0.54, 0.76)	54/67 (81%)	0.78 (0.67, 0.87)	0.12 (0.00, 0.24), 0.02	4.84 (1.13, 25.80)
18-month	102/120 (85%)	0.84 (0.76, 0.91)	100/125 (80%)	0.80 (0.71, 0.87)	-0.04 (-0.12, 0.06), 0.83	0.49 (0.12, 2.10)
19-month	51/64 (80%)	0.81 (0.71, 0.89)	46/54 (85%)	0.84 (0.74, 0.92)	0.03 (-0.09, 0.13), 0.33	1.23 (0.19, 7.20)
PnPs 5						
6-month	52/72 (72%)	0.72 (0.63, 0.82)	55/68 (81%)	0.77 (0.67, 0.86)	0.05 (-0.05, 0.15), 0.18	2.01 (0.46, 9.29)
7-month	48/69 (70%)	0.69 (0.58, 0.80)	49/67 (73%)	0.71 (0.60, 0.81)	0.01 (-0.11, 0.15), 0.40	1.20 (0.23, 6.97)
18-month	97/120 (81%)	0.80 (0.72, 0.88)	104/125 (83%)	0.83 (0.74, 0.89)	0.02 (-0.07, 0.10), 0.32	1.35 (0.29, 5.93)
19-month	49/64 (77%)	0.77 (0.66, 0.86)	48/54 (89%)	0.83 (0.73, 0.91)	0.06 (-0.04, 0.18), 0.14	2.59 (0.41, 16.91)
PnPs 6A						
6-month	52/72 (72%)	0.76 (0.64, 0.86)	58/68 (85%)	0.86 (0.77, 0.93)	0.09 (0.00, 0.21), 0.02	7.01 (0.78, 125.73)
7-month	48/69 (70%)	0.73 (0.61, 0.84)	59/67 (88%)	0.86 (0.76, 0.93)	0.13 (0.02, 0.25), 0.01	17.70 (1.22, 361.97)
18-month	119/120 (99%)	0.99 (0.95, 1.00)	125/125 (100%)	1.00 (0.98, 1.00)	0.01 (0.00, 0.04), 0.05	14.83 (0.31, 2,068.32)
19-month	64/64 (100%)	0.99 (0.95, 1.00)	54/54 (100%)	1.00 (0.97, 1.00)	0.00 (-0.01, 0.04), 0.14	10.83 (0.11, 1,520.84)
PnPs 6B						
6-month	10/72 (14%)	0.18 (0.11, 0.29)	20/68 (29%)	0.27 (0.18, 0.39)	0.08 (-0.01, 0.19), 0.04	6.55 (0.76, 65.88)
7-month	12/69 (17%)	0.21 (0.12, 0.32)	21/67 (31%)	0.29 (0.19, 0.41)	0.07 (-0.03, 0.20), 0.09	6.55 (0.52, 90.10)
18-month	95/120 (79%)	0.78 (0.69, 0.86)	114/125 (91%)	0.90 (0.82, 0.95)	0.11 (0.04, 0.20), 0.01	23.43 (2.99, 316.41)
19-month	47/64 (73%)	0.77 (0.66, 0.86)	51/54 (94%)	0.90 (0.82, 0.97)	0.13 (0.03, 0.24), 0.01	37.32 (1.81, 1,289.19)
PnPs 7F						
6-month	68/72 (94%)	0.96 (0.90, 0.99)	67/68 (99%)	0.97 (0.93, 1.00)	0.02 (-0.02, 0.07), 0.15	3.67 (0.36, 77.91)
7-month	63/69 (91%)	0.92 (0.84, 0.97)	63/67 (94%)	0.93 (0.85, 0.98)	0.01 (-0.07, 0.09), 0.42	1.15 (0.07, 26.12)
18-month	114/120 (95%)	0.96 (0.91, 0.98)	123/125 (98%)	0.98 (0.94, 1.00)	0.02 (-0.02, 0.06), 0.11	7.50 (0.42, 193.37)
19-month	60/64 (94%)	0.96 (0.90, 0.99)	54/54 (100%)	0.98 (0.93, 1.00)	0.02 (-0.02, 0.07), 0.12	9.46 (0.29, 2,602.07)
PnPs 9V						
6-month	49/72 (68%)	0.70 (0.59, 0.80)	54/68 (79%)	0.78 (0.67, 0.88)	0.08 (-0.03, 0.19), 0.08	2.54 (0.55, 12.39)
7-month	41/69 (59%)	0.63 (0.51, 0.74)	50/67 (75%)	0.73 (0.63, 0.82)	0.10 (-0.02, 0.24), 0.06	3.34 (0.72, 17.01)
18-month	95/120 (79%)	0.79 (0.70, 0.86)	105/125 (84%)	0.85 (0.77, 0.91)	0.06 (-0.02, 0.15), 0.07	2.19 (0.60, 10.28)
19-month	44/64 (69%)	0.72 (0.62, 0.82)	45/54 (83%)	0.84 (0.73, 0.91)	0.11 (0.01, 0.23), 0.02	5.40 (0.95, 34.99)
PnPs 14						
6-month	50/72 (69%)	0.75 (0.64, 0.84)	65/68 (96%)	0.92 (0.84, 0.96)	0.17 (0.07, 0.27), 0.00	10.97 (2.52, 60.20)
7-month	52/69 (75%)	0.76 (0.66, 0.86)	63/67 (94%)	0.92 (0.83, 0.97)	0.15 (0.04, 0.25), 0.00	11.01 (1.69, 68.61)
18-month	111/120 (92%)	0.91 (0.84, 0.96)	115/125 (92%)	0.92 (0.86, 0.97)	0.01 (-0.06, 0.07), 0.33	1.41 (0.30, 5.82)
19-month	56/64 (88%)	0.88 (0.80, 0.94)	52/54 (96%)	0.95 (0.88, 0.99)	0.07 (-0.01, 0.15), 0.04	6.91 (0.78, 61.86)
PnPs 18C						
6-month	56/72 (78%)	0.76 (0.65, 0.85)	61/68 (90%)	0.87 (0.78, 0.94)	0.11 (0.02, 0.21), 0.01	6.08 (1.21, 42.10)
7-month	48/69 (70%)	0.69 (0.57, 0.79)	58/67 (87%)	0.85 (0.75, 0.93)	0.16 (0.04, 0.27), 0.01	11.49 (1.53, 94.64)
18-month	76/120 (63%)	0.64 (0.52, 0.74)	91/125 (73%)	0.72 (0.62, 0.82)	0.09 (-0.03, 0.18), 0.06	2.94 (0.63, 12.36)
19-month	37/64 (58%)	0.59 (0.47, 0.71)	38/54 (70%)	0.66 (0.52, 0.78)	0.06 (-0.07, 0.20), 0.19	1.77 (0.34, 11.14)
PnPs 19A						
6-month	39/72 (54%)	0.55 (0.44, 0.67)	45/68 (66%)	0.62 (0.50, 0.73)	0.07 (-0.05, 0.18), 0.12	2.18 (0.47, 8.27)
7-month	32/69 (46%)	0.46 (0.34, 0.58)	34/67 (51%)	0.46 (0.35, 0.57)	0.00 (-0.12, 0.12), 0.48	0.96 (0.19, 4.37)
18-month	75/120 (62%)	0.61 (0.51, 0.72)	83/125 (66%)	0.66 (0.55, 0.76)	0.05 (-0.05, 0.14), 0.18	1.58 (0.44, 6.06)
19-month	37/64 (58%)	0.57 (0.45, 0.69)	41/54 (76%)	0.68 (0.55, 0.80)	0.11 (-0.01, 0.23), 0.04	3.51 (0.67, 18.23)
PnPs 19F						
6-month	67/72 (93%)	0.94 (0.88, 0.98)	66/68 (97%)	0.97 (0.93, 0.99)	0.03 (-0.01, 0.09), 0.07	3.82 (0.55, 40.66)
7-month	63/69 (91%)	0.93 (0.86, 0.97)	65/67 (97%)	0.96 (0.90, 0.99)	0.03 (-0.03, 0.10), 0.18	3.57 (0.35, 62.14)
18-month	114/120 (95%)	0.95 (0.90, 0.99)	124/125 (99%)	0.99 (0.96, 1.00)	0.03 (0.00, 0.08), 0.03	13.64 (1.17, 264.02)
19-month	55/64 (86%)	0.90 (0.82, 0.96)	53/54 (98%)	0.97 (0.92, 1.00)	0.07 (0.00, 0.15), 0.02	20.07 (1.21, 520.83)
PnPs 23F						
6-month	32/72 (44%)	0.47 (0.37, 0.59)	38/68 (56%)	0.55 (0.44, 0.67)	0.08 (-0.05, 0.20), 0.10	2.24 (0.46, 11.03)

7-month	26/69 (38%)	0.41 (0.30, 0.54)	34/67 (51%)	0.47 (0.36, 0.59)	0.05 (-0.08, 0.19), 0.21	1.55 (0.28, 7.59)
18-month	87/120 (72%)	0.72 (0.62, 0.80)	105/125 (84%)	0.84 (0.76, 0.91)	0.12 (0.03, 0.22), 0.00	7.89 (1.64, 46.72)
19-month	41/64 (64%)	0.66 (0.55, 0.76)	45/54 (83%)	0.80 (0.70, 0.89)	0.14 (0.03, 0.27), 0.01	8.05 (1.35, 60.84)

All models adjust for and standardise over sex, birth order, breastfeeding status, delivery method, family history of atopic disease, and parental income at baseline.

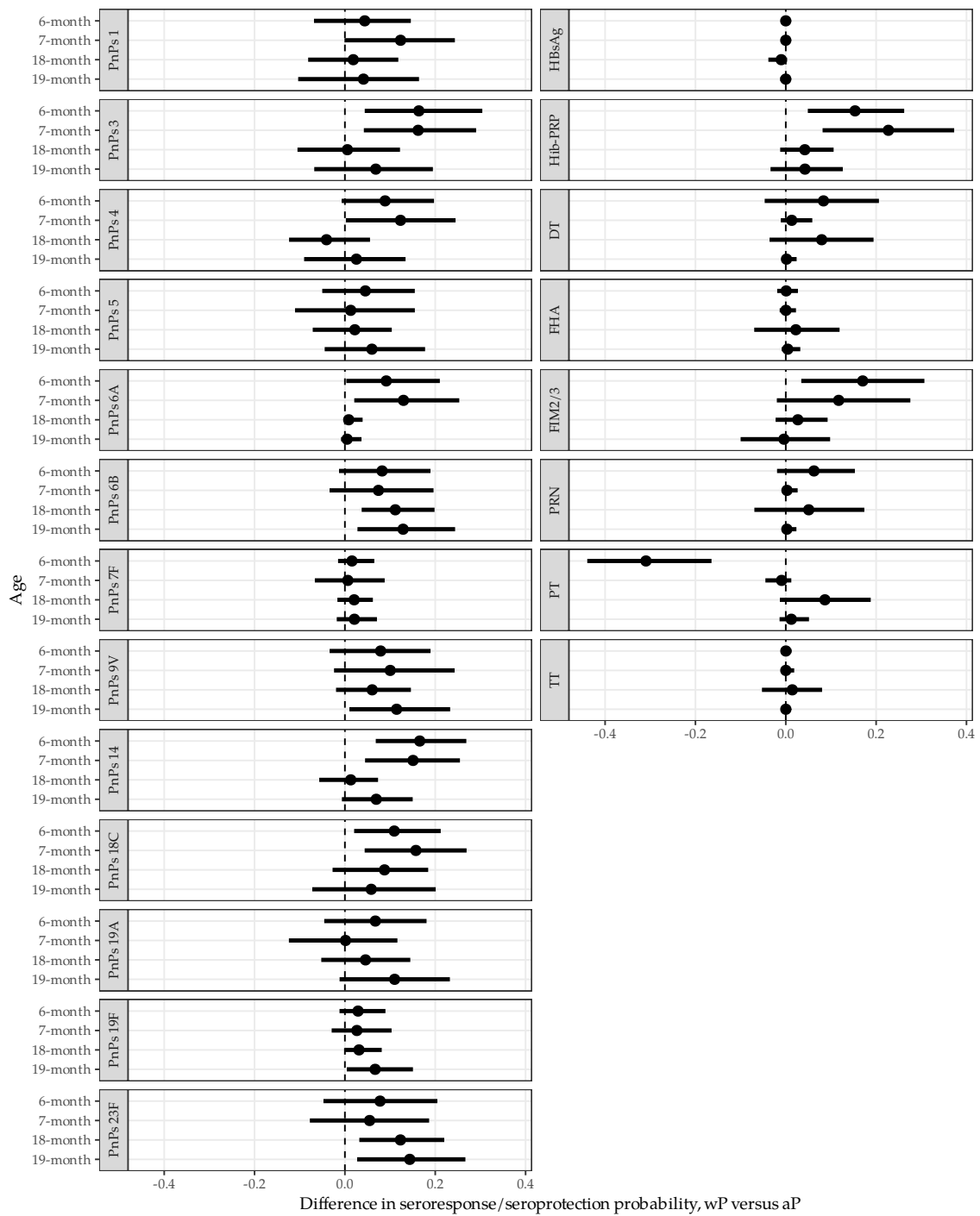


Figure 31: Posterior difference in seroresponse and seroprotection (where applicable) probability (median and 95% CrI) for wP vs. aP by age and antigen.

7.3.5 Four-fold rise

Model estimated differences in the probability of a 4-fold rise from 18-month vaccination to 21–35 days after vaccination are summarised in Table 75.

Table 75: Model estimated differences (wP compared to aP) in probability of 4-fold rise in IgG concentration from 18-months (pre-vaccination) to 21–35 days after vaccination by antigen.

	aP		wP		Difference	Conditional odds ratio
	y/n (%)	Median (95% CrI)	y/n (%)	Median (95% CrI)	Median (95% CrI), Pr(<0)	Median (95% CrI)
Hib-PRP	101/104 (97%)	0.97 (0.93, 0.99)	101/105 (96%)	0.96 (0.92, 0.99)	-0.01 (-0.05, 0.04), 0.61	0.75 (0.17, 3.38)
DT	104/104 (100%)	1.00 (0.98, 1.00)	106/106 (100%)	1.00 (0.99, 1.00)	0.00 (-0.01, 0.01), 0.51	0.92 (0.01, 43.48)
FHA	93/104 (89%)	0.89 (0.82, 0.94)	100/106 (94%)	0.94 (0.89, 0.97)	0.05 (-0.02, 0.12), 0.08	2.06 (0.77, 5.65)
FIM2/3	18/64 (28%)	0.28 (0.19, 0.38)	17/54 (31%)	0.31 (0.20, 0.43)	0.03 (-0.12, 0.18), 0.34	1.18 (0.50, 2.79)
PRN	103/104 (99%)	0.99 (0.96, 1.00)	105/106 (99%)	0.99 (0.96, 1.00)	0.00 (-0.03, 0.03), 0.51	0.92 (0.08, 8.37)
PT	100/104 (96%)	0.96 (0.91, 0.98)	104/106 (98%)	0.98 (0.95, 1.00)	0.02 (-0.02, 0.07), 0.12	1.96 (0.40, 10.96)
TT	87/104 (84%)	0.84 (0.77, 0.90)	103/106 (97%)	0.97 (0.92, 0.99)	0.13 (0.05, 0.20), 0.00	6.04 (1.91, 24.66)

All models adjust for and standardise over sex, birth order, breastfeeding status, delivery method, family history of atopic disease, and parental income at baseline.

8 Additional Information

8.1 Primary outcome listing

Table 76: Listing of infants with OFC outcome by 18 months of age and corresponding SPT positive allergens.

Record ID	Treatment	OFC positive foods ¹	Foods with positive SPT
4629-32	wP	egg	whole egg
4629-228	aP	peanut	peanut
4630-30	aP	egg	whole egg
4630-45	wP	peanut	peanut
5785-39	aP	peanut, tahini	peanut
5785-78	aP	peanut, cashew, cooked egg	peanut, cashew, whole egg
5785-98	aP	whole egg cooked	whole egg
5785-112	aP	egg	whole egg
5785-122	aP	peanut	peanut
5785-123	aP	egg	whole egg
5785-133	aP	peanut, cows milk	peanut, cow's milk
5785-169	aP	cows milk	cow's milk
5785-192	wP	cow milk	cow's milk
5785-242	wP	peanut, cows milk	peanut, cow's milk
5785-266	aP	well cooked scrambled egg	whole egg
5785-295	aP	cows milk	cow's milk
5785-319	wP	cashew	cashew
5785-320	aP	egg	whole egg
5785-333	wP	peanut, cows milk, egg	peanut, cow's milk, whole egg
5785-367	wP	peanut, cooked egg	peanut, whole egg
8644-14	wP	hazelnut	hazelnut
8644-24	wP	cashew	cashew

¹These were from free-text fields on the outcome report. Responses are reported as documented.

Table 77: Listing of infants with food anaphylaxis outcome by 18 months of age and corresponding SPT positive allergens.

Record ID	Treatment	Anaphylactic triggers ¹	Foods with positive SPT
136	wP	cows milk	cow's milk
4629-92	aP	cows milk	cow's milk
5785-133	aP	scrambled egg	whole egg

¹These were from free-text fields on the outcome report. Responses are reported as documented.

Table 78: Listing of infants with highly probable IgE-mediated food allergy outcome by 18 months of age and corresponding SPT positive allergens.

Record ID	Treatment	Suspected food allergen ¹	Foods with positive SPT ²
17	wP	peanut	peanut
21	aP	egg	whole egg
56	wP	egg	whole egg
73	wP	peanut	peanut
115	wP	cows milk, peanut, egg	cow's milk, peanut, whole egg
117	aP	cows milk	cow's milk
4629-5	aP	cashews, almonds and coconut	cashew, almond, coconut
4629-60	wP	egg	whole egg
4629-87	aP	egg	whole egg
4629-92	aP	pecan, egg allergy	pecan, whole egg
4629-181	wP	egg	whole egg
4629-189	wP	egg	whole egg
4629-191	wP	peanut	peanut
4629-214	wP	egg	whole egg
4629-226	wP	egg, peanut	peanut, egg yolk, egg white
4629-253	wP	egg powder	whole egg
4630-7	aP	soy, dairy, egg, beef	cow's milk, egg yolk, egg white, soy, beef
4630-17	aP	egg	egg yolk, egg white
4630-57	wP	egg	whole egg
5785-1	wP	dairy	cow's milk
5785-22	aP	egg	whole egg
5785-89	wP	well cooked scrambled egg	whole egg
5785-93	wP	egg	whole egg
5785-94	aP	cooked egg	whole egg
5785-98	aP	cow milk formula	cow's milk
5785-124	wP	cows milk	cow's milk
5785-145	aP	egg	whole egg
5785-200	wP	cows milk	cow's milk
5785-203	aP	egg, milk	whole egg, cow's milk
5785-242	wP	baked egg	whole egg
5785-301	wP	cashew, peanut	cashew, peanut
5785-362	wP	egg	whole egg
5785-367	wP	peanut	peanut
5785-372	wP	peanut	peanut
5785-380	wP	egg	whole egg
5785-384	aP	egg	whole egg
8644-14	wP	whole egg, peanut	whole egg, peanut
8644-34	aP	sesame, peanut	sesame, peanut

¹These were from free-text fields on the outcome report. Responses are reported as documented.

²In some instances, separate SPTs for egg yolk and egg white were reported rather than one SPT for whole egg.

8.2 Secondary outcome listings

Table 79: Listing of infants with eczema diagnosis and positive SPT to any allergen.

Record ID	Treatment	Age at first reaction (months)
17	wP	13
56	wP	8
68	aP	5
73	wP	4
115	wP	6
117	aP	12
149	wP	5
4629-21	aP	6
4629-31	wP	4
4629-32	wP	10
4629-33	aP	16
4629-39	wP	6
4629-49	wP	12
4629-60	wP	6
4629-61	aP	6
4629-87	aP	5
4629-92	aP	5
4629-117	aP	7
4629-132	aP	4
4629-145	aP	7
4629-168	wP	3
4629-191	wP	3
4629-205	wP	13
4629-212	aP	9
4629-214	wP	12
4629-215	wP	14
4629-228	aP	9
4629-239	wP	6
4629-245	aP	16
4629-253	wP	7
4629-260	aP	5
4630-3	aP	4
4630-4	aP	5
4630-7	aP	2
4630-14	wP	12
4630-17	aP	4
4630-20	aP	6
4630-22	aP	4
4630-23	aP	6
4630-27	aP	3
4630-28	wP	10
4630-32	wP	4
4630-33	aP	5
4630-43	aP	11
4630-45	wP	4
4630-54	wP	4
4630-57	wP	4
5785-1	wP	11
5785-2	wP	4
5785-7	wP	3
5785-22	aP	3
5785-39	aP	3
5785-44	wP	8
5785-49	aP	9
5785-53	wP	5
5785-70	wP	3
5785-78	aP	5

5785-91	wP	5
5785-93	wP	3
5785-98	aP	4
5785-112	aP	4
5785-122	aP	2
5785-123	aP	3
5785-124	wP	7
5785-133	aP	9
5785-145	aP	6
5785-169	aP	3
5785-170	wP	4
5785-192	wP	7
5785-203	aP	6
5785-207	wP	6
5785-211	aP	6
5785-218	aP	6
5785-265	aP	6
5785-266	aP	6
5785-275	aP	4
5785-283	wP	2
5785-288	aP	14
5785-295	aP	5
5785-301	wP	3
5785-319	wP	3
5785-320	aP	8
5785-332	wP	6
5785-349	aP	6
5785-358	wP	9
5785-370	aP	3
5785-372	wP	8
5785-380	wP	6
5785-384	aP	2
8644-14	wP	2
8644-19	aP	7
8644-24	wP	18
8644-27	wP	3
8644-34	aP	4
8644-41	aP	3
8644-44	wP	4
8644-51	wP	2
8644-58	wP	15

8.3 Additional IgG concentration summaries

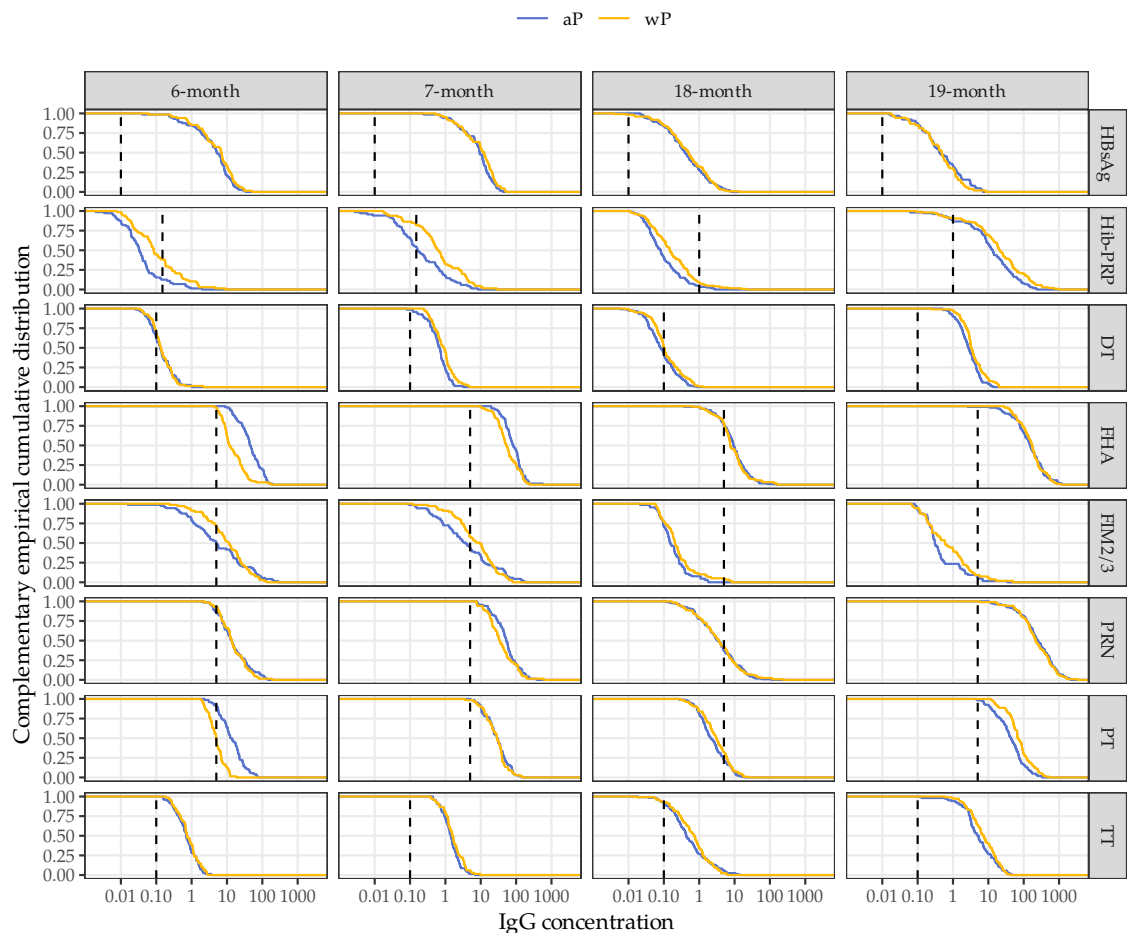


Figure 32: Complementary empirical cumulative distribution of serotype-specific IgG concentrations by age and assigned schedule. Vertical line indicates seropositivity reference value.

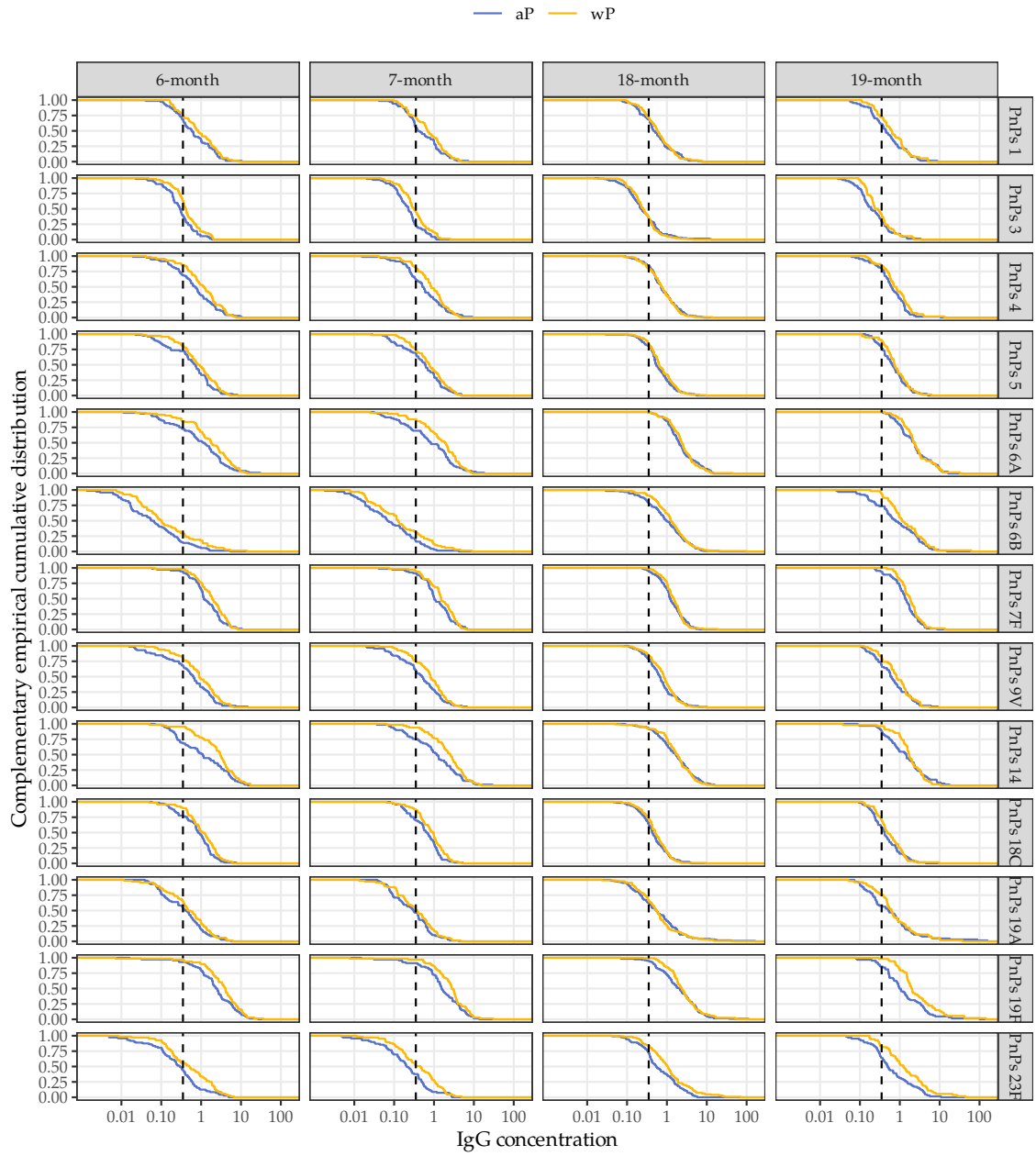


Figure 33: Complementary empirical cumulative distribution of serotype-specific IgG concentrations by age and assigned schedule. Vertical line indicates 0.35 µg/mL.

References

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